Class 9
Chemistry

Table of Content

1.	Nitrogen and Oxygen	1 - 14
2.	Chemical Bonding	15 – 33
3.	Gaseous State	34 – 41
4.	Carbon and Its Compounds	42 – 63
5.	Chemical Reactions And Equations	64 – 74
6.	Stoichiometry	75 – 102

1.Nitrogen and Oxygen Solutions

Level – 1

Single Answer type Questions

Nitrogen

1. **The correct answer is: (C)**.

Ammonium nitrite (NH_4NO_2) decomposes on heating to produce nitrogen gas and water vapor. This decomposition reaction occurs at temperatures around 60-70°C. The balanced chemical equation for this reaction is: $NH_4NO_2 \rightarrow N_2 + 2H_2O$

This reaction is exothermic and can be potentially explosive if large quantities are heated rapidly. Ammonium nitrite is unstable at room temperature and slowly decomposes, which is why it's not commonly stored or used.

2. **The correct answer is: (B).**

Earth's atmosphere is primarily composed of nitrogen and oxygen, with these two gases making up the vast majority of dry air. The approximate composition of dry air by volume is about 78% nitrogen (N_2) and 21% oxygen (O_2), with the remaining 1% consisting of other gases like argon, carbon dioxide, and trace amounts of various other elements.

3. **The correct answer is: (A).**

Nitrogen gas is typically collected in the laboratory by the method of downward displacement of water. This method is effective because nitrogen is less dense than air and insoluble in water.

When nitrogen is produced or released in a reaction, it's directed through a delivery tube into an inverted container filled with water. As the nitrogen enters, it displaces the water downwards, gradually filling the container from top to bottom.

4. **The correct answer is: (D).**

Nitrogen (N_2) is an inert gas under normal conditions. It is neither combustible itself nor does it support combustion of other substances. This property is due to the strong triple bond between the two nitrogen atoms in a molecule, making it very stable and unreactive at room temperature and pressure.

Nitrogen's inertness is why it's often used to create inert atmospheres in various industrial processes and applications where oxidation or combustion needs to be prevented.

5. **The correct answer is: (A).**

Nitrogen (N_2) has a lower boiling point than oxygen (O_2) and carbon dioxide (CO_2). Nitrogen boils at -196°C (-321°F), while oxygen boils at -183°C (-297°F). When liquid air is warmed, nitrogen, having the lowest boiling point among the components of air, will boil off first.

This separation process takes advantage of the different boiling points of the gases in air to isolate nitrogen from other components.

6. **The correct answer is: (C)**.

The atomic number of an element is defined as the number of protons in the nucleus of an atom of that element. For nitrogen, this number is 7. This means that every nitrogen atom, regardless of its isotope, has 7 protons in its nucleus.

The atomic number 7 for nitrogen determines its position in the periodic table, placing it in the second period and the 15th group (also known as the nitrogen group

or pnictogens). This atomic structure gives nitrogen its characteristic properties, including its ability to form triple covalent bonds in molecular nitrogen (N_2)

7. The correct answer is: (D).

In the gaseous state, nitrogen can exist in various forms, including nitrous oxide (N_2O), nitric oxide (NO), and ammonia (NH_3). Each of these compounds represents a different oxidation state of nitrogen and has unique properties and applications:

1. Nitrous oxide (N_2O): Also known as laughing gas, it's used as an aesthetic and in whipped cream dispensers.
2. Nitric oxide (NO): A signalling molecule in the body and an air pollutant.
3. Ammonia (NH_3): Used in fertilizers and cleaning products.

8. The correct answer is: (B).

Nitrogen gas is sparingly soluble in water. Meaning it dissolves in water to a very limited extent. The solubility of nitrogen in water is relatively low compared to other gases like oxygen or carbon dioxide. At room temperature and standard pressure, the solubility of nitrogen in water is approximately 0.02 grams per litre.

This low solubility is due to the non-polar nature of nitrogen molecules, which don't interact strongly with polar water molecules.

9. The correct answer is: (A).

Nitrogen gas (N_2) is a diatomic molecule that exists as a colourless and odorless gas under normal atmospheric conditions. It makes up about 78% of Earth's atmosphere by volume, making it the most abundant gas in our air. Nitrogen's lack of colour and odor is due to its strong triple covalent bond between the two nitrogen atoms, which makes the molecule very stable and non-reactive at room temperature.

10. The correct answer is: (B).

The atomicity of nitrogen is 2, which means that nitrogen exists as a diatomic molecule in its natural gaseous state. This is represented by the chemical formula N_2.

The nitrogen atom has 5 valence electrons, and to achieve a stable octet configuration, two nitrogen atoms share three pairs of electrons, forming a triple covalent bond.

This strong triple bond (N≡N) makes nitrogen gas very stable and non-reactive under normal conditions.

Compounds of Nitrogen

11. The correct answer is: (C).

Ammonia (NH_3) is collected by upward displacement of air due to its unique properties. Ammonia is a gas at room temperature and pressure, but it's less dense than air. Its molecular weight is about 17 g/mol, compared to air's average of about 29 g/mol.

This lower density causes ammonia to rise in air. Additionally, ammonia is highly soluble in water, so it can't be collected over water like some other gases. When preparing ammonia in a laboratory, the gas is typically directed into an inverted container. As the ammonia enters from the bottom, it pushes the air upwards and out of the container, gradually filling it from the bottom up.

12. The correct answer is: (A).

Nitric acid (HNO_3) is a strong oxidizing agent, means it readily accepts electrons from other substances, causing them to be oxidized. This property makes nitric acid a powerful and versatile chemical in various industrial and laboratory applications. As an oxidizing agent, nitric acid can oxidize many metals, non-metals, and organic compounds. For example, it can oxidize copper to copper (II) nitrate:

$3Cu + 8HNO_3 \rightarrow 3Cu(NO_3)_2 + 2NO + 4H_2O$

13. The correct answer is: (B).

Nitric acid has a unique effect on certain metals like nickel, cobalt, and iron, causing them to become passive. This phenomenon, known as passivation, occurs when concentrated nitric acid reacts with these metals, forming a thin, protective oxide layer on their surface. This oxide layer acts as a barrier, preventing further reaction between the metal and the acid.

For example, with iron:

$Fe + HNO_3 \rightarrow Fe_2O_3$ (thin layer) + nitrogen oxides + H_2O

14. The correct answer is: (B).

Ammonia is a reducing agent due to its ability to donate electrons in chemical reactions. This property stems from the lone pair of electrons on the nitrogen atom. Ammonia can reduce metal ions to their elemental state and participate in various redox reactions. For example, it can reduce copper (II) ions to copper metal:

$2Cu^{2+} + 4NH_3 + H_2O \rightarrow 2Cu + 4NH^{4+} + O_2$

In this reaction, ammonia acts as the reducing agent by donating electrons to the copper ions, reducing them from Cu^{2+} to Cu.

15. The correct answer is: (B).

Fresh nitric acid (HNO_3) is a colourless liquid when pure. It's a strong, corrosive mineral acid with important industrial applications. Over time, nitric acid can decompose and develop a pale yellow or brownish colour due to the formation of nitrogen dioxide (NO_2). This occurs when the acid is exposed to light or heat, following the equation: $4\ HNO_3 \rightarrow 4\ NO_2 + 2\ H_2O + O_2$

16. The correct answer is: (C).

When ammonium hydroxide is added to copper sulphate solution, it initially forms a pale blue precipitate of copper (II) hydroxide. This reaction can be represented by the equation: $CuSO_4 + 2NH_4OH \rightarrow Cu\ (OH)_2 + (NH_4)_2SO_4$

The pale blue precipitate of Cu (OH)2 is the characteristic colour observed in this reaction. As more ammonium hydroxide is added, the precipitate may dissolve to form a deep blue complex ion $[Cu\ (NH_3)_4]^{2+}$, but the initial precipitate is pale blue.

17. The correct answer is (D).

The correct product of the reaction between ammonia (NH_3) and water (H_2O) is ammonium hydroxide (NH_4OH). When ammonia dissolves in water, it forms ammonium ions (NH^{4+}) and hydroxide ions (OH^-). The reaction can be represented as $NH_3 + H_2O \rightarrow NH_4OH$

18. The correct answer is: (B).

Nitric acid can dissolve silver metal through a chemical reaction that forms silver nitrate, nitrogen dioxide, and water. This process is an example of an oxidation-reduction reaction, where silver is oxidized, and nitric acid is reduced. The chemical equation for this reaction is:

Ag (silver) + $2HNO_3$ (nitric acid) $\rightarrow$ $AgNO_3$ (silver nitrate) + NO_2 (nitrogen dioxide) + H_2O (water)

19. **The correct answer is (D).**

Tungsten (W) – Does not continue to burn in nitrogen because it is much less reactive and does not readily form a nitride under these conditions.

LEVEL – II

SINGLE OPTION CORRECT TYPE

1. The correct answer is (B).

When a rod dipped in ammonia is brought near a jar containing chlorine gas, a chemical reaction occurs, forming dense white fumes. These fumes are composed of

ammonium chloride (NH_4Cl). This reaction demonstrates the strong affinity between ammonia and chlorine. The process can be represented by the following equation:
$NH_3 + HCl \rightarrow NH_4Cl$

2. **The correct answer is (A**).
The single largest use of nitrogen is in the manufacture of ammonia. This process, known as the Haber-Bosch process, combines atmospheric nitrogen with hydrogen to produce ammonia. The equation for this reaction is:
$N_2 + 3H_2 \rightarrow 2NH_3$
Ammonia is a crucial compound used in the production of fertilizers, which are essential for modern agriculture.

3. The correct answer is (C).
Noble metals, such as gold and platinum, are highly resistant to chemical attack and do not react with most acids individually. However, they can be dissolved in a specific mixture of hydrochloric and nitric acids known as aqua regia.
This mixture, with a ratio of 3 parts HCl to 1-part HNO_3, creates a powerful oxidizing environment that can dissolve noble metals. The reaction produces complex chloride compounds of the metal. For example, with gold:
$Au + HNO_3 + 4HCl \rightarrow HAuCl_4 + NO + 2H_2O$

4. **The correct answer is: (A).**
When ammonia (NH_3) reacts with excess chlorine (Cl_2), it undergoes a series of reactions. Initially, NH_3 reacts with Cl_2 to form nitrogen trichloride (NCl_3) and hydrogen chloride (HCl). The overall reaction can be represented as:
$NH_3 + 3Cl_2 \rightarrow NCl_3 + 3HCl$

5. **The correct answer is: (C)**.
When ammonium nitrate is heated, it decomposes to produce nitrous oxide and water vapor. This reaction is an endothermic process and can be represented by the following equation:
$NH_4NO_3 \rightarrow N_2O + 2H_2O$
Nitrous oxide, also known as laughing gas, is a colorless gas with a slightly sweet odor. This decomposition reaction is important in various applications, including the production of nitrous oxide for medical and industrial uses.

6. The correct answer is: (C).
Calcium oxide (CaO), also known as quicklime, is used to dry ammonia gas. This is because ammonia reacts with most common drying agents, including concentrated sulfuric acid and phosphorus pentoxide. The reaction between ammonia and calcium oxide is relatively mild and doesn't lead to unwanted side reactions.
$CaO + H_2O \rightarrow Ca(OH)_2$
Calcium oxide absorbs moisture from the ammonia gas, forming calcium hydroxide. This reaction effectively removes water without consuming the ammonia.

7. **The correct answer is: (D)**.
Ammonia is a versatile chemical compound with numerous industrial applications. It's used in the manufacture of fertilizers, explosives, and nitric acid, among other products.
In fertilizer production, ammonia is a key component in making ammonium nitrate and urea. For explosives, it's used to produce ammonium nitrate, a common ingredient in many industrial explosives. In nitric acid production, ammonia is oxidized in the Ostwald process:
$4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$
$2NO + O_2 \rightarrow 2NO_2$

$3NO_2 + H_2O \rightarrow 2HNO_3 + NO$

8. The correct answer is: (C).

In the laboratory preparation of nitrogen gas, the most common method involves the thermal decomposition of ammonium nitrite or the reaction between ammonium chloride and sodium nitrite. The latter reaction is more frequently used and produces nitrogen gas along with water and sodium chloride as byproducts. The reaction can be represented as:

$NH_4Cl + NaNO_2 \rightarrow N_2 + 2H_2O + NaCl$

9. The correct answer is: (A).

When magnesium (Mg) reacts with nitrogen (N_2), it forms magnesium nitride (Mg_3N_2). This reaction occurs at high temperatures, typically above 600°C. The balanced equation for this reaction is: $3Mg + N_2 \rightarrow Mg_3N_2$

10. The correct answer is: (D).

Both reactions (b) and (c) produce ammonia as a product:

(b) $2NH_4Cl + CaO \rightarrow 2NH_3 + CaCl_2 + H_2O$

This is a displacement reaction where calcium oxide (quicklime) reacts with ammonium chloride to produce ammonia gas.

(c) $N_2 + 3H_2 \rightleftharpoons 2NH_3$

This is the Haber process, a reversible reaction where nitrogen and hydrogen gases combine to form ammonia under high pressure and temperature with an iron catalyst.

11. The correct answer is: (C).

When ammonia reacts with a limited amount of chlorine, the main product is ammonium chloride (NH_4Cl). The reaction can be represented as:

$NH_3 + Cl_2 \rightarrow NH_4Cl + HCl$

With excess chlorine, the reaction proceeds further to form nitrogen trichloride (NCl_3):

$NH_3 + 3Cl_2 \rightarrow NCl_3 + 3HCl$

12. The correct answer is: (A).

Ammonium nitrate (NH_4NO_3) is widely used in the production of explosives due to its strong oxidizing properties and ability to rapidly decompose under certain conditions. When heated or detonated, it breaks down to release a large volume of gases:

$NH_4NO_3 \rightarrow N_2O + 2H_2O$

This reaction produces nitrous oxide and water vapor, accompanied by a significant energy release. Ammonium nitrate's explosive potential is harnessed in various applications, including mining and construction.

13. The correct answer is: (D).

When dilute nitric acid (HNO_3) is added to sodium hydroxide (NaOH), a neutralization reaction occurs. This reaction produces sodium nitrate ($NaNO_3$) and water. The chemical equation for this reaction is:

$HNO_3 + NaOH \rightarrow NaNO_3 + H_2O$

Sodium nitrate is a salt formed from the strong base (NaOH) and strong acid (HNO_3). This reaction doesn't produce sodium carbonate or sodium nitrite. Sodium nitrate is a white, crystalline solid that's highly soluble in water.

14. The correct answer is (A).

When phosphorus reacts with concentrated nitric acid (HNO_3), it forms phosphoric acid (H_3PO_4). This reaction is an oxidation process where nitric acid, being a strong oxidizing agent, oxidizes phosphorus to its highest oxidation state (+5). The general reaction can be represented as:

$P + 5HNO_3 \rightarrow H_3PO_4 + 5NO_2 + H_2O$

The nitric acid not only oxidizes the phosphorus but also provides the oxygen atoms necessary to form the oxyacid. This reaction is exothermic and produces nitrogen dioxide gas (NO_2) as a byproduct.

15. The correct answer is: (D).

This principle is used in the fractional distillation of liquid air to separate nitrogen and oxygen. Liquid air is slowly warmed in a tall column. Nitrogen, having a lower boiling point (-195.8°C) than oxygen (-183°C), vaporizes first and rises to the top of the column. Oxygen, with a higher boiling point, remains liquid longer and collects at the bottom. This difference of about 13°C in boiling points allows for efficient separation. The process is crucial in industrial gas production, providing pure nitrogen and oxygen for various applications.

16. The correct answer is: (C).

Magnesium is the most reactive metal among the given options and will readily liberate hydrogen when treated with very dilute nitric acid (HNO_3).

This reaction occurs because magnesium is above hydrogen in the reactivity series, allowing it to displace hydrogen from acids. The reaction can be represented by the equation: $Mg + 2HNO_3 \rightarrow Mg(NO_3)_2 + H_2$

17. The correct answer is: (D).

When slaked lime (calcium hydroxide, $Ca(OH)_2$) reacts with ammonium chloride (NH_4Cl), a double displacement reaction occurs. This reaction produces calcium chloride ($CaCl_2$) and ammonia (NH_3). The process is often used in laboratories to generate ammonia gas. The reaction can be represented by the following equation:
$Ca(OH)_2 + 2NH_4Cl \rightarrow CaCl_2 + 2NH_3 + 2H_2O$

18. The correct answer is: (A).

When ammonia (NH_3) reacts with potassium (K), it forms potassium amide (KNH_2). This reaction is an example of a metal-ammonia reaction where the highly reactive alkali metal potassium displaces one hydrogen atom from ammonia. The reaction can be represented by the following equation: $2K + 2NH_3 \rightarrow 2KNH_2 + H_2$

One or more than one option :

19. The correct answer is: (A,B,C).

Nitrogen is found in several forms in nature:

A) Atmosphere

About 78% of the Earth's atmosphere consists of nitrogen gas (N_2).

B) Ammoniumsalts

Nitrogen is present in compounds such as ammonium chloride (NH_4Cl) and ammonium sulphate ($(NH_4)_2SO_4$).

C) Nitrates in soil

Nitrogen occurs as nitrates like sodium nitrate ($NaNO_3$) and potassium nitrate (KNO_3).

20. The correct answer is: (A,B,C).

Heating ammonium nitrite

Ammonium nitrite decomposes as:

$\boxed{NH_4NO_2 \rightarrow N_2 + 2H_2O}$

This is the common laboratory method for preparing nitrogen.

B) Decomposition of sodium azide

$\boxed{2NaN_3 \rightarrow 2Na + 3N_2}$

his reaction produces nitrogen gas and is used in applications such as airbags.

C) Fractional distillation of liquid air

This is the industrial method of obtaining nitrogen from air.

21. **The correct answer is: (A,B)**

A) Nitrogen molecules contain a strong **triple bond (N≡N)**.

B) The N≡N bond has very high bond dissociation energy, making it difficult to break.

22. **The correct answer is: (A,B,D)**

A) It is chemically **inert** under ordinary conditions.

B) It does **not support combustion**, making it useful for preserving food and preventing fires.

D) It is **stable** because of the strong triple bond.

23. The correct answer is: (A,C)

A) Correct.

$\boxed{NH_4NO_2 \rightarrow N_2 + 2H_2O}$

B) Oxygen is **not** produced.

C) Ammonium nitrite is unstable and may decompose rapidly if not carefully controlled. Hence, the reaction can become hazardous.

Assertion-Reason Type Questions

24. Both the assertion (A) and reason (R) are incorrect.

Assertion (A) is incorrect because the simple burning of ammonia does not yield only nitrogen and water. When ammonia burns in oxygen, it primarily produces nitrogen monoxide (NO) and water. The reaction can be represented as:

$4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$

Reason (R) is also incorrect. Ammonia is not acidic; it is actually a weak base. In water, ammonia acts as a proton acceptor, forming ammonium ions (NH^{4+}) and hydroxide ions (OH^-):

$NH_3 + H_2O \rightleftharpoons NH^{4+} + OH^-$

This reaction gives water containing ammonia a basic p_H.

25. The correct answer is that both the assertion (A) and reason (R) are true, but R is not the correct explanation for A.

Assertion (A) is true: Ammonium nitrate (NH_4NO_3) can be used to prepare nitrogen gas, typically through thermal decomposition at high temperatures:

$NH_4NO_3 \rightarrow N_2O + 2H_2O$

$2N_2O \rightarrow 2N_2 + O_2$

Reason (R) is also true: Ammonium nitrite (NH_4NO_2) is indeed unstable and decomposes readily: $NH_4NO_2 \rightarrow N_2 + 2H_2O$

26. Both the Assertion (A) and Reason (R) are correct, and R is the correct explanation for A.

Freshly prepared nitric acid (HNO_3) is indeed colorless. However, when exposed to light or left standing for some time, it undergoes partial decomposition. This decomposition produces nitrogen dioxide (NO_2), a brown gas, according to the reaction:

$4HNO_3 \rightarrow 4NO_2 + 2H_2O + O_2$

The NO_2 formed remains partially dissolved in the acid, imparting a yellowish-brown tinge to the solution. This process is accelerated by exposure to light and heat.

27. P. Atmosphere → b. 78% by volume

Explanation:

Nitrogen is the most abundant gas in the Earth's atmosphere, making up about 78% by volume. P → b

Q. Soil → c. Nitrates and ammonium salts
Explanation:
In the soil, nitrogen is present mainly in the form of nitrates (NO_3^-) and ammonium salts (NH_4^+), which are absorbed by plants.
Q → c
R. Living organisms → a. Proteins and DNA
Explanation:
Nitrogen is an essential component of:
Proteins (made up of amino acids)
DNA and RNA (genetic material)
Hence, nitrogen is present in all living organisms.
R → a
S. Fertilizers → d. Urea and ammonium sulfate
Explanation:
Nitrogen is supplied to crops through nitrogen-rich fertilizers such as:
Urea [$CO(NH_2)_2$]
Ammonium sulfate [$(NH_4)_2SO_4$]
These fertilizers promote healthy plant growth.S → d

Comprehension type:

28. In the laboratory, nitrogen gas is prepared by heating ammonium nitrite (NH_4NO_2). It decomposes into nitrogen gas and water.

$\boxed{NH_4NO_2 \xrightarrow{\Delta} N_2 + 2H_2O}$

Ammonium chloride (NH_4Cl) does not produce nitrogen on heating.
Sodium nitrate ($NaNO_3$) gives sodium nitrite and oxygen on heating.
Ammonium sulfate ($(NH_4)_2SO_4$) does not produce nitrogen gas on heating.

29. From the reaction:
$NH_4NO_2 \xrightarrow{\Delta} N_2 + 2H_2O$
The products formed are:
Nitrogen gas (N_2)
Water (H_2O)
No oxygen gas is produced.

30. Heating ammonium nitrite is the laboratory method for preparing nitrogen.
The industrial method is fractional distillation of liquid air.

31. The composition of dry air is approximately:
Nitrogen = **78%** ,Oxygen = **21%** ,Argon = **0.93%** ,Carbon dioxide = **0.04%**

32. In a nitrogen molecule (N_2), the two nitrogen atoms are joined by a **triple covalent** bond.$\boxed{N \equiv N}$
A triple bond consists of:
1 sigma (σ) bond
2 pi (π) bonds
Total number of bonds = 3

33. The atomic number of nitrogen is 7, which means a nitrogen atom contains:
7 protons
7 electrons (in a neutral atom)
Electronic configuration:2, 5

OXYGEN

LEVEL – I

1. The correct answer is: (C).

Oxygen is the most abundant element in the Earth's crust, making up approximately 46.6% of its mass. It is primarily found in various oxide compounds and silicate minerals that form the bulk of rocks and soil. Although oxygen is typically thought of as a gas in the atmosphere, in the Earth's crust it is bound in solid mineral forms.

2. The correct answer is: (A).

Alloying is the process of mixing a metal with one or more other elements, which can be metals or non-metals, to form a homogeneous mixture called an alloy. This mixture is designed to have properties superior to those of the individual components.

The elements are typically melted together and then cooled to form the alloy. Alloying can improve various characteristics such as strength, corrosion resistance, ductility, or electrical properties. Examples of common alloys include steel (iron and carbon), brass (copper and zinc), and bronze (copper and tin).

3. The correct answer is: (D).

Mercury (II) oxide is commonly used for the laboratory preparation of oxygen due to its relatively low decomposition temperature and the ease of the reaction. When heated, HgO decomposes to produce oxygen gas and mercury metal. This reaction is represented by the equation: $2HgO \rightarrow 2Hg + O_2$

4. The correct answer is: (B).

Liquid oxygen is used as a supporter of combustion in rockets and spaceships. It acts as an oxidizer, providing the necessary oxygen to burn the fuel, typically liquid hydrogen or another fuel source, in the rocket engines. This combination produces a high-energy reaction that generates the thrust needed to propel rockets into space. The chemical reaction can be represented as:

$2H_2$ (liquid hydrogen) + O_2 (liquid oxygen) $\rightarrow 2H_2O$ (water) + energy

5. The correct answer is: (B).

Magnesium continues to burn in an atmosphere of carbon dioxide (CO_2). This is because magnesium is highly reactive and can strip oxygen from the carbon dioxide molecule. When magnesium burns in CO_2, it reacts with the carbon dioxide to form magnesium oxide (MgO) and carbon (C). The chemical equation for this reaction is:

$2Mg + CO_2 \rightarrow 2MgO + C$

6. The correct answer is (B).

Soda water, also known as carbonated water, contains an excess amount of carbon dioxide (CO_2) dissolved in water. When CO_2 is dissolved in water, it forms carbonic acid (H_2CO_3), which gives soda water its characteristic fizz and slightly acidic taste. The dissolution process can be represented by the following chemical equation:

$CO_2 + H_2O \rightarrow H_2CO_3$

7. The correct answer is: (A).

Deep-sea diving can lead to excess intake of oxygen, a condition known as oxygen toxicity or hyperoxia. This occurs when divers breathe pressurized air or oxygen-enriched mixtures at depth.

As a diver descends, the increased pressure causes more oxygen to dissolve in the blood and tissues. At high partial pressures, oxygen becomes toxic, potentially causing seizures, lung damage, and other serious health issues.

8. The correct answer is (B).

Ozone (O_3) exhibits a dual nature depending on its location in the atmosphere. In the troposphere, the lowest layer of Earth's atmosphere, ozone is considered a pollutant.

It's a key component of smog and can cause respiratory issues and damage vegetation. Tropospheric ozone forms from reactions between nitrogen oxides and volatile organic compounds in the presence of sunlight.

9. The correct answer is (B).

Oxygen is typically collected by the downward displacement of water because it is not very soluble in water. This method ensures that the gas is captured efficiently. The reaction to generate oxygen in the lab, for example, from the decomposition of hydrogen peroxide (H_2O_2), can be represented as:

$2H_2O_2 \rightarrow 2H_2O + O_2$

Carbon dioxide is collected by the upward displacement of air because it is denser than air. This method allows the heavier CO2 to displace the lighter air in the collection container. An example reaction is the reaction of calcium carbonate with hydrochloric acid: $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$

10. The correct answer is: (B).

Calcium (Ca) burns in oxygen with a brilliant golden yellow flame. When calcium is heated, it reacts vigorously with oxygen to form calcium oxide (CaO). The reaction can be represented by the following chemical equation: $2Ca + O_2 \rightarrow 2CaO$

11. The correct answer is: (B).

CO_2 (Carbon dioxide)

Carbon dioxide is the major greenhouse gas responsible for global warming due to increased emissions from burning fossil fuels, deforestation, and industrial activities.

12. The correct answer is: (D).

Oxygen was historically called the acid former because it was once believed to be present in all acids.

13. The correct answer is: (B).

Ozone (O_3) present in the stratosphere plays a crucial role in protecting life on Earth's surface by absorbing harmful ultraviolet (UV) rays from the Sun. The stratospheric ozone layer, also known as the ozone layer, acts as a shield, preventing most of the Sun's high-energy UV radiation from reaching the Earth's surface. Without this protective layer, life on Earth would be exposed to increased levels of UV radiation, leading to a higher incidence of skin cancer, cataracts, and other health issues, as well as negative effects on ecosystems.

14. The correct answer is: (A).

Zinc is commonly used in alloying with iron to create galvanized steel, which is highly resistant to corrosion. This process involves coating iron or steel with a layer of zinc through galvanization, where the zinc layer acts as a sacrificial anode. When exposed to corrosive elements, such as moisture and oxygen, zinc corrodes preferentially instead of the underlying iron or steel.

15. **The correct answer is: (B).**

The hole in the ozone layer is primarily caused by chlorofluorocarbons (CFCs), which were widely used as refrigerants in air conditioners and refrigerators. When released, CFCs rise to the stratosphere where ultraviolet radiation breaks them down, releasing chlorine atoms. These chlorine atoms catalyse the destruction of ozone molecules through a chain reaction:

$Cl + O_3 \rightarrow ClO + O_2$

$ClO + O \rightarrow Cl + O_2$

16. The correct answer is: (D).

Acid rain, caused by the emission of sulphur dioxide and nitrogen oxides that react with water in the atmosphere, has significant negative impacts on aquatic

ecosystems. When acid rain falls into lakes, rivers, and streams, it lowers the pH of the water. This increased acidity can be lethal to many aquatic organisms, including fish, amphibians, and various invertebrates.

The chemical reaction for acid rain formation can be simplified as:

$SO_2 + H_2O \rightarrow H_2SO_3$ (sulphurous acid)

17. The correct answer is: (C).

O_2 (Oxygen)

Oxygen is the **most abundant element** in the Earth's crust

18. The correct answer is: (D).

Chromium is added to iron to produce stainless steel.

It forms a protective chromium oxide layer, preventing corrosion.

19. The correct answer is: (A).

Fountain ink pens tend to leak at higher altitudes due to the decrease in atmospheric pressure as elevation increases. This phenomenon occurs because the air pressure inside the pen's ink reservoir, which is initially equal to the atmospheric pressure at lower altitudes, becomes higher than the external air pressure at higher elevations. This pressure difference causes the ink to be pushed out of the pen, resulting in leaks.

20. The correct answer is: (B).

Older air conditioners released CFCs, which are the main pollutants responsible for ozone depletion.

21. The correct answer is: (D).

Death of aquatic animals

Acid rain lowers the pH of lakes and rivers, making the water acidic and causing the death of fish and other aquatic organisms.

22. The correct answer is: (C).

A gas collected by the upward displacement of air implies that the gas is denser (heavier) than air. Additionally, the fact that the gas cannot be collected by displacement of water, either downward or upward, indicates that the gas is highly soluble in water. When a gas is highly soluble in water, it dissolves readily and cannot be collected using water displacement methods.

23. The correct answer is (B).

This order represents the steps in the Ostwald process for industrial nitrogen fixation:

1. Formation of nitric oxide (3): $N_2 + O_2 \rightarrow 2NO$

This occurs at high temperatures in the presence of air.

2. Formation of nitrogen dioxide (2): $2NO + O_2 \rightarrow 2NO_2$

Nitric oxide further oxidizes to form nitrogen dioxide.

3. Formation of nitric acid (4): $3NO_2 + H_2O \rightarrow 2HNO_3 + NO$

Nitrogen dioxide reacts with water to produce nitric acid.

4. Formation of metal nitrates (1): HNO_3 + Metal → Metal nitrate + H_2

The nitric acid then reacts with metals to form metal nitrates.

LEVEL – II

1. The correct answer is: (D).

Respiration is the process by which organisms break down glucose to release energy for cellular activities. In aerobic respiration, which is the most common and efficient form, glucose is completely oxidized using oxygen. The overall equation for this process is:

$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + Energy (ATP)

The end products are:
1. Carbon dioxide (CO_2): Released as a waste gas.
2. Water (H_2O): Produced as a byproduct.
3. Energy: Primarily in the form of ATP (adenosine triphosphate), which cells use for various functions.

2. The correct answer is: (A).

Carbon dioxide (CO_2) is the primary greenhouse gas responsible for trapping heat in the Earth's atmosphere, contributing significantly to the greenhouse effect. While other gases like methane and water vapor also contribute, CO_2 is the most prevalent and long-lasting.

CO_2 molecules absorb and re-emit infrared radiation, which is heat energy from the Earth's surface trying to escape into space. This process traps heat in the lower atmosphere, warming the planet. The greenhouse effect is natural and necessary for life, but human activities have dramatically increased CO_2 levels, enhancing this effect.

Solar radiation → Earth's surface → Infrared radiation → CO2 absorption → Heat trapped in atmosphere

3. The correct answer is: (B).

Excess carbon dioxide (CO_2) in the atmosphere is primarily responsible for global warming. As a greenhouse gas, CO_2 traps heat in the Earth's atmosphere, leading to an overall increase in global temperatures. This process occurs because:
1. Solar radiation passes through the atmosphere and warms the Earth's surface.
2. The Earth emits infrared radiation back towards space.
3. CO_2 molecules absorb this infrared radiation and re-emit it in all directions, including back towards the Earth's surface.

Solar energy in - Energy radiated to space = Energy trapped in the Earth system.

4. The correct answer is (D).

Plants and animals require a combination of rainfall, sunlight, and oxygen for their survival. Rainfall provides the necessary water for hydration and nutrient transport within plants and animals.

Sunlight is crucial for photosynthesis in plants, which produces the oxygen and food needed by both plants and animals. Oxygen is essential for cellular respiration in both plants and animals, enabling them to convert food into energy. Each of these elements plays a vital role in maintaining the life processes and overall health of ecosystems.

5. The correct answer is: (C).

Nitrogen is the most abundant gas in the Earth's atmosphere, making up about 78% by volume. It is followed by oxygen, which comprises around 21%. Carbon dioxide is present in much smaller quantities, approximately 0.04%, while lithium is not found in the atmosphere as a gas. Nitrogen's high concentration is due to its chemical stability and the nitrogen cycle, which recycles nitrogen between the atmosphere, living organisms, and the soil.

This abundance of nitrogen is crucial for various biological processes, including the synthesis of proteins and nucleic acids in living organisms.

6. The correct answer is: (B).

Green plants use carbon dioxide (CO_2) as a key ingredient in photosynthesis, the process by which they produce their own food. During photosynthesis, plants take in CO_2 from the air through tiny pores in their leaves called stomata. They combine

this CO_2 with water (H_2O) and light energy from the sun to produce glucose ($C_6H_{12}O_6$) and oxygen (O_2).

$6CO_2 + 6H_2O$ + light energy → $C_6H_{12}O_6 + 6O_2$

7. The correct answer is (C).

Oxygen is the essential gas used by human beings for breathing. It is crucial for the process of cellular respiration, where it helps in the breakdown of glucose to produce energy, water, and carbon dioxide. The equation for cellular respiration is:

$C_6H_{12}O_6$ (glucose) + $6O_2$ (oxygen) → $6CO_2$ (carbon dioxide) + $6H_2O$ (water) + energy (ATP)

Inhaled oxygen is transported by the blood to the cells, where it is used to generate energy necessary for various bodily functions. Without oxygen, human cells cannot perform vital processes, leading to life-threatening consequences.

8. The correct answer is (A,B,C,D).

Oxygen can be prepared in the laboratory by several methods:

Heating mercuric oxide (HgO) produces oxygen.

$2HgO \xrightarrow{\Delta} 2Hg + O_2$

Decomposition of hydrogen peroxide (H_2O_2) in the presence of manganese dioxide (MnO_2) gives oxygen.

$2H_2O_2 \xrightarrow{MnO_2} 2H_2O + O_2$

Electrolysis of water also produces oxygen at the anode.

Heating potassium nitrate (KNO_3) produces oxygen.

$2KNO_3 \xrightarrow{\Delta} 2KNO_2 + O_2$

Since the question asks which methods prepare oxygen, **all four are correct.**

9. The correct answer is (A,B,D).

Carbon burns to form **CO_2**, an acidic oxide. $C + O_2 \rightarrow CO_2$

Sulphur burns to form **SO_2**, an acidic oxide. $S + O_2 \rightarrow SO_2$

Magnesium burns to form **MgO**, a **basic oxide**. $2Mg + O_2 \rightarrow 2MgO$

Phosphorus burns to form **P_4O_{10} (or P_2O_5)**, an acidic oxide. $P_4 + 5O_2 \rightarrow P_4O_{10}$

10. The correct answer is (A,B)

Rusting occurs only when **both oxygen and moisture** are present.

$4Fe + 3O_2 + xH_2O \rightarrow 2Fe_2O_3 \cdot xH_2O$

11. The correct answer is (A,B,C)

Respiration

Supports combustion

Artificial respiration (oxygen cylinders)

12. The correct answer is (A,B,C)

Reaction:$2H_2O_2 \xrightarrow{MnO_2} 2H_2O + O_2$

MnO_2 acts as catalyst

Oxygen gas is produced

Water is formed

13. Answer: C

AssertionOxygen supports combustion. True

ReasonOxygen itself burns easily. False

Oxygen **supports burning** but **does not burn itself**

14. Answer: A

Assertion: Oxygen is collected by downward displacement of water.(True)

Reason: Oxygen is only slightly soluble (almost insoluble) in water.(True)

Because oxygen is only sparingly soluble in water, it can be collected by water displacement.

15. **P → d ;Q → a ;R → c ;S → b**

16. **Answer: B) MnO_2**
MnO_2 acts as catalyst.

17. **Answer: B) Burning**
Oxygen supports burning.

18. **Answer: C) Downward displacement of water**
Because oxygen is only slightly soluble in water, it is collected by downward displacement of water.

19. Percentage of oxygen in atmospheric air (by volume)
Composition of air:
Nitrogen = 78%
Oxygen = 21%
Other gases = 1%

20. Formula:CO_2
One molecule contains:
1 Carbon atom
2 Oxygen atoms

2. Chemical Bonding

Solutions

LEVEL – I

1. The correct answer is: (B).

These elements are all noble gases, which occupy Group 18 (formerly Group 8A) of the periodic table. Noble gases are characterized by their full outer electron shells, which make them extremely stable and unreactive under normal conditions. The complete list of noble gases includes:

1. Helium (He)
2. Neon (Ne)
3. Argon (Ar)
4. Krypton (Kr)
5. Xenon (Xe)
6. Radon (Rn)

2. The correct answer is: (D).

Noble gases are the only elements that exist as independent atoms in nature under normal conditions. This unique property is due to their full outer electron shells, which make them extremely stable and unreactive. The noble gases include helium (He), neon (Ne), argon (Ar), krypton (Kr), xenon (Xe), and radon (Rn).

Their electron configurations result in:

1. No tendency to gain or lose electrons.
2. Very weak intermolecular forces.
3. Existence as monatomic gases at room temperature.

3. The correct answer is: (C).

Helium is a noble gas that exists as independent atoms in nature under normal conditions. This unique property is due to its completely filled outer electron shell, which makes it extremely stable and unreactive. Helium's electron configuration ($1s^2$) results in:

1. No tendency to form chemical bonds.
2. Existence as a monatomic gas at room temperature.
3. Inertness in most chemical environments.

4. The correct answer is (A).

Oxygen cannot exist as an independent atom under normal conditions. Instead, it naturally occurs as a diatomic molecule, O_2. This is due to oxygen's electron configuration, which Favors the formation of a covalent bond between two oxygen atoms to achieve a stable octet structure.

The formation of O_2 can be represented as: $O + O \rightarrow O_2$

In contrast, Helium (He), Argon (Ar), and Krypton (Kr) are all noble gases.

5. The correct answer is (B).

Oxygen exists in atmospheric air as diatomic molecules, O_2. This form is known as molecular oxygen or dioxygen. In the Earth's atmosphere, oxygen makes up approximately 21% of the air by volume.

The formation of O_2 can be represented as: $O + O \rightarrow O_2$

This molecular structure is crucial for life processes and many chemical reactions.

1. It's essential for respiration in most organisms
2. It participates in combustion reactions
3. It's produced by plants during photosynthesis

6. The correct answer is: (B).

Hydrogen exists in Earth's atmosphere as diatomic molecules, H2, also known as molecular hydrogen. However, it's important to note that atmospheric hydrogen is present in extremely small quantities, making up only about 0.00005% of the air by volume.

The formation of H_2 can be represented as: $H + H \rightarrow H_2$

1. It's the lightest molecule and tends to escape Earth's gravity.
2. Most atmospheric H_2 is produced by human activities and natural processes.
3. It's not a significant component of air compared to nitrogen, oxygen, or even carbon dioxide.

7. The correct answer is: (B).

Nitrogen exists in atmospheric air primarily as diatomic nitrogen molecules (N_2). This form is extremely stable due to the strong triple covalent bond between the two nitrogen atoms.

N_2 makes up approximately 78% of Earth's atmosphere by volume, making it the most abundant gas in our air.

8. The correct answer is: (B).

The equilibrium distance, also known as the bond length, is the optimal separation between two atomic nuclei at which a chemical bond forms. At this distance, the attractive and repulsive forces between the atoms are balanced, resulting in the lowest energy state for the bonded pair.

This distance represents the point where the potential energy of the system is at its minimum. It's not the least possible distance, as the nuclei would repel each other too strongly if brought closer. It's also not an infinite or arbitrary distance, as the atoms wouldn't interact to form a bond if too far apart. The term "inter nuclear distance" is a general description and doesn't specifically refer to the bonding distance.

9. The correct answer is: (A).

Atoms X and Y in an XY molecule are held together primarily by attractive forces. These attractive forces are typically electromagnetic in nature and arise from the interactions between the electrons and nuclei of the atoms.

The most common types of attractive forces in chemical bonding are covalent bonds (sharing of electrons) and ionic bonds (electrostatic attraction between oppositely charged ions).

While repulsive forces do exist between nuclei and between electrons, it's the net attractive force that allows the formation and stability of the molecule. Gravitational forces are negligible at the atomic scale, and magnetic forces are not typically responsible for holding atoms together in most molecules.

10. The correct answer is: (D).

In a molecule where sodium (Na) is the positive species (cation) and chlorine (Cl) is the negative species (anion), they form an ionic compound known as sodium chloride. In this compound, sodium donates one electron to chlorine, resulting in a sodium ion (Na^+) and a chloride ion (Cl^-).

These ions are held together by the electrostatic force of attraction between the positively charged sodium ion and the negatively charged chloride ion, forming the formula NaCl. $Na + Cl \rightarrow NaCl$

11. The correct answer is: (C).

The attractive forces that hold atoms together in a molecule are primarily due to the electrostatic attraction between the negatively charged electrons of one atom and the positively charged nuclei of another atom. This type of attraction is fundamental to the formation of chemical bonds.

For example, in a hydrogen molecule (H_2), the electrons of one hydrogen atom are attracted to the nucleus of the other hydrogen atom, leading to the formation of a covalent bond: $H + H \rightarrow H_2$

12. The correct answer is: (A).

Repulsive forces exist primarily between the nuclei of two atoms when they come close together. This repulsion is due to the positive charges of the protons in both nuclei, following Coulomb's law of electrostatic interactions.

As two atoms approach each other, their electron clouds begin to overlap, and the positively charged nuclei start to repel each other strongly. This nuclear repulsion plays a crucial role in determining the equilibrium bond length in molecules. It balances the attractive forces between electrons and nuclei, preventing atoms from collapsing into each other.

13. The correct answer is: (C).

This relationship indicates that stability is inversely proportional to energy. In chemical systems, lower energy states are generally more stable. As the energy of a system decreases, its stability increases.

This principle is fundamental in understanding chemical reactions, bond formation, and molecular structures. Stable molecules or configurations tend to have lower potential energy, making them less likely to undergo spontaneous changes. Conversely, high-energy states are less stable and more reactive.

14. The correct answer is: (C).

A system with low energy possesses high stability. In physics and chemistry, stability refers to the tendency of a system to remain in its current state without changing. Systems naturally tend to move towards a state of lower energy because it is more stable.

For example, a molecule with low potential energy is less likely to react and change its structure compared to a molecule with high potential energy. Similarly, a physical object at a low energy position, like a ball at the bottom of a hill, is in a stable state and less likely to move spontaneously. Therefore, a system with low energy is more stable.

15. The correct answer is: (A).

A system is said to be more stable if it possesses Less energy. In physics and chemistry, stability is often associated with the energy state of a system. A system with lower energy is more stable because it is less likely to undergo spontaneous changes or reactions.

This principle is evident in various contexts, such as chemical bonds, where molecules with lower potential energy are more stable and less reactive. For instance, the formation of a water molecule (H_2O) from hydrogen (H_2) and oxygen (O2) releases energy, resulting in a more stable system: $2H_2 + O_2 \rightarrow 2H_2O$ + energy

16. The correct answer is: (A).

Potential energy is the energy a system possesses due to its position or configuration. This form of energy is stored and can be converted into other forms of energy when the system's position or shape changes.

In chemistry, potential energy is crucial in understanding molecular structures, chemical bonds, and reaction energetics. It's the energy that exists in a system due to forces acting on it, such as gravitational, electrostatic, or elastic forces.

17. The correct answer is: (B).

When a system's potential energy decreases, it tends to become more stable. This principle is fundamental in chemistry and physics. Systems naturally move towards lower energy states, which are inherently more stable. As potential energy is lost, it's often converted into other forms of energy (like kinetic or thermal energy) or work, but the overall system becomes more stable.

This concept explains why chemical reactions often proceed spontaneously towards products with lower potential energy, why atoms form bonds to achieve more stable electron configurations, and why physical objects tend to move to positions of lower gravitational potential energy.

18. The correct answer is: (B).

The integral energy of a molecule is less than the sum of the integral energies of its individual atoms. This principle is fundamental to understanding chemical bonding and molecular stability.

When atoms combine to form a molecule, energy is released in the process of bond formation. This release of energy results in the molecule having a lower total energy than its constituent atoms would have separately.

This energy difference is known as the binding energy or bond energy. The lower energy state of the molecule explains why it's more stable than its separate atoms.

19. The correct answer is: (A).

Among the given options, atomic hydrogen (H) is the most unstable. This is because a single hydrogen atom has one unpaired electron in its outer shell, making it highly reactive and eager to form bonds to achieve a more stable electron configuration. In contrast, H_2 (hydrogen gas), O_2 (oxygen gas), and N_2 (nitrogen gas) are all stable diatomic molecules. These molecules have achieved lower energy states through covalent bonding, sharing electrons to fill their outer shells.

20. The correct answer is: (C).

Hydrogen chloride (HCl) is the most stable. As a covalent compound, HCl forms a strong bond between hydrogen and chlorine atoms, resulting in a lower energy state and greater stability compared to its constituent atoms.

Both H (hydrogen) and Cl (chlorine) as individual atoms are highly reactive due to their unpaired electrons, making them unstable in their atomic forms. Similarly, O (oxygen) as a single atom is unstable, as it seeks to fill its outer shell by forming bonds.

HCl's stability is evident in its existence as a common compound in both gaseous and aqueous forms. Its stability allows it to be widely used in various industrial and laboratory applications, unlike the reactive atomic forms of hydrogen, chlorine, or oxygen.

LEVEL - II

SINGLE OPTION CORRECT TYPE

1. The correct answer is: (D).

This statement is accurate because the formation of a covalent bond between two hydrogen atoms to create an H_2 molecule releases energy, making the molecule more stable than two separate hydrogen atoms. The bond formation is an exothermic process, lowering the overall energy of the system.

The bond dissociation energy of H_2 is approximately 436 kJ/mol, which means this much energy is required to break the bond. Conversely, this same amount of energy is released when the bond forms, indicating that H_2 is in a lower, more stable energy state than two individual hydrogen atoms.

2. The correct answer is: (D).

The incorrect statement is The attractive forces operate between the atoms, when the distance of atoms is more than equilibrium distance.

(A) Noble gases are stable elements. This is correct because noble gases have a complete valence shell, making them chemically inert and stable.

(B) If a substance has low energy, then it is highly stable. This is correct because stability is generally associated with lower energy states.

(C) Stability is inversely proportional to energy. This is correct because higher stability corresponds to lower energy.

(D) The attractive forces operate between the atoms, when the distance of atoms is more than equilibrium distance. This is incorrect because, at distances greater than the equilibrium distance, repulsive forces usually dominate, leading to instability.

3. The correct answer is: (C).

Chlorine (Cl) is the only unstable element among the given options. As a halogen, chlorine exists as a diatomic molecule (Cl_2) in its natural state, not as a single atom. Single chlorine atoms are highly reactive due to their unpaired electron in the outermost shell, making them unstable and eager to form bonds to achieve a stable electron configuration.

He (helium), Ne (neon), and Ar (argon) are all noble gases.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

4. The correct answer is: (A).

Matter is fundamentally composed of atoms, which are the smallest units of an element that retain its chemical properties. Atoms consist of a nucleus containing protons and neutrons, surrounded by electrons. Molecules are groups of atoms bonded together, compounds are substances made of different elements chemically combined, and elements are pure substances made of only one type of atom. The atomic structure can be represented by simple equations like:

Atom = Protons + Neutrons + Electrons

5. The correct answer is: (A, B).

Molecules are formed by (A) atoms of the same kind and (B) atoms of different kinds. A molecule is a group of two or more atoms held together by chemical bonds. These atoms can be of the same element, as in the case of an oxygen molecule (O_2):

$O + O \rightarrow O_2$

or of different elements, as in the case of a water molecule (H_2O): $2H + O \rightarrow H_2O$

6. The correct answer is: (A).

Chemical bonds are primarily formed between atoms. These bonds are the fundamental forces that hold atoms together to create molecules and compounds. While ions and molecules can participate in bonding, the most basic level of chemical bonding occurs between individual atoms.

This process involves the sharing, transfer, or attraction of electrons between atoms. The three main types of chemical bonds are ionic, covalent, and metallic bonds. Examples of bond formation can be represented by simple equations:

$Na + Cl \rightarrow NaCl$ (ionic bond)

$H + H \rightarrow H_2$ (covalent bond)

7. The correct answer is: (C)

When two neutral atoms approach each other, electrostatic forces exist between them. These forces include both attractive and repulsive components.

The attractive forces arise due to the attraction between the positively charged nuclei of one atom and the negatively charged electrons of the other atom. Repulsive forces occur due to the repulsion between the electron clouds of both atoms as they approach each other. The balance between these attractive and repulsive

electrostatic forces determines the distance at which the atoms will interact, potentially forming chemical bonds if the attractive forces dominate.

8. The correct answer is: (C).

Forces of attraction in a chemical context primarily refer to the electrostatic forces between the positively charged nuclei of one atom and the negatively charged electrons of another atom.

These attractive forces are fundamental to the formation of chemical bonds, such as ionic and covalent bonds, which hold atoms together in molecules and compounds.

For example, in a water molecule (H_2O), the hydrogen nuclei are attracted to the oxygen's electrons, creating a stable bond. This nucleus-electron attraction is crucial for the stability and structure of matter at the atomic level.

9. The correct answer is: (D)

The graph typically represents the potential energy of a system as a function of the distance between two atoms. As the atoms approach each other, their potential energy decreases, reaching a minimum point where the molecule is most stable.

This minimum represents the equilibrium bond length, indicating the distance at which the potential energy is lowest and the molecule is formed. The difference in energy from the initial state (separated atoms) to this minimum represents the loss in potential energy after bond formation, signifying the energy released when the bond forms.

10. The correct answer is: (A).

When the distance between two hydrogen atoms decreases below their equilibrium bond length (typically around 74 pm for H_2), the repulsive forces between the atoms increase significantly. This is due to the increased overlap of the electron clouds and the stronger electrostatic repulsion between the positively charged nuclei. As a result, the potential energy of the system also increases.

11. The correct answer is: (A, B, C, D).

A) Potential energy is typically plotted on the Y-axis, showing energy changes as atoms approach or separate.

B) Internuclear distance is conventionally shown on the X-axis, representing the distance between atomic nuclei.

C) Potential energy is commonly measured in kJ/mol, a unit that's convenient for expressing bond energies.

D) Internuclear distance is often measured in meters, though Angstroms (1 Å = 10^{-10} m) are also frequently used for atomic-scale distances.

12. The correct answer is: (C).

This accurately represents the equilibrium bond length of a hydrogen molecule (H_2). At this distance, the attractive and repulsive forces between the two hydrogen atoms balance, resulting in the most stable configuration of the molecule. This equilibrium distance minimizes the total potential energy of the system.

H-H bond length = 74 pm

13. The correct answer is: (A, B, C).

A) The potential energy of the H2 molecule at its equilibrium bond length is indeed -432 kJ/mol. This represents the bond dissociation energy.

B) The equilibrium bond length of the H2 molecule is 74 pm, which is the internuclear distance at the lowest point of the potential energy curve.

C) The lowest point on the potential energy curve corresponds to the most stable configuration of the H2 molecule, which occurs at the equilibrium bond length.

These can be represented as: H_2 bond energy = -432 kJ/mol

H-H bond length = 74 pm

Assertion and reason type:

14. **The correct answer is: (C)**.
Systems in nature tend to move from a higher energy state to a lower energy state.True
According to the **principle of minimum energy**, natural systems tend to attain the **lowest possible energy state** because it is the most stable state. For example, atoms form chemical bonds to lower their energy and become more stable.
Reason (R):
Lower energy states are less stable than higher energy states.False
Explanation:
This statement is incorrect. **Lower energy states are more stable**, not less stable. Higher energy states are unstable and tend to lose energy to become more stable.

15. **The correct answer is: (A)**.
Assertion (A):
Atoms with completely filled outer shells are stable.True
Explanation:
Atoms with completely filled valence shells (such as noble gases) are chemically stable because they have little tendency to gain, lose, or share electrons.
Examples:
- Helium (2 electrons)
- Neon (8 electrons)
- Argon (8 electrons)

Reason (R):
Atoms with unfilled outer shells form chemical bonds to achieve stability.True
Explanation:
Atoms having incomplete valence shells tend to:
- lose electrons,
- gain electrons, or
- share electrons

to achieve the stable electronic configuration of the nearest noble gas (octet or duplet).
This explains why atoms with filled shells are already stable.

16. **The correct answer is: (C)**.
Noble gases such as helium, neon, and argon have completely filled valence shells. Therefore, they are chemically inert and exist as **single atoms (monoatomic)** under normal conditions.
- Sodium is highly reactive.
- Oxygen exists as O_2 molecules.
- Carbon does not exist as isolated atoms under normal conditions.

17. **The correct answer is: (B)**.
When two or more atoms combine chemically, they form a **molecule**.
Examples:
- H_2
- O_2
- H_2O
- CO_2

Electrons, protons, and neutrons are subatomic particles and are **not formed by combining atoms**.

18. **The correct answer is: (C)**.
The attractive force that holds atoms or ions together in a compound is called a **chemical bond**.
Types of chemical bonds include:
- Ionic bond
- Covalent bond
- Metallic bond

Magnetic, gravitational, and frictional forces do not hold atoms together in chemical compounds.

INTEGER TYPE

19. One chlorine molecule consists of 2 chlorine atoms.
Chlorine is a diatomic element, meaning its molecules are composed of two atoms of the same element bonded together. This can be represented by the chemical formula: Cl_2
The formation of a chlorine molecule from two chlorine atoms can be shown as:
$Cl + Cl \rightarrow Cl_2$
This covalent bond between the two chlorine atoms is formed by sharing a pair of electrons. The resulting molecule is more stable than the individual atoms.

20. Correct Option 1
One molecule of a noble gas contains only one atom. Noble gases, such as helium (He), neon (Ne), argon (Ar), krypton (Kr), xenon (Xe), and radon (Rn), exist as single atoms in their elemental form under normal conditions.
These atoms are stable with complete outer electron shells, making them inert and non-reactive. Due to their stable electronic configurations, noble gases do not typically form molecules consisting of two or more atoms bonded together. Therefore, a molecule of a noble gas consists of exactly one atom, distinguishing them from most other elements which often form molecules containing multiple atoms.

21. Correct Option 3
In an ozone molecule, the number of oxygen atoms is 3.
Ozone is a triatomic molecule composed entirely of oxygen atoms. Its chemical formula is: O_3
The formation of ozone from oxygen can be represented as: $3O_2 \rightarrow 2O_3$
This reaction typically occurs in the upper atmosphere due to the action of ultraviolet light on oxygen molecules.

22. Correct Option 1
In an iron molecule, the number of iron atoms is one. Iron (Fe) is an element that exists as individual atoms in its elemental form. Each iron molecule consists of a single iron atom.
This atomic form of iron is stable and does not naturally bond with other iron atoms to form diatomic or polyatomic molecules under standard conditions.

Definition of Chemical Bond

LEVEL – I

1. The correct answer is: (B).
A chemical bond is the attractive force that holds atoms or ions together in a chemical species, such as a molecule or compound. This force arises from the electromagnetic interaction between electrons and protons of the participating atoms. Chemical bonds can be of various types, including:
1. Covalent bonds: Sharing of electrons between atoms
2. Ionic bonds: Electrostatic attraction between oppositely charged ions

3. Metallic bonds: Attraction between metal ions and delocalized electrons
The formation of a chemical bond can be represented generally as:A + B → A-B
Where A and B are atoms or ions, and A-B represents the bonded species.

2. The correct answer is: (D).

When atoms combine to form a molecule, the process can be either exothermic (releasing energy) or endothermic (absorbing energy), depending on the specific atoms involved and the type of bond being formed.
Exothermic bond formation: A + B → AB + energy
Endothermic bond formation: A + B + energy → AB

3. The correct answer is: (B).

Atoms combine to form chemical bonds primarily to achieve a more stable electron configuration, typically resembling that of the nearest noble gas. This concept is known as the octet rule, though there are exceptions.
In this process, atoms may:
1. Share electrons (covalent bonding)
2. Transfer electrons (ionic bonding)
3. Pool electrons (metallic bonding)

For example: Na (2,8,1) + Cl (2,8,7) → Na^+ (2,8) + Cl^- (2,8,8)
Here, sodium loses an electron and chlorine gains one, both achieving stable configurations.

4. The correct answer is: (B).

The octet rule states that atoms tend to gain, lose, or share electrons to achieve a stable electron configuration with eight electrons in their outermost shell, similar to noble gases. This rule applies to many elements, particularly those in the main group of the periodic table.
Examples:
- Chlorine atom: 2,8,7 → Chloride ion: 2,8,8
- Sodium atom: 2,8,1 → Sodium ion: 2,8
Element + x electrons → Element with 8 valence electrons
Where x is the number of electrons needed to reach 8.

5. The correct answer is: (A).

When atoms come together to form a chemical bond, there are simultaneous attractive and repulsive forces at play. The attraction occurs between the positively charged nuclei and negatively charged electrons.
Repulsion exists between the positively charged nuclei of different atoms and between the electron clouds. The balance of these forces determines the bond formation and strength.
As atoms approach each other, attraction initially dominates. However, at very close distances, repulsion becomes significant. The equilibrium bond length is reached when the net force (attraction minus repulsion) is zero, resulting in a stable chemical bond.

6. The correct answer is: (A).

When atoms come together to form a chemical bond, the system's potential energy decreases. This decrease in potential energy is what makes bond formation energetically favourable.
As the atoms approach each other, attractive forces begin to dominate, causing the release of energy and lowering the system's overall potential energy. The potential energy reaches its minimum at the equilibrium bond length, where the bond is most stable.
This energy release during bond formation is typically expressed as the bond energy or bond enthalpy. The general equation for this process can be written as:

$A + B \rightarrow AB + Energy$

Where A and B are the bonding atoms, and AB is the resulting molecule.

7. The correct answer is: (C).

The Lewis electron dot structure you've described (Cl with a dot) represents the chloride ion (Cl^-). In this structure, chlorine has gained an electron, giving it a full octet of 8 valence electrons.

A) Cl atom: ·Cl· (7 valence electrons)

B) Cl^+ ion: Cl (6 valence electrons)

C) Cl^- ion: Cl: (8 valence electrons)

D) $2Cl^+$ is not a valid Lewis structure

8. Correct answer is: (D)

9. The correct answer is: (A).

Valency refers to the combining capacity of an atom, which is its ability to form chemical bonds with other atoms. It represents the number of electrons an atom can gain, lose, or share when forming chemical compounds. Valency is typically determined by the number of electrons in the outermost shell (valence electrons) and the atom's tendency to achieve a stable electron configuration.

10. The correct answer is (B)

In the compound NH_3 (ammonia), the valency of nitrogen (N) is 3. This means nitrogen is forming three bonds with hydrogen atoms.

1. Nitrogen has 5 valence electrons
2. In Ammonia, nitrogen shares one electron with each of the three hydrogen atoms.
3. This results in three covalent bonds (N-H).

The chemical equation for the formation of ammonia can be written as:

$N + 3H \rightarrow NH_3$

11. The correct answer is: (C)

In PCl_5 (phosphorus pentachloride), the valency of phosphorus (P) is 5. This compound demonstrates phosphorus's ability to expand its octet and form five covalent bonds.

1. Phosphorus has 5 valence electrons.
2. In PCl_5, phosphorus shares one electron with each of the five chlorine atoms.
3. This results in five covalent bonds (P-Cl).

The formation of PCl_5 can be represented as: $P + 5Cl \rightarrow PCl_5$

12. The correct answer is: (C).

In CO_2 (carbon dioxide), the valency of carbon (C) is 4. Although carbon forms only two bonds in this molecule, each bond is a double bond, utilizing all four of carbon's valence electrons.

1. Carbon has 4 valence electrons
2. In CO_2, carbon forms two double bonds with oxygen atoms
3. Each double bond uses two of carbon's valence electrons

The structure of CO_2 can be represented as O=C=O

The formation of CO_2 can be written as: $C + 2O \rightarrow CO_2$

13. The correct answer is: (C)

The valency of an element in a compound is determined by its combining capacity, which corresponds to the number of bonds it can form with other atoms. In Cl_2O_7, chlorine (Cl) is in the compound dichloride heptoxide. The prefix "di-" indicates two chlorine atoms, and "hept-" indicates seven oxygen atoms.

In Cl_2O_7, chlorine typically exhibits a valency of 7. This is because each chlorine atom in Cl_2O_7 forms one covalent bond with oxygen atoms, and chlorine can accommodate a total of 7 bonds due to its position in the periodic table and electron configuration.

14. The correct answer is: (B).

In Na_2O (sodium oxide), the valency of sodium (Na) is 1. This compound demonstrates sodium's tendency to lose one electron to form a stable ionic compound.

1. Sodium has 1 valence electron
2. In Na_2O, each sodium atom loses one electron to oxygen
3. Two sodium atoms contribute one electron each to one oxygen atom

$2Na + O \rightarrow Na_2O$

15. The correct answer is; (C).

The electronic theory of valency was independently developed by Walter Kossel and Gilbert Newton Lewis in 1916. This theory explains chemical bonding and valency in terms of electron configurations:

1. Kossel focused on ionic bonding, explaining how atoms transfer electrons to achieve stable electron configurations.
2. Lewis concentrated on covalent bonding, introducing the concept of electron sharing between atoms.

16. The correct answer is: (B).

Core electrons are the electrons present in the inner energy levels of an atom, closer to the nucleus. These electrons:

1. Are not involved in chemical bonding
2. Are tightly bound to the nucleus
3. Have lower energy compared to valence electrons
4. Are less affected by the chemical environment of the atom

17. The correct answer is: (A).

Valence electrons are the electrons present in the outermost energy level (or shell) of an atom. These electrons:

1. Determine the atom's chemical properties
2. Participate in chemical bonding
3. Are responsible for the element's reactivity
4. Determine the atom's valency

For example, in sodium (Na):

$1s^2\ 2s^2\ 2p^6\ 3s^1$

The single electron in the 3s orbital is the valence electron.

18. The correct answer is: (A).

Helium is the only stable atom that does not follow the octet rule, which states that atoms tend to have eight electrons in their valence shell for maximum stability. Helium has a complete valence shell with only two electrons:

He: $1s^2$

This two-electron configuration is stable for helium because:

1. It matches the first energy level's maximum capacity (2 electrons)
2. It gives helium a full s-subshell

19. The correct answer is: (C).

An electrovalent bond, also known as an ionic bond, is formed by the complete transfer of electrons from one atom to another. This process typically occurs between:

A metal (electron donor) and a non-metal (electron acceptor)

Elements with significantly different electronegativities

For example, in sodium chloride (NaCl): $Na \rightarrow Na^+ + e^-$

$Cl + e^- \rightarrow Cl^-$

$Na^+ + Cl^- \rightarrow NaCl$

Sodium loses its single valence electron to chlorine, forming Na+ and Cl- ions.

20. The correct answer is: (D)

Ions are formed from neutral atoms through either the loss or gain of electrons, depending on the element's electron configuration and chemical properties:

1. Cations (positively charged ions) are formed by losing electrons:

Na → Na+ + e- (sodium loses one electron)

2. Anions (negatively charged ions) are formed by gaining electrons:

Cl + e- → Cl- (chlorine gains one electron)

21. The correct answer is: (D).

Sodium chloride exhibits electrovalent bonding, also known as ionic bonding. This type of bond forms through the complete transfer of electrons between atoms with significantly different electronegativities, typically a metal and a non-metal.

In NaCl: $Na \rightarrow Na^+ + e^-$ (sodium loses one electron)

$Cl + e^- \rightarrow Cl^-$ (chlorine gains one electron)

$Na^+ + Cl^- \rightarrow NaCl$

22. The correct answer is: (C)

Covalent bonds can form between:

1. Similar atoms:

H• + •H → H-H (hydrogen molecule)

Cl• + •Cl → Cl-Cl (chlorine molecule)

2. Dissimilar atoms:

H• + •Cl → H-Cl (hydrogen chloride)

C•• + 4H• → CH4 (methane)

Covalent bonding involves the sharing of electrons to achieve a stable electron configuration.

23. The correct answer is: (B).

N_2 contains a triple covalent bond between the two nitrogen atoms. This can be represented as:

N≡N

Each nitrogen atom contributes three unpaired electrons, resulting in three shared electron pairs and a very strong bond.

24. The correct answer is: (D)

In a triple bond, two atoms share a total of 6 electrons. This consists of three pairs of electrons, each pair forming a covalent bond between the atoms. The first pair forms a sigma (σ) bond, while the second and third pairs form two pi (π) bonds.

Triple bond = σ bond + π bond + π bond

Total electrons shared = 2 + 2 + 2 = 6 electrons

25. The correct answer is: (D).

The contracted octet rule applies to elements in the third period and below of the periodic table (period 3 and beyond), which can accommodate more than eight electrons in their valence shell due to the availability of d orbitals for bonding. Among the given options, PCl_5 follows the contracted octet rule.

In PCl_5 (phosphorus pentachloride), phosphorus (P) is in the third period and can expand its valence shell to accommodate ten electrons (five bonds with chlorine atoms), exceeding the traditional octet rule of eight electrons. This expanded valence shell configuration is characteristic of elements that follow the contracted octet rule.

LEVEL - II

SINGLE OPTION CORRECT OPTION

1. The correct answer is: (B)

If stability were attained with 6 electrons instead of 8, fluorine would aim to lose an electron rather than gain one. Fluorine's electron configuration is $1s^2 2s^2 2p^5$, meaning

it has 7 valence electrons. To reach a stable 6-electron configuration in this hypothetical scenario, it would need to lose one electron.

F (7 electrons) → F+ (6 electrons) + e-

2. The correct answer is: (A).

To determine the formula, we need to consider the valence electrons and likely charges of X and Y:

X: 2,8,8,2 - This element has 2 valence electrons and is likely to form a +2 ion.

Y: 2,8,7 - This element has 7 valence electrons and is likely to form a -1 ion.

For a neutral compound, the charges must balance. We can represent this as:

$X^{2+} + 2Y^{-} \rightarrow XY_2$

Element X contributes a +2 charge, which is balanced by two Y ions, each with a -1 charge.

3. The correct answer is (D).

Sulphur (S) is an element located in group 16 of the periodic table, also known as the oxygen group or chalcogens. Elements in this group typically have 6 valence electrons. Valence electrons are the electrons in the outermost shell of an atom, responsible for the atom's chemical properties and interactions with other atoms.

Number of valence electrons present in a sulphur atom is (D) 6.

4. The correct answer is (C)

Hydrogen atoms typically combine to form diatomic hydrogen molecules, represented as H_2. This is the most stable and common form of molecular hydrogen.

The formation of H_2 can be represented by the equation:

$H + H \rightarrow H_2$

In this molecule, the two hydrogen atoms share their single electrons, forming a covalent bond.

5. The correct answer is: (B)

An atom with atomic number 20 is calcium (Ca), which has an electron configuration of $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2$. Calcium readily loses its two 4s electrons to form a Ca^{2+} ion.

Element 16 is sulphur (S), with configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^4$. It tends to gain two electrons to form S^{2-}.

These ions can combine to form a stable ionic compound:

$Ca \rightarrow Ca^{2+} + 2e^{-}$

$S + 2e^{-} \rightarrow S^{2-}$

$Ca^{2+} + S^{2-} \rightarrow CaS$

6. The correct answer is: (A)

Element A, with 3 electrons in its outermost orbit, tends to lose these electrons to achieve a stable configuration, forming a +3 ion (A^{3+}).

Element B, with 6 electrons in its outermost orbit, tends to gain 2 electrons to achieve a stable octet, forming a -2 ion (B^{2-}).

To form a neutral compound, the charges must balance:

$2(A^{3+}) + 3(B^{2-}) \rightarrow A_2B_3$

Represent this process as:

$2A \rightarrow 2A^{3+} + 6e^{-}$

$3B + 6e^{-} \rightarrow 3B^{2-}$

$2A^{3+} + 3B^{2-} \rightarrow A_2B_3$

7. The correct answer is: (D)

To determine the number of core electrons in a fluorine atom, we need to consider its electron configuration and distinguish between core and valence electrons.

Fluorine's atomic number is 9, and its electron configuration is $1s^2\ 2s^2\ 2p^5$.

Core electrons are those in completely filled inner shells, which for fluorine is only the 1s orbital. The 2s and 2p electrons are considered valence electrons.
Core electrons ($1s^2$): 2
Valence electrons ($2s^2\ 2p^5$): 7

8. The correct answer is (B).
Carbon (C):
Atomic number 6
Electron configuration: $1s^2\ 2s^2\ 2p^2$
Valence electrons: 4 ($2s^2\ 2p^2$)
Sulphur (S):
Atomic number 16
Electron configuration: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^4$
Valence electrons: 6 ($3s^2\ 3p^4$)
Fluorine (F):
Atomic number 9
Electron configuration: $1s^2\ 2s^2\ 2p^5$
Valence electrons: 7 ($2s^2\ 2p^5$).

ONE OR MORE THAN ONE OPTION CORRECT TYPE

9. The correct answer is: (B, D).
The octet rule states that atoms tend to gain, lose, or share electrons to achieve a stable configuration with eight electrons in their outermost shell (except for hydrogen, which achieves stability with two electrons).
(A) CO_2 (carbon dioxide) is not an exception to the octet rule. Carbon (C) forms double bonds with two oxygen (O) atoms, achieving a stable octet configuration for each atom.
(B) $BeCl_2$ (beryllium chloride) is an exception to the octet rule. Beryllium (Be) typically forms only two bonds, resulting in four electrons in its outer shell, not eight.
(C) CH_4 (methane) follows the octet rule, where carbon (C) shares electrons with four hydrogen (H) atoms, completing its octet.
(D) BCl_3 (boron trichloride) is also an exception to the octet rule. Boron (B) forms three bonds with chlorine (Cl) atoms, resulting in only six electrons in its outer shell.

10. The correct answer is (B).
The octet rule states that atoms tend to gain, lose, or share electrons until they have a full outer shell of eight electrons (or two electrons for hydrogen).
In CO_2:
- Carbon (C) has 4 valence electrons and forms double bonds with two oxygen (O) atoms, sharing electrons to complete its octet.
- Each oxygen atom has 6 valence electrons and forms a double bond with carbon, achieving an octet by sharing electrons.

11. The correct answer is: (A, B, C).
These compounds follow the expanded octet rule
A) BrF5: Bromine has 12 electrons around it (expanded octet)
B) SF6: Sulphur has 12 electrons around it (expanded octet)
C) IF7: Iodine has 14 electrons around it (expanded octet)
D) CO2: Carbon follows the standard octet rule with 8 electrons

12. The correct answer is: (B, C).
B) N_2 (Nitrogen molecule):
N≡N
Nitrogen forms a triple bond between the two atoms, sharing 6 electrons.
C) O_2 (Oxygen molecule):
O=O

Oxygen forms a double bond between the two atoms, sharing 4 electrons.
H_2 (Hydrogen molecule) and Cl_2 (Chlorine molecule) do not contain multiple bonds.
Assertion and reason:

13. **The correct answer is: (A).**
Assertion (A):Only valence electrons participate in chemical bond formation.
True
Explanation:
Valence electrons are the electrons present in the **outermost shell** of an atom. These electrons are responsible for:

- Gaining electrons
- Losing electrons
- Sharing electrons

Thus, they participate in the formation of **ionic** and **covalent bonds**.
Reason (R):Core electrons are present in the inner energy levels and generally do not take part in bonding.
True
Explanation:
Core (inner-shell) electrons are held tightly by the nucleus. They remain unaffected during chemical reactions and **do not participate in bond formation**. Since only valence electrons are available for bonding, the reason correctly explains the assertion.

14. **The correct answer is: (D).**
Assertion (A):
All stable molecules follow the octet rule.
False
Explanation:
The octet rule is useful for many molecules, but **not all stable molecules follow it**.
Examples of exceptions:

- **Incomplete octet:** $BeCl_2$, BF_3
- **Expanded octet:** PCl_5, SF_6
- **Odd-electron molecules:** NO, NO_2

Therefore, the assertion is false.
Reason (R):
Certain molecules may contain less than eight or more than eight electrons around the central atom.
True
Explanation:
Some molecules are stable even though they do not satisfy the octet rule.

- **Less than eight electrons:** BF_3 (Boron has 6 electrons)
- **More than eight electrons:** SF_6 (Sulphur has 12 electrons)

Thus, the reason is true.
MATRIX MATCHING TYPE

15. **a-R, b-S, c-Q, d-P**
a) Magnesium has 2 valence electrons (Electronic configuration: 2,8,2).
b) Chlorine has 7 valence electrons (2,8,7). It has three pairs and one single electron.
c) Sodium has 1 valence electron (2,8,1).
d) Sulphur has 6 valence electrons (2,8,6). It has two pairs and two single electrons.

16. a-Q, b-R, c-S, d-P

a) F (Fluorine): Atomic number 9, configuration Q) 2,7
b) K (Potassium): Atomic number 19, configuration R) 2,8,8,1
c) Mg (Magnesium): Atomic number 12, configuration S) 2,8,2
d) Cl (Chlorine): Atomic number 17, configuration P) 2,8,7

COMPREHENSION TYPE

Passage - I:

17. The correct answer is: (A).

Walter Kossel, a German physicist and chemist, proposed the electronic theory of valence in 1916. This theory was a significant advancement in understanding chemical bonding and molecular structure.

Key points of Kossel's theory:

1. Valence is related to the electron configuration of atoms.

Atoms tend to gain, lose, or share electrons to achieve a stable electron configuration similar to that of the nearest noble gas.

The theory explains ionic bonding through electron transfer and covalent bonding through electron sharing.

18. The correct answer is: (B)

Octet theory states that atoms tend to gain, lose, or share electrons in order to achieve a stable electron configuration similar to that of the nearest noble gas, which typically has eight electrons in its valence shell.

Noble gases naturally possess a complete valence shell with eight electrons (except for helium, which has two), making them chemically stable and unreactive. This full valence shell configuration prevents noble gases from readily forming chemical bonds, resulting in their inertness.

19. The correct answer is: (A)

Valence electrons are the electrons present in the outermost shell, or valence shell, of an atom. These electrons are important because they are involved in chemical bonding and reactions.

For instance, in a sodium (Na) atom, the valence electron configuration is $3s^1$, indicating one valence electron in the third shell. Valence electrons determine the chemical properties and reactivity of an element.

They are different from core electrons, which are the electrons in the inner shells that do not typically participate in chemical bonding.

Passage - II:

20. The correct answer is: (D).

In the compound phosphine (PH_3), phosphorus (P) forms three single covalent bonds with three hydrogen (H) atoms. Each hydrogen atom contributes one electron to form a covalent bond with phosphorus.

Phosphorus has five valence electrons, and by forming three bonds with hydrogen, it uses three of its valence electrons for bonding, leaving two electrons as a lone pair. The number of bonds formed by an atom in a molecule indicates its valence.

21. The correct answer is: (B).

The valence, or oxidation state, of sulphur (S) in sulphur trioxide (SO_3) is 6. To determine this, we consider the oxidation states of the atoms in the compound. Oxygen (O) typically has an oxidation state of -2. In SO_3, there are three oxygen atoms, contributing a total of -6 to the oxidation state.

The equation for the sum of the oxidation states in a neutral molecule is:

$S + 3(-2) = 0$

Solving for sulphur (S):

S - 6 = 0
S = +6

22. The correct answer is (B)

To determine the valence of phosphorus (P) in P_2O_5, consider the compound's structure. Each oxygen (O) atom typically has a valence of 2. In P_2O_5, there are 5 oxygen atoms contributing a total of 5*2=10 valence electrons. Since the total valence for the compound must be zero and each phosphorus (P) atom must balance the oxygen atoms' valence, each phosphorus atom has a valence of 5.

INTEGER TYPE

23. Correct answer is 2

The number of valence electrons present in a Beryllium atom is 2.

Beryllium (Be) is an element in Group 2 of the periodic table, also known as the alkaline earth metals. The atomic number of Beryllium is 4, meaning it has 4 protons and, in a neutral atom, 4 electrons. These electrons are arranged in the electron configuration $1s^2\ 2s^2$.

The electrons in the outermost shell of an atom are called valence electrons, and they are important because they are involved in chemical bonding. For Beryllium, the outermost shell is the second shell ($2s^2$), which contains 2 electrons.

24. Correct answer is 3

To determine the number of unpaired electrons in a phosphorus atom, we first need to understand its electron configuration. Phosphorus has an atomic number of 15, meaning it has 15 electrons. The electron configuration of phosphorus is:

$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$

In the 3p subshell, there are three electrons. According to Hund's rule, these electrons will occupy separate orbitals within the same subshell before pairing up. This rule helps minimize electron-electron repulsions and stabilizes the atom.

25. Correct answer is 3.

The valency of an element is the measure of its ability to combine with other atoms. In the compound aluminum oxide (Al2O3), we need to determine the valency of aluminum (Al).

Aluminum oxide consists of two aluminum atoms (Al) and three oxygen atoms (O). The valency of oxygen is typically -2. To maintain charge neutrality in the compound, the total positive charge from aluminum atoms must balance the total negative charge from oxygen atoms.

- Each oxygen atom has a valency of -2.
- Three oxygen atoms contribute a total charge of 3 x -2 = -6.

To balance these -6 charges, the two aluminum atoms must collectively contribute a +6 charge. Hence, each aluminum atom must have a valency of +3.

26. Correct answer is 2.

Sulphur is in Group 16, which means it has six valence electrons. When sulphur is in its elemental form (as in a molecule like S8), each sulphur atom uses two of its valence electrons to form two single bonds with adjacent sulphur atoms, creating a ring structure. This leaves four valence electrons as lone pairs.

1. Total valence electrons: 6
2. Electrons used in bonding: 2 (one for each single bond)
3. Remaining valence electrons: 6 - 2 = 4

Since each lone pair consists of two electrons, the sulphur atom will have 4 / 2 = 2 lone pairs of electrons.

The number of lone pairs present in a sulphur atom is 2.

27. Correct answer is 3

In a nitrogen molecule (N_2), two nitrogen atoms are bonded together through a triple bond. Each nitrogen atom has five valence electrons, and they need three more electrons to complete their octet. By sharing three pairs of electrons, both nitrogen atoms achieve a stable electronic configuration.

The Lewis structure of the nitrogen molecule can be represented as follows:

N ≡ N

In this structure, the triple bond consists of three shared pairs of electrons (bond pairs). Thus, the number of bond pairs in a nitrogen (N_2) molecule is 3.

Previous year questions

1. . A) CH_3-$\underline{CH_2}$-$CH=CH_2$: The underlined carbon is bonded to two Carbons and two Hydrogens via single bonds. This is sp^3.

B) CH_3-$\underline{CH_2}$-NH_2: Similar to above, this carbon has 4 sigma bonds. This is also sp^3.

D) CH_3-$\underline{CH_2}$-CN: The underlined carbon has 4 sigma bonds. This is sp^3.

2. C**)** O_2^- (1.5) + N_2 (3) + O_2^+ (2.5) = 7

3. A) NH_3 to NH_4^+: NH_3 is sp^3 (3 bond pairs + 1 lone pair). NH_4^+ is sp^3 (4 bond pairs). No change.

4. Total: 5 molecules. Correct Option: C

- ClF_3: 10 electrons around Cl. (Yes)
- SF_6: 12 electrons around S. (Yes)
- CH_4: 8 electrons around C. (No)
- NH_3: 8 electrons around N. (No)
- SF_4: 10 electrons around S. (Yes)
- XeF_4: 12 electrons around Xe. (Yes)
- PCl_5: 10 electrons around P. (Yes)

5. Correct Option: D (0 and -1)

The structure of Ozone is a resonance hybrid. In a single Lewis structure:

The central oxygen has a formal charge of +1.

One terminal oxygen (double bonded) has a charge of 0.

The other terminal oxygen (single bonded) has a charge of -1.

6. Correct Option: A

(ii) SF_6 is sp^3d hybridised: Incorrect**.** SF6 is sp^3d^2 (octahedral).

(iii) $S_2O_3^{2-}$ is a linear ion: Incorrect**.** Thiosulfate is tetrahedral around the central sulfur.

(iv) There is no pi-bonding in SO_4^{2-} ion: Incorrect. SO_4^{2-} has $d\pi - p\pi$ bonding.

7. Correct Answer: B

sing Molecular Orbital Theory (Total electrons):

1. NO^+ (14 electrons): Same as N_2, Bond Order = 3.0
2. NO (15 electrons**):** Bond Order = 2.5
3. NO^- (16 electrons): Bond Order = 2.0

8. Correct Answer: B (sp^2, sp)

Based on the structure provided in the prompt (H_3C-$CH=C=CH_2$):

C_2: Involved in one double bond (CH=). Hybridization = sp^2

- C_3**:** Central carbon of the allene system bonded to two double bonds (=C=). Hybridization = sp

9. Correct Answer: C) H_2S, CH_4, SO_2:

H_2S: The bond angle is approximately 92°. (The angle is much smaller than H_2O because sulfur is larger and uses orbitals with more p-character).

CH_4: 109.5°.

SO_2: Bent geometry (due to one lone pair and double bonds), the angle is approximately 119°.

Order: 92° < 109.5° < 119° (This is increasing order).

10. Correct Answer:C

According to Valence Bond Theory (VBT) for coordination compounds:

1. sp^3: Involves one 4s and three 4p orbitals (e.g., in tetrahedral complexes like $[NiCl_4]^{2-}$.
2. dsp^2: Involves one inner 3d, one 4s, and two 4p orbitals (e.g., in square planar complexes like $[Ni(CN)_4]^{2-}$).
3. d^2sp^3: Involves two inner 3d, one 4s, and three 4p orbitals (e.g., in inner-orbital octahedral complexes like $[NH_3)_6]^{3+}$

While sp^3d^2 (outer-orbital octahedral) is also possible using 4d orbitals, the options provided specifically highlight the use of the 3d (inner) and 4s (outer) orbitals.

3. Liquid State and Gaseous State

LEVEL – I

1. **Answer: B)** energy or work

- Explanation: Pressure (P) is defined as Force per unit Area ($\frac{F}{A}$), and Volume (V) is Area × Length.

  $PV = \left(\frac{\text{Force}}{\text{Area}}\right) \times (\text{Area} \times \text{Length}) = \text{Force} \times \text{Length}$

  Since Force × Distance is the formula for work done or energy, its unit is the Joule (J).

2. **Answer: B or D** (0 °C and 1 atm)

Explanation: STP stands for Standard Temperature and Pressure. Traditionally, it is defined as a temperature of 0°C (273.15 K) and a pressure of 1 atm (or 1 bar in modern IUPAC standards).

3. **Answer: A, B,** and C (Volume, Temperature, Pressure)

- Explanation: Vapour density is mathematically defined as:

  $\text{Vapour Density} = \frac{\text{Molar Mass of Gas}}{2}$

  Because it depends purely on the molecular weight of the substance, changes in experimental conditions like temperature, pressure, or the size of the container (volume) do not affect it.

4. Answer: A) Boyle's law

- Explanation: Boyle's Law states that pressure is inversely proportional to volume ($P \propto \frac{1}{V}$) only when the temperature and the number of moles (n) are kept constant.

  In a real balloon, you are adding more air (increasing n), so it doesn't actually break physics, but on paper, a simultaneous increase in both P and V contradicts the inverse relationship of Boyle's Law.

5. Answer: B) pressure

- Explanation: Torr is a non-SI unit of pressure named after Evangelista Torricelli. 1 atm = 760 torr = 760 mm Hg.

6. Answer: C) No fixed shape and volume

- Explanation: Gas particles have negligible intermolecular forces and high kinetic energy, meaning they move rapidly in all directions to occupy whatever space is available to them.

7. Answer: B) Liquid

- Explanation: Liquids take the shape of the container they are poured into but maintain a fixed volume.

8. Answer: B) Gas < Liquid < Solid

- Explanation: Solids have the strongest forces holding particles tightly together, liquids have intermediate forces allowing movement, and gases have the weakest (almost negligible) forces.

9. Answer: C) Large intermolecular spaces

- Explanation: Gas particles are separated by vast empty spaces. Applying external pressure easily forces these particles closer together.

10. Answer: B) Liquid

- Explanation: When heated, liquids expand more than solids because their intermolecular bonds are weaker, but significantly less than gases, which expand drastically.

11. Answer: C) Gas

- Explanation: If both the melting point and boiling point are below room temperature (e.g., 25°C), the substance has already transitioned past its boiling threshold and exists as a gas.

12. Answer: B) Volume, pressure, temperature

- Explanation: These are the state variables (along with the number of moles) featured in the ideal gas equation ($PV = nRT$).

13. Answer: B) Barometer

- Explanation: A barometer measures atmospheric pressure, while a manometer measures the pressure of an enclosed gas sample.

14. Answer: D) Joule

- Explanation: The Joule is a unit of energy/work, not pressure. Pascal, atm, and Torr are all pressure units.

15. Answer: B) 300 K

- Explanation: Kelvin =° C + 273 = 27 + 273 = 300 K.

16. Answer: 2×10^5 Pa

Explanation: Since 1 atm ≈ 1.013×10^5 Pa, doubling it yields 2 atm ≈ 2.026×10^5 Pa.

17. Answer: A) 0 °C

- Explanation: Standard temperature is defined exactly at 0°C or 273.15 K.

18. Answer: C) Due to collision of molecules with container walls

- Explanation: Gas pressure is the macroscopic result of billions of tiny gas particles constantly bouncing off the internal surfaces of their container.

19. Answer: B) 2 L

- Explanation: Using Boyle's Law ($P_1V_1 = P_2V_2$):

$2 \text{ atm} \times 4 \text{ L} = 4 \text{ atm} \times V_2 \Longrightarrow V_2 = \frac{8}{4} = 2 \text{ L}$

20. Answer: B) Becomes half

- Explanation: From the ideal gas law rearrangement, density $d = \frac{PM}{RT}$. At a constant temperature (T), density is directly proportional to pressure ($d \propto P$). If density is halved, the pressure drops by half as well.

LEVEL – II

1. Answer: A) Intermolecular force of attraction

- Explanation: According to the Kinetic Molecular Theory, the attractive forces between ideal gas particles are so weak that they are considered completely negligible.

2. Answer: (B) dyne/cm²

- Explanation: In the CGS system, the unit of force is the *dyne* and area is cm^2. Therefore, pressure ($\frac{\text{Force}}{\text{Area}}$) is measured in dyne/cm^2.

3. Answer: C) 48 mL

- Explanation: Using Boyle's Law ($P_1V_1 = P_2V_2$):

$432 \times 80 = 720 \times V_2 \Longrightarrow V_2 = \frac{34560}{720} = 48 \text{ mL}$

4. Answer: C) Absolute temperature

- Explanation: Charles's Law dictates $V \propto T$, given that the temperature must be measured on the absolute scale (Kelvin).

5. Answer: B) 25 °C

- Explanation: Using the conversion formula $^\circ\text{C} = (\circ \text{F} - 32) \times \frac{5}{9}$:

$^{\circ}\text{C} = (77 - 32) \times \frac{5}{9} = 45 \times \frac{5}{9} = 25^{\circ}\text{C}$

6. Answer: B) 160 mL

- Explanation: Applying Boyle's law ($P_1V_1 = P_2V_2$):

$0.720 \times 200 = 0.900 \times V_2 \Longrightarrow V_2 = \frac{144}{0.900} = 160 \text{ mL}$

7. Answer: D) 0.8 bar

- Explanation: Assuming the temperature remains constant:

$P_1V_1 = P_2V_2 \Longrightarrow 1.2 \times 120 = P_2 \times 180 \Longrightarrow P_2 = \frac{144}{180} = 0.8 \text{ bar}$

8. Answer: D) h = P/dg

- Explanation: Hydrostatic pressure is calculated by $P = hdg$ (where g is acceleration due to gravity). Rearranging for height gives $h = \frac{P}{dg}$.

9. Answer: D) 18 L

- Explanation: $P_1V_1 = P_2V_2 \Longrightarrow 3 \text{ atm} \times 6 \text{ L} = 1 \text{ atm} \times V_2 \Longrightarrow V_2 = 18 \text{ L}$.

10. Answer: C) Absolute zero

- Explanation: Absolute zero (0 K or −273.15°C) is the theoretical temperature where an ideal gas drops to zero volume as all molecular motion stops.

One or More than One Correct Answer Questions

11. Answers: A and B

- Explanation: According to the formula $d = \frac{PM}{RT}$, density is directly proportional to both its pressure (P) and its molar mass (M).

12. Answer: D) At altitudes, water will boil at a temperature greater than 100°C

- Explanation: Atmospheric pressure drops at higher altitudes. Because boiling occurs when vapor pressure equals atmospheric pressure, water boils at a lower temperature than 100°C in the mountains, making statement D completely incorrect.

13. Answers: B and D

- Explanation: * B is correct: Lower atmospheric pressure at high altitudes lowers water's boiling point.
  - D is correct: Water vapor (H_2O, molar mass ≈ 18 g/mol) is lighter than dry air (mostly N_2 and O_2, average molar mass ≈ 29 g/mol). Replacing dry air with moisture lowers overall density.

Assertion and Reason Type Questions

14. Assertion (A): At constant pressure, the volume of given mass of gas is directly proportional to its absolute temperature. *(True — This is Charles's Law)*

- Reason (R): If the temperature is doubled from Celsius, the volume does not double. *(False — The doubling rules only apply to the absolute Kelvin scale).*
- Answer: C) Assertion is true, but Reason is false.

15. Assertion (A): At high altitudes, ears pop. *(True)*

- **Reason (R):** At high altitude, the pressure is less and hence to balance the pressure, ears pop. *(True)*
- **Answer: A)** Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

Matrix Match Type Questions

16.

- (a) Solid → (R) Fixed shape and definite volume
- (b) Liquid → (Q) Fixed volume but no definite shape

- (c) Gas → (P) No fixed shape and high compressibility
- (d) Plasma → (S) It is created when extreme heat or strong electromagnetic fields are applied to a gas

Comprehension Type Questions

17. Answer: B) Particles are far apart

18. Answer: B) Gases fill the entire container

19. Answer: C) Homogeneous mixture

Kinetic Theory of Gases

Level – I

1. Answer: Ideal Gas Equation: $PV = nRT$ **or Combined Gas Law:** $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$

- **Explanation:** Though the specific options are omitted in your text, the fundamental mathematical descriptions governing ideal gas behavior are either the ideal gas equation ($PV = nRT$) or the combined state relationship ($\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$).

2. Answer: A) exert no attractive forces on each other

- **Explanation:** In an ideal gas, intermolecular forces are completely absent. During free (unrestricted) expansion into a vacuum, no work is performed against attractive forces, meaning internal energy remains constant and no temperature drop (cooling) occurs.

3. Answer: B) H_2 and Cl_2 mixture

Explanation:

- Hydrogen (H_2) and Chlorine (Cl_2) react chemically with one another to form Hydrogen Chloride (HCl) gas:
- H_2 (g) + Cl_2 (g) → 2HCl (g)
- Since H_2 and Cl_2 undergo a chemical reaction that changes the total number of gas molecules present in the container, **Dalton's law cannot be applied to their mixture**.

4. Answer: B) 4.92 atm

- **Explanation:** 1. Find the number of moles (n):

  $$n = \frac{\text{Given mass}}{\text{Molar mass}} = \frac{120\ \text{g}}{40\ \text{g/mol}} = 3\ \text{moles}$$

  2. Use the ideal gas formula $PV = nRT \Longrightarrow P = \frac{nRT}{V}$:

  $$P = \frac{3 \times 0.0821 \times 400}{20} = 3 \times 0.0821 \times 20 = 4.926\ \text{atm}$$

5. Answer: C) 2/5 the total pressure

- **Explanation:** By Dalton's Law, Partial Pressure = Mole Fraction × Total Pressure.

  Mole fraction of Ar (χ_{Ar}) $= \frac{n_{\text{Ar}}}{n_{\text{He}} + n_{\text{Ar}}} = \frac{2}{3+2} = \frac{2}{5}$

  Thus, $P_{\text{Ar}} = \frac{2}{5}P_{\text{total}}$.

6. Answer: C) the kinetic energy of the molecules remains the same

- **Explanation:** The average kinetic energy of gas molecules depends solely on the absolute temperature (KE $= \frac{3}{2}RT$). Since temperature is held constant, the kinetic energy does not change.

7. Answer: D) All of these

- **Explanation:** The Kinetic Molecular Theory (KMT) provides the theoretical groundwork used to derive and prove all basic empirical gas relationships, including Boyle's, Charles's, and Avogadro's laws.

8. Answer: B) 900 cal

- **Explanation:** The internal translational kinetic energy for 1 mole of a monoatomic gas is given by:

  $E = \frac{3}{2}RT$

  Using $R \approx 2$ cal $mol^{-1}K^{-1}$and $T = 300$ K:

  $E = \frac{3}{2} \times 2 \times 300 = 900$ cal

9. Answer: B) double

- **Explanation:** Kinetic energy is directly proportional to absolute temperature (KE $\propto T$). Since the temperature is exactly doubled (300 K → 600 K), the kinetic energy doubles as well.

10. Answer: B) in a straight line path

- **Explanation:** KMT postulates that gas molecules experience no intermolecular attractions or repulsions. Consequently, they move at a constant speed in a straight line until a collision redirects them.

11. Answer: B) most probable speed

- **Explanation:** By definition, the **most probable speed** (v_{mp}) marks the peak of the Maxwell-Boltzmann distribution curve, representing the speed value held by the largest fraction of molecules.

12. Answer: B) The square of the mean speed of the molecules is equal to the mean squared speed at a certain temperature

- **Explanation:** Mathematically, the square of the average value is not equal to the average of the squared values ($(\overline{v})^2 \neq \overline{v^2}$). The root-mean-square speed (v_{rms}) is always slightly larger than the mean speed ($\overline{v}$).

13. Answer:D) CO

Explanation:

1. The Formula: The root-mean-square (rms) speed (v_{rms}) of a gas molecule is mathematically defined by the expression:

$v_{rms} = \sqrt{\frac{3RT}{M}}$

Where:

- R is the universal gas constant
- T is the absolute temperature in Kelvin
- M is the molar mass of the gas

2. The Relationship: Since all the given gases are at the exact same temperature (0°C), the values of 3, R, and T are constant. Therefore, the rms speed is inversely proportional to the square root of the molar mass of the gas:

$v_{rms} \propto \frac{1}{\sqrt{M}}$

This means the gas with the lowest molar mass (the lightest gas) will have the highest rms speed.

3. Calculating Molar Masses:

- A) O_3 (Ozone): $3 \times 16 = 48$ g mol^{-1}
- B) CO_2 (Carbon Dioxide): $12 + (2 \times 16) = 12 + 32 = 44$ g mol^{-1}
- C) SO_3 (Sulfur Trioxide): $32 + (3 \times 16) = 32 + 48 = 80$ g mol^{-1}
- D) CO (Carbon Monoxide): $12 + 16 = 28$ g mol^{-1}

Conclusion:

Carbon Monoxide (CO) has the smallest molar mass (28 g mol^{-1}) among the options. Consequently, its molecules move the fastest, giving it the highest rms speed.

14. Answer: A) 327°C

- **Explanation:** Equating the v_{rms}values:

$$\sqrt{\frac{3RT_{He}}{M_{He}}} = \sqrt{\frac{3RT_{H_2}}{M_{H_2}}} \Rightarrow \frac{T_{He}}{4} = \frac{27+273}{2}$$

$$\frac{T_{He}}{4} = \frac{300}{2} \Rightarrow T_{He} = 600 \text{ K}$$

Converting back to Celsius: $600 - 273 = 327°C$.

15. Answer: A) 5611.95 J mol^{-1}

- **Explanation:**

$$E = \frac{3}{2}RT = \frac{3}{2} \times 8.314 \text{ J mol}^{-1}\text{K}^{-1} \times 450 \text{ K} = 5611.95 \text{ J mol}^{-1}$$

LEVEL – II

1. Answer: (D) The product of pressure and volume always remains constant

- **Explanation:** This is the exact statement of Boyle's Law ($PV = k$).

2. Answer: (A) 233.91 cm^3

- **Explanation:** Convert initial pressure to atm: $P_1 = \frac{800}{760} = 1.0526$ atm. Apply Boyle's Law ($P_1V_1 = P_2V_2$):

$$1.0526 \times 200 = 0.9 \times V_2 \Rightarrow V_2 = \frac{210.52}{0.9} = 233.91 \text{ cm}^3$$

3. Answer: (A) 10 cc

- **Explanation:** Using $P_1V_1 = P_2V_2$:

$1 \text{ atm} \times 100 \text{ cc} = 10 \text{ atm} \times V_2 \Rightarrow V_2 = 10$ cc

4. Answer: (C) Increases from V to 1.084 V

- **Explanation:** According to Charles's Law ($\frac{V_1}{T_1} = \frac{V_2}{T_2}$):

$T_1 = 273$ K, $T_2 = 273 + 23 = 296$ K

$$V_2 = V \times \frac{296}{273} = 1.084 \text{ V}$$

5. Answer: (C) 8.314 J $mol^{-1}K^{-1}$

- **Explanation:** This is the standard value of the universal gas constant (R) in SI units.

6. Answer: B) Pressure of moist gas minus pressure of dry gas... depends upon the nature of the gas

- **Explanation:** The difference between moist and dry gas pressure equals the **aqueous tension** (vapor pressure of water), which is a function of temperature alone and remains entirely independent of the nature of the gas mixed with it.

7. Answer: A and B (Increase in average molecular speed & Increased rate of collision)

- **Explanation:** Higher temperatures supply kinetic energy, causing molecules to move faster (A). As a result, they strike the container walls more frequently per unit time (B), which scales up the total measured pressure.

Assertion and Reason Type Questions

8.

- **Assertion (A):** The pressure of a gas is due to the continuous bombardment of gas molecules on the walls of the container. (True)
- **Reason (R):** Gas molecules move randomly with very high velocities in all directions. (True)
- **Answer: (A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.**

Matrix Matching Type Questions

9.

- (A) Root mean square speed →(r) $\sqrt{\frac{3RT}{M}}$
- (B) Average speed →(q) $\sqrt{\frac{8RT}{\pi M}}$
- (C) Most probable speed →(p) $\sqrt{\frac{2RT}{M}}$
- (D) Kinetic energy per mole →(s) $\frac{3}{2}RT$

10.

- **(A) Avogadro's number →(r)** 6.022×10^{23}
- **(B) Molecules of gas →(p)** Perfectly elastic
- **(C) Collisions of molecules with walls →(s)** Pressure of gas
- **(D) 1 mole of** CO_2 **→(q)** 44 g

Comprehension Type Questions

11. Answer: (D) CO_2

- **Explanation:**

  $$\text{Molar mass} = \frac{\text{Mass}}{\text{Volume at STP}} \times 22.4 \text{ L} = \frac{4.4}{2.24} \times 22.4 = 44 \text{ g/mol}$$

  The molar mass of $CO_2$is $12 + (16 \times 2) = 44$ g/mol.

12. Answer: (D) NO, C_2H_6

- **Explanation:** By Graham's Law, gases with identical molecular weights diffuse at identical rates. Both NO($14 + 16 = 30$ g/mol) and C_2H_6($(12 \times 2) + 6 = 30$ g/mol) share the same molar mass.

13. Answer: (C) P/RT

- **Explanation:** Rearranging $PV = nRT$yields $\frac{n}{V} = \frac{P}{RT}$.

Integer Type Questions

14. Answer: 2

15. Answer: 2

- **Explanation:** $n = \frac{\text{Mass}}{\text{Molar mass of } O_2} = \frac{64 \text{ g}}{32 \text{ g/mol}} = 2$ moles.

Previous Year Questions

1. Answer: B) Both statement-I and statement-II are not correct

- **Explanation:** * Statement-I is false: If intermolecular forces dominate thermal energy, the substance condenses into a **solid or liquid**, not a gas.
  - Statement-II is false: There are **11** gaseous elements at room temperature (H, He, N, O, F, Ne, Cl, Ar, Kr, Xe, Rn), not 10.

2. Answer: C) High temperature and low pressure

- **Explanation:** At high temperature (high kinetic energy overcoming attractions) and low pressure (molecules are spread very far apart), real gases behave ideally.

3. Answer: A) Both statement-I and statement-II are correct

- **Explanation:** Positive deviation implies the cohesive forces between unlike components (A-B) are weaker than pure components (A-A and B-B), yielding an endothermic mixing phase where $\Delta H_{\text{mix}} > 0$.

4. Answer: C) 1.00

- **Explanation:** Concentration $(C) = \frac{n}{V} = \frac{P}{RT} = \frac{16.4}{0.082 \times 200} = \frac{16.4}{16.4} = 1.00$ mol L^{-1}.

5. Answer: B) b, c, a only

Explanation: Look at lines **b, c, and a**: From the very beginning (as pressure increases from zero), these curves go straight up and remain completely **above the** $Z = 1.0$ **line** at all pressure points. They never dip below it. Therefore, gases **b, c, and a** show **only positive deviation**.

Look at lines **d, e, and f**: These curves dip below the $Z = 1.0$ line first (showing negative deviation at lower pressures) before eventually crossing back above it at high pressures.

6. Answer: $d = \frac{PM}{RT}$

7. Answer: B) There is a complete disorder of molecules in gases

- Explanation: Unlike structured solids, gas particles undergo continuous, uncoordinated chaotic motions.

8. Answer:D) n**-**$\mathbf{C_6H_{14}}$ **+** n**-**$\mathbf{C_7H_{16}}$

Explanation:

An ideal solution is a solution that obeys Raoult's Law strictly across the entire range of concentrations and temperatures. For a mixture of component A and component B to behave ideally, two main conditions must be satisfied:

1. The molecular size and structure of both components must be highly similar.
2. The intermolecular forces of attraction between the mixed components (A-B interactions) must be nearly identical to the pure components (A-A and B-B interactions).

9. Answer: C) greater than one

- Explanation: Hydrogen exhibits purely repulsive behavior modifications at these standard limits, keeping $Z > 1$.

10. Answer: C) (i), (ii) and (iii)

- Explanation: All options correctly list the core foundational postulates of the Kinetic Molecular Theory for ideal gases.

4. Carbon and Its Compounds

LEVEL-I

1. The correct answer is C.

Diamond is composed of carbon atoms arranged in a crystal structure called a diamond cubic. Each carbon atom is covalently bonded to four other carbon atoms in a tetrahedral configuration, creating a highly stable and pure form of carbon.

While graphite, coal, and lamp black are also carbon-based materials, they contain impurities or different structural arrangements.

2. The correct answer is B.

(A) Carbon dioxide: This is a gaseous compound of carbon and oxygen (CO_2). It is not a liquid state of carbon.

(B) Petroleum: This is a complex mixture of hydrocarbons, which are compounds primarily made up of carbon and hydrogen. While petroleum is not pure carbon, it is a liquid mixture containing significant amounts of carbon in various forms.

(C) Charcoal: This is a solid form of carbon, typically used as a fuel or for filtration.

(D) Lamp black: This is a fine black powder consisting of carbon, produced by burning oil or gas with insufficient air supply, and it is solid.

3. The correct answer is A.

Allotropic forms have the same chemical properties (P) and different physical properties (Q).

Allotropes are different structural forms of the same element. They have identical chemical composition but differ in their physical properties due to variations in atomic or molecular arrangements.

P: Chemical properties are the same because allotropes are composed of the same element and have the same number of protons in their nuclei.

Q: Physical properties differ because of the distinct structural arrangements of atoms or molecules in each allotropic form.

4. The correct answer is B.

Coal is a complex, heterogeneous material formed from the decomposition of plant matter over millions of years. Its carbon content varies widely depending on the type of coal, which is determined by factors such as age, pressure, and temperature during formation.

he main types of coal, in order of increasing carbon content, are:

1. Lignite (28-40% carbon)
2. Sub-bituminous coal (40-50% carbon)
3. Bituminous coal (60-80% carbon)
4. Anthracite (86-98% carbon)

This wide range in carbon content (28-98%) reflects the diverse nature of coal and its various stages of formation, making it an impure but valuable carbon-rich resource.

5. The correct answer is D.

Diamond is an allotrope of carbon characterized by its crystalline structure, where each carbon atom is covalently bonded to four other carbon atoms in a tetrahedral arrangement. This unique structure contributes to diamond's high density, as well as its exceptional hardness and thermal conductivity.

The density of a material is calculated using the formula:

Density = Mass / Volume

For diamond, this high density of 3.52 g cm^{-3} results from the tight packing of carbon atoms in its crystal lattice. This compact arrangement allows for a large mass of carbon to be contained within a relatively small volume.

6. The correct answer is A.

Black diamonds, also known as carbonados.

Carbonados are a unique and rare type of diamond that are typically found in alluvial deposits in Central African Republic and Brazil. Unlike traditional diamonds, which are transparent or exhibit various colours, carbonados are opaque and have a black or dark greycolour. They are composed of numerous small diamond crystals that are bonded together, giving them a porous, polycrystalline structure.

7. The correct answer is C.

Black diamonds, also known as carbonados, are polycrystalline diamonds with a unique structure that contributes to their exceptional hardness. While all diamonds are extremely hard, ranking 10 on the Mohs scale of mineral hardness, black diamonds are considered the hardest variant.

Key points about black diamonds:

1. Composed of many small diamond crystals bonded together
2. Higher resistance to cleavage than single-crystal diamonds
3. More durable and less prone to chipping or breaking

8. The correct answer is C.

The weight of diamonds is traditionally expressed in carats (C), and one carat is equal to 0.2 grams.

Key points:

1. Carat is a unit of mass specifically used for gemstones and pearls.
2. The term "carat" comes from the Greek word for carob seed, which was used as a standard for weighing small quantities.
3. The modern carat was standardized in 1907 to exactly 200 milligrams or 0.2 grams.

The conversion between carats and grams can be expressed as:

1 carat = 0.2 grams

1 gram = 5 carats

9. The correct answer is D.

In a regular diamond crystal, each carbon atom is held firmly in place by 4 bonds of neighbouring carbon atoms.

Diamond has a unique crystalline structure where each carbon atom forms covalent bonds with four other carbon atoms in a tetrahedral arrangement. This structure, known as the diamond cubic crystal system, gives diamond its exceptional hardness and other remarkable properties.

Key points:

1. Each carbon atom in diamond uses all four of its outer electrons to form single covalent bonds.
2. The bonding angle between any two bonds is approximately 109.5°.
3. This tetrahedral arrangement repeats throughout the crystal, creating a strong, rigid 3D network.

10. The correct answer is B.

A pure diamond is characterized by its transparency. Diamonds are crystalline solids composed of carbon atoms arranged in a tetrahedral lattice structure, resulting in their transparency to visible light. This crystalline structure gives

diamonds their clarity and brilliance, allowing light to pass through and refract within the gemstone, giving it its characteristic sparkle.

11. The correct answer is D.

The carbon described in the question where carbon atoms are arranged in hexagonal rings stacked on top of each other in flat planes is graphite.

Graphite is a crystalline form of carbon where each carbon atom is bonded to three others in a flat hexagonal lattice. These lattices are arranged in layers that can slide over each other, giving graphite its characteristic slippery feel and lubricating properties. The layers are held together by relatively weak van der Waals forces, allowing them to be easily separated, which is why graphite is used in pencils and as a lubricant.

12. The correct answer is C.

Graphite is commonly known as (C) Black lead. This name originates from historical usage, although it's important to note that graphite is not chemically related to lead. Graphite is a form of carbon that has a layered structure, with each layer made up of carbon atoms arranged in a hexagonal lattice. It is known for its lubricating properties, electrical conductivity, and resistance to high temperatures.

13. The correct answer is B.

Fullerenes, a class of carbon molecules with a hollow sphere or tube structure, were discovered by H.W. Kroto, R.F. Curl, and R.E. Smalley. They made this groundbreaking discovery in 1985, for which they were awarded the Nobel Prize in Chemistry in 1996.

The discovery was accidental and occurred during experiments to simulate the chemistry in the atmosphere of giant stars using laser vaporization of graphite. The resulting clusters of carbon atoms, observed through mass spectrometry, revealed the presence of C60 and C70, which are among the most well-known fullerenes.

14. The correct answer is C.

A mixture of carbon monoxide (CO) and hydrogen (H_2) is commonly called Water gas.

Water gas is produced by passing steam over red-hot coke (carbon) in the presence of limited oxygen. The reaction can be represented as:

$C + H_2O \rightarrow CO + H_2$

In this reaction, carbon (C) reacts with water vapor (H_2O) to produce carbon monoxide (CO) and hydrogen gas (H_2). Water gas is used as a fuel and as a source of hydrogen in various industrial processes, such as in the production of ammonia and methanol.

15. The correct answer is D.

The form of carbon that matches the given properties of being soft, porous, and floating on water is Wood charcoal.

Wood charcoal is produced by heating wood in the absence of oxygen, resulting in a lightweight and porous material. It is soft to the touch and can easily break apart. Due to its porous nature, wood charcoal has a large surface area, making it effective for adsorption and filtration applications. The porosity allows it to trap gases and other impurities, which is why it is commonly used in water purification processes and air filters.

16. The correct answer is A.

(A) Oxygen II. 400°C III. without flame IV. carbon dioxide

When wood charcoal is heated in air or oxygen, it catches fire at around 400°C and burns without a visible flame, producing carbon dioxide gas and liberating a large

amount of heat energy. This process is an example of combustion, where carbon (in the form of charcoal) reacts with oxygen to produce carbon dioxide and energy in the form of heat. The absence of a visible flame at the initial stage of combustion is due to the gradual release of heat, which is sufficient to sustain the reaction but may not yet be intense enough to produce a visible flame. As combustion continues and more heat is released, it may eventually lead to a visible flame as the temperature rises further.

17. The correct answer is D.

The different layers in graphite are held together by van der Waals forces.

Graphite consists of layers of carbon atoms arranged in a hexagonal lattice, where each carbon atom forms strong covalent bonds within a layer. However, between these layers, the bonding is primarily due to van der Waals forces, which are weak attractive forces between molecules or atoms. In graphite, these forces arise from temporary dipoles induced in adjacent layers of carbon atoms. These forces allow the layers to slide past each other easily, giving graphite its lubricating properties and ability to be used as a solid lubricant.

18. The correct answer is B.

The main constituent of LPG (liquefied petroleum gas) is propane.

Propane (C_3H_8) is a hydrocarbon gas that is a major component of LPG. It is colourless and odourless in its pure form but is typically odorized for safety reasons. Propane is widely used as a fuel for heating, cooking, and in vehicles. It is produced during the refining of crude oil and natural gas processing.

19. The correct answer is C.

High calorific value means that a fuel can produce a large amount of heat energy per unit mass or volume when burned completely. This is desirable because it indicates greater efficiency in energy generation for various applications, whether for heating, power generation, or transportation.

20. The correct answer is C.

In the reaction $CuO + C \rightarrow Cu + CO$, Copper acts as oxidising agent

In this reaction, copper oxide (CuO) is reduced to copper (Cu), while carbon (C) is oxidized to carbon monoxide (CO). Copper oxide serves as the oxidizing agent because it accepts electrons and is reduced in the process.

The redox (reduction-oxidation) reactions can be broken down as follows:

Reduction: $CuO + 2e^- \rightarrow Cu + O^{2-}$

Oxidation: $C + O^{2-} \rightarrow CO + 2e^-$

An oxidizing agent is a substance that causes oxidation by accepting electrons.

21. The correct answer is B.

The chemical reaction $CO + N_2$ is known as producer gas. Producer gas is produced by passing air or oxygen over red-hot coke or coal, resulting in a mixture primarily composed of carbon monoxide (CO) and nitrogen (N_2), along with other gases like hydrogen (H_2) and methane (CH_4). This gas mixture has been historically used as a fuel in industrial processes, particularly in gasification processes where solid carbonaceous materials are converted into gas.

22. The correct answer is D.

In the reaction:

$4HNO_3 + C \text{ Boiling} \rightarrow CO_2 + 2H_2O + 2NO_2$

- Statement (A), "Wood charcoal reduces nitric acid to nitrogen dioxide," is true because in the presence of wood charcoal (C), nitric acid (HNO_3) is reduced to nitrogen dioxide (NO_2).

- Statement (B), "The reaction occurs on boiling," is true as indicated by "Boiling" in the reaction conditions.
- Statement (C), "The substance reduced is nitric acid," is also true because nitric acid (HNO_3) undergoes reduction to produce nitrogen dioxide (NO_2) in the presence of carbon (C).

23. The correct answer is B.

Wood charcoal, produced by heating wood in the absence of oxygen (a process called pyrolysis), is primarily composed of carbon. This high carbon content gives wood charcoal some interesting properties:

1. Electrical conductivity: The carbon atoms in charcoal form a network structure with delocalized electrons, allowing for the flow of electric current. While not as conductive as metals, charcoal is significantly more conductive than wood.
2. Thermal conductivity: Charcoal conducts heat better than wood due to its denser structure and carbon content. However, its thermal conductivity is lower than that of most metals.

24. The correct answer is A.

Bones are complex structures composed of both inorganic and organic components. The inorganic portion, which makes up about 60-70% of bone tissue, is primarily calcium phosphate in the form of hydroxyapatite. The chemical formula for hydroxyapatite is: $Ca_{10}(PO_4)_6(OH)_2$

The remaining 30-40% of bone tissue consists of organic matter, primarily collagen (a protein that provides flexibility and tensile strength) and other proteins, as well as water and living cells in fresh bone.

The approximate composition can be represented as:

Bone = 60% Calcium Phosphate + 40% Organic Matter

25. The correct answer is A.

The ingredient commonly used in ayurvedic dental powder is Wood charcoal.

Wood charcoal is a popular ingredient in many traditional ayurvedic dental powders due to its unique properties and perceived benefits for oral health. Here's why it's used:

1. Mild abrasiveness: Wood charcoal can help remove surface stains from teeth without being overly harsh on the enamel.
2. Adsorptive properties: Charcoal can absorb (bind to its surface) various substances, potentially helping to remove toxins and bacteria from the mouth.
3. p_H balancing: It may help neutralize acids in the mouth, promoting a healthier oral environment.
4. Antimicrobial properties: Some studies suggest charcoal has mild antimicrobial effects, which could be beneficial for oral health.

26. The correct answer is B.

The carbon present in bone ash is removed by dissolving it in concentrated HCl (Hydrochloric Acid).

Concentrated hydrochloric acid (HCl) is used to remove carbon and other impurities from bone ash. This process is important in the purification of bone ash, which is primarily composed of calcium phosphate.

The reaction can be represented as: $Ca_3(PO_4)_2 + 6HCl \rightarrow 3CaCl_2 + 2H_3PO_4$

In this reaction: - Calcium phosphate from the bone ash reacts with HCl.
- Calcium chloride ($CaCl_2$) and phosphoric acid (H_3PO_4) are formed.
- Carbon and other organic impurities are separated from the solution.

27. The correct answer is C.

Wood charcoal acts as an excellent reducing agent. A reducing agent is a substance that causes another substance to gain electrons, leading to a decrease in the oxidation state of the substance being reduced. Wood charcoal, primarily composed of carbon, has a high affinity for oxygen.

When heated in the presence of a substance that can be oxidized, such as metal oxides or other compounds, charcoal donates electrons to reduce the compound, typically forming carbon dioxide and water vapour as by-products.

For example, in the reduction of iron oxide (Fe_2O_3) to iron (Fe) using charcoal:

$Fe_2O_3 + 3C \rightarrow 2Fe + 3CO_2$

28. The correct answer is C.

The main source of energy in living organisms is carbohydrates. Carbohydrates are organic compounds composed of carbon, hydrogen, and oxygen, and they serve as a primary energy source due to their ability to be broken down into glucose through various metabolic processes.

Glucose is a simple sugar that can be readily utilized by cells to produce ATP (adenosine triphosphate), which is the energy currency of the cell. ATP provides the necessary energy for cellular functions such as metabolism, growth, repair, and movement.

29. The correct answer is B.

Fossil fuels are natural resources formed from the remains of ancient organisms over millions of years.

Fossil fuels primarily consist of coal and petroleum. Coal forms from the compressed remains of plants that lived millions of years ago, while petroleum (crude oil) originates from ancient marine organisms. These fuels are vital energy sources globally, powering electricity generation, transportation, and industrial processes.

30. The correct answer is D.

The poisonous gas that is released when petrol (gasoline) burns in a car engine is carbon monoxide (CO).

During combustion in an internal combustion engine, such as in a car, the carbon in the petrol combines with oxygen (O_2) from the air to form carbon dioxide (CO2) as the primary product. However, incomplete combustion due to insufficient oxygen supply or improper combustion conditions can lead to the formation of carbon monoxide instead of carbon dioxide.

Carbon monoxide is highly toxic because it binds strongly to haemoglobin in red blood cells, reducing their ability to carry oxygen, which can lead to carbon monoxide poisoning.

LEVEL-II
SINGLE ANSWER QUESTIONS

1. The correct answer is B.

The nature of the chemical bond in diamond is covalent.

Diamond is composed entirely of carbon atoms arranged in a tetrahedral lattice structure. Each carbon atom forms strong covalent bonds with four other carbon atoms, resulting in a three-dimensional network or crystal lattice. In this structure, each carbon atom shares electrons with its neighbouring atoms, forming very stable covalent bonds.

Covalent bonds are characterized by the sharing of electron pairs between atoms, where each atom contributes electrons to form a stable configuration.

2. The correct answer is D.

This sequence represents the coal formation process, also known as coalification:
Organic matter → Peat → Lignite → bituminous → Anthracite
Approximate carbon content ranges: Peat: 50-60% carbon
Lignite: 60-70% carbon
Bituminous: 70-85% carbon
Anthracite: 85-95% carbon

3. The correct answer is D.

The most reactive form of carbon among the options provided is option charcoal.
Charcoal is highly reactive due to its porous structure, which provides a large surface area for chemical reactions. It is produced by heating wood or other organic materials in the absence of oxygen, resulting in a material rich in carbon with a network of pores.
These pores can absorb gases and liquids, making charcoal useful in various applications like water filtration, air purification, and as an adsorbent in chemical processes.

4. The correct answer is D.

Carborundum is not an allotrope of carbon. It's actually silicon carbide (SiC), a compound of silicon and carbon. The other options - graphite, diamond, and coal - are all allotropes of carbon.
Allotropes are different structural forms of the same element. Carbon has several allotropes due to its ability to form different bonding arrangements. Graphite consists of layers of hexagonally arranged carbon atoms.
Diamond has a tetrahedral crystal structure. Coal, while not a pure form of carbon is primarily composed of carbon atoms arranged in complex structures.

5. The correct answer is C.

The property of graphite that is used for making electrodes is good conductor of electricity.
Graphite is composed of layers of carbon atoms arranged in a hexagonal lattice structure, where each carbon atom is covalently bonded to three other carbon atoms. These layers are held together by weak van der Waals forces, allowing them to slide over each other easily, which give graphite its characteristic soft and slippery feel.

6. The correct answer is B.

Dry ice is solid carbon dioxide. Carbon dioxide (CO_2) ordinarily exists as a gas at room temperature and pressure. However, under certain conditions of low temperature and high pressure, it can be transformed directly into a solid without passing through a liquid phase. This solid form of carbon dioxide is commonly known as dry ice.
Dry ice is used in various applications, including as a cooling agent in food transportation, medical preservation, and special effects due to its ability to sublimate directly from a solid to a gas at temperatures above -78.5°C (-109.3°F) without leaving a liquid residue.

7. The correct answer is A.

Lime water (calcium hydroxide solution) turns milky when it reacts with carbon dioxide (CO_2). This reaction forms calcium carbonate ($CaCO_3$), which is insoluble in water and appears as a white precipitate: $Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$

8. The correct answer is D.

Despite its name, a "lead pencil" doesn't actually contain any lead. The writing core of a pencil is made of graphite, mixed with clay as a binder. This misconception about lead content likely arose from the early use of lead in writing instruments and the similar appearance of graphite to lead.

Graphite, an allotrope of carbon, has the chemical formula C (graphite).

9. The correct answer is B.

Diamond and graphite are allotropes of carbon. Allotropes are different structural forms of the same element occurring in the same physical state. Both diamond and graphite are composed entirely of carbon atoms, but their unique properties stem from different atomic arrangements.

In diamond, each carbon atom is covalently bonded to four other carbon atoms in a tetrahedral structure.

10. The correct answer is A.

The incomplete combustion of organic compounds occurs when there is insufficient oxygen present to completely oxidize the carbon and hydrogen atoms in the compound.

Typically, incomplete combustion results in the formation of carbon monoxide (CO) as a primary product along with carbon dioxide (CO_2) and water vapour (H_2O) in varying proportions.

Among the options provided:

- (A) CO is carbon monoxide, a common product of incomplete combustion. It forms when there is limited oxygen available, preventing complete oxidation to CO2.
- (B) CO_2 is carbon dioxide, which is the product of complete combustion when sufficient oxygen is present.
- (C) C_2H_2 is acetylene, which is not typically a product of combustion but rather a compound itself.
- (D) C_2H_4 is ethylene, another organic compound that is not typically a direct product of incomplete combustion.

11. The correct answer is B.

The gas that gives a tangy taste to soft drinks is carbon dioxide.

Carbon dioxide is dissolved in soft drinks under pressure, which creates carbonation. When you open a bottle or can of soda, the release of pressure causes carbon dioxide to come out of solution, forming bubbles. When you drink the soda, these bubbles burst on your tongue, and the dissolved carbon dioxide reacts with water in your mouth to produce carbonic acid (H_2CO_3). Carbonic acid is mildly acidic and gives a tangy or slightly sour taste sensation, which contributes to the refreshing taste of carbonated beverages.

12. The correct answer is D.

Graphite is known for its layered structure where carbon atoms are arranged in hexagonal rings and are linked together in planes stacked on top of each other. This structure allows for easy sliding of layers over each other, making graphite an excellent lubricant. In fact, graphite is widely used as a lubricant due to its low friction properties.

13. The correct answer is C.

Anthracite coal has the highest calorific value among the options provided. Calorific value refers to the amount of heat energy released when a unit mass of fuel is completely burned. It's typically measured in units of energy per mass, such as

kJ/kg or kcal/kg. The order of calorific values for the given options, from highest to lowest, is generally: 1. Anthracite coal
2. Bituminous coal
3. Lignite coal
4. Wood

Anthracite coal's high calorific value is due to its high carbon content and low impurities. It typically contains:
Carbon: 86-98%
Hydrogen: 2-5%
Oxygen: 1-2%

14. The correct answer is C.

Kajal, also known as kohl, is a form of carbon-black. Carbon-black is a fine, powdery form of elemental carbon produced by the incomplete combustion of heavy petroleum products or vegetable oils.

Traditional Kajal is made by collecting the soot (carbon-black) produced from burning oil lamps or other organic materials. This process can be represented as:
Organic material + insufficient O_2 → C (soot) + other products

ONE OR MORE THAN ONE CORRECT ANSWER QUESTIONS

15. The correct answer is D.

Allotropes are different structural forms of the same element in the same physical state. Their formation is influenced by multiple factors:

1. Atomic arrangement: This is the primary factor. Different spatial arrangements of atoms lead to different allotropes. For example, in carbon:
Diamond: C (tetrahedral)
Graphite: C (hexagonal layers)

2. Experimental conditions: Temperature and pressure can determine which allotrope forms. For instance:
C (graphite) → C (diamond) (under high pressure and temperature)

3. Physical properties: Allotropes exhibit different physical properties due to their structural differences. These properties can influence their formation and stability under various conditions.

16. The correct answer is A, B, C, D.

Carbon is a crucial element found in all living organisms and is a fundamental building block of organic compounds. Among the options provided, carbon is a key constituent of all choices: cellulose (A), proteins (B), fats (C), and carbohydrates (D).

- Cellulose (A): Cellulose is a polysaccharide composed of glucose units linked together. Carbon forms the backbone of these glucose molecules.
- Proteins (B): Proteins are composed of amino acids, each containing carbon atoms in their structure, forming the peptide bonds that link amino acids together.
- Fats (C): Fats, or triglycerides, consist of glycerol and fatty acids. Fatty acids are long hydrocarbon chains with carboxylic acid groups, primarily made of carbon and hydrogen atoms.
- Carbohydrates (D): Carbohydrates are organic compounds like sugars and starches, consisting of carbon, hydrogen, and oxygen atoms in various ratios, with carbon serving as the primary structural element.

17. The correct answer is A, B, D.

(A) Diamond is stable towards heat: This statement is correct. Diamond has a very high melting point and is stable at high temperatures, making it resistant to heat.

(B) Diamond is chemically inert: This statement is generally correct. Diamond is resistant to most chemical reactions at standard conditions, although it can burn in oxygen at high temperatures.

(C) Diamond is a good conductor of electricity: This statement is incorrect. Diamond is an electrical insulator; it does not conduct electricity due to its tetrahedral structure and lack of free electrons.

(D) Diamond is a bad conductor of heat: This statement is correct. Diamond has a very high thermal conductivity, making it an excellent conductor of heat.

Therefore, the correct statements are:

- (A) Diamond is stable towards heat.
- (B) Diamond is chemically inert.
- (D) Diamond is a bad conductor of heat.

18. The correct answer is A, C, D.

Graphite, an allotrope of carbon, has several distinct physical properties. To identify the false statements about graphite, Examine each option:

(A) It is heavier than diamond.

- This statement is false. Graphite is not heavier than diamond. In fact, diamond is denser and heavier due to its more tightly packed crystal structure.

(B) It has a density of about 2.25 g/cm^3.

- This statement is true. Graphite has a density of approximately 2.25 g/cm^3, which is less than the density of diamond.

(C) It is transparent to light.

- This statement is false. Graphite is opaque and not transparent to light. In contrast, diamond is transparent.

(D) It has a very low melting point.

- This statement is false. Graphite has a very high melting point, exceeding 3600°C, due to the strong covalent bonds within its layers.

19. The correct answer is A, C.

Consider the oxidation states:

C: 0 → +2 (in CO)

H: +1 → +1 (unchanged)

O: -2 → -2 (unchanged)

(A) Wood charcoal (C) undergoes oxidation as its oxidation state increases from 0 to +2.

(C) Wood charcoal acts as a reducing agent because it reduces the oxidation state of oxygen in H_2O from -2 to -2 in CO and 0 in H_2.

(B) is incorrect because wood charcoal is being oxidized, not acting as an oxidizing agent.

(D) Is incorrect because steam (H_2O) is being reduced, not oxidized.

ASSERTION & REASON TYPE QUESTIONS

20. The correct answer is D.

Assertion: Carbon can form different allotropes.

This statement is correct. Carbon is known to exist in several allotropic forms, including diamond, graphite, fullerenes (such as C60 buckyballs), carbon nanotubes, and graphene.

Reason: Carbon exhibits the property of allotropy.

This statement is also correct. Allotropy is the property of an element to exist in two or more different forms in the same physical state. Carbon indeed exhibits this property.

The assertion and reason are directly related. Carbon's ability to form different allotropes (assertion) is a direct result of its property of allotropy (reason). This property arises from carbon's unique electronic structure and its ability to form different types of bonds and structural arrangements with itself.

21. The correct answer is D.

The assertion "Diamond is well known for its transparency and brilliant nature" is correct. Diamonds are highly valued for their ability to transmit light and their sparkling brilliance.

The reason "Diamond has high refractive index" is also correct. The high refractive index of diamond (2.42) means that it can bend light significantly as it passes through the stone. This high refractive index is responsible for the diamond's brilliance and the way it disperses light into various colours, contributing to its sparkling appearance.

22. The correct answer is D.

The assertion states that graphite, despite being a non-metal, is a good conductor of heat and electricity. This is correct. Graphite, an allotrope of carbon, has a unique structure where each carbon atom is bonded to three other carbon atoms, forming hexagonal layers. These layers can slide over each other, making graphite slippery and a good lubricant.

The reason provided is that the fourth valence electron of each carbon atom remains loosely bound between the hexagonal planes, allowing it to conduct heat and electricity. This explanation is also correct. In graphite, three of the four valence electrons of each carbon atom are involved in covalent bonding within the hexagonal layers, while the fourth electron is free to move between the layers. This delocalized electron can conduct electrical current and contribute to thermal conductivity.

MATCHING TYPE QUESTIONS

23.

(a) Diamond - (S) Abrasive
(b) Water gas - (R) $CO + H_2$
(c) Producer gas - (Q) $CO + N_2$
(d) Graphite - (P) Lubricant
(a) Diamond is extremely hard and is used as an abrasive in cutting and polishing tools.
(b) Water gas is a mixture of carbon monoxide (CO) and hydrogen (H_2). It's produced by passing steam over hot carbon:

$C + H_2O \rightarrow CO + H_2$

(c) Producer gas is primarily a mixture of carbon monoxide (CO) and nitrogen (N_2), with some hydrogen.

(d) Graphite's layered structure, with weak van der Waals forces between layers, allows it to act as a solid lubricant. The layers can easily slide over each other, reducing friction between surfaces.

24.

The correct matches between Column I and Column II are:
(a) CO_2 - (Q) Global warming
(b) CO - (P) Carboxy haemoglobin
(c) Fossil fuels - (S) Natural gas, Petroleum
(d) Diamond - (R) Carats
(a) Carbon dioxide (CO_2) is a major greenhouse gas contributing to global warming.

It traps heat in the Earth's atmosphere, leading to an increase in global average temperatures.

(b) Carbon monoxide (CO) forms Carboxy haemoglobin when it binds to haemoglobin in blood. This binding is much stronger than that of oxygen, reducing the blood's oxygen-carrying capacity and potentially leading to carbon monoxide poisoning.

(c) Fossil fuels include natural gas and petroleum, among others like coal. These are formed from the remains of ancient plants and animals over millions of years.

(d) Diamond weight and quality are often measured in carats. One carat is equal to 200 milligrams (0.2 grams).

25.

The correct matches between Column I and Column II are:

(a) Present in Natural gas - (Q) CH_4

(b) Carbon dioxide gas - (S) Carbon and oxygen

(c) Present in chalk, marble - (R) Obtained by heating sodium bicarbonate and limestone

(d) Used as 'dry ice' - (P) CO_2

(a) Methane (CH_4) is the primary component of natural gas. It's a simple hydrocarbon and the main constituent of natural gas.

(b) Carbon dioxide (CO_2) is composed of carbon and oxygen atoms. Its molecular structure consists of one carbon atom covalently bonded to two oxygen atoms.

(c) Chalk and marble are primarily composed of calcium carbonate ($CaCO_3$). When heated, calcium carbonate decomposes to form calcium oxide and carbon dioxide.

(d) 'Dry ice' is the solid form of carbon dioxide (CO_2). It sublimes directly from solid to gas at −78.5 °C under normal atmospheric conditions.

COMPREHENSION TYPE QUESTIONS

Passage-I

26. The correct answer is B.

Diamond and graphite are both allotropes of carbon. Allotropes are different structural forms of the same element that exist in the same physical state. In this case, both diamond and graphite are solid forms of pure carbon, but they have different crystal structures and properties.

Diamond has a tetrahedral crystal structure where each carbon atom is covalently bonded to four other carbon atoms. This results in a rigid, three-dimensional network that gives diamond its extreme hardness and high melting point.

Graphite has a layered structure. Each layer (called graphene) consists of carbon atoms arranged in a hexagonal pattern. These layers are held together by weak van der Waals forces.

27. The correct answer is C.

Diamond is well-known as the hardest naturally occurring substance on Earth. Its extreme hardness is due to its unique crystal structure, where each carbon atom is covalently bonded to four other carbon atoms in a tetrahedral arrangement. This creates a rigid, three-dimensional network that is extremely difficult to deform or scratch.

On the Mohs scale of mineral hardness, which ranges from 1 (softest) to 10 (hardest), diamond is rated at 10.

28. The correct answer is D.

Graphite is a form of carbon where each carbon atom is bonded to three other carbon atoms in a planar hexagonal lattice, forming layers.

In graphite, each carbon atom forms three sigma bonds with neighbouring carbon atoms, resulting in a trigonal planar structure with a bond angle of 120 degrees. The fourth electron of each carbon atom is in a p-orbital, which overlaps with p-orbital's of adjacent carbon atoms to form a delocalized π-electron cloud above and below the plane of carbon atoms. This delocalization allows electrons to move freely within the layers, giving graphite its characteristic electrical conductivity and lubricating properties. The electrons in graphite are not localized on individual carbon atoms but are spread out over the entire structure.

INTEGER TYPE QUESTIONS

29. Correct answer: 4

In a diamond structure, each carbon atom is covalently bonded to four other carbon atoms. This bonding arrangement forms a tetrahedral structure where each carbon atom shares electrons with four neighbouring carbon atoms. These bonds are strong covalent bonds, creating a three-dimensional network structure known as a diamond lattice.

This tetrahedral bonding configuration is essential for the unique properties of diamond, including its hardness, high thermal conductivity, and optical clarity. The strong covalent bonds ensure that diamond is very resistant to chemical attack and has a high melting point.

30. Correct answer: 4

The valency of a carbon atom, like any other element, refers to its ability to form chemical bonds with other atoms. Carbon typically forms four covalent bonds in most of its stable compounds. This valency arises because carbon has four electrons in its outer shell, allowing it to form stable molecules by sharing electrons with other atoms.

In methane (CH_4), carbon forms four single covalent bonds with hydrogen atoms, utilizing its entire valency:

CH_4: Carbon (C) forms 4 single bonds with Hydrogen (H).

31. Correct answer: 3

The maximum number of bonds between two carbon atoms in an organic compound is three. This occurs in compounds containing a triple bond, also known as an alkyne functional group.

In organic chemistry, carbon atoms can form up to four covalent bonds due to their electronic configuration. When two carbon atoms share three pairs of electrons, they create a triple bond. This is the strongest and shortest type of covalent bond between carbon atoms.

The simplest example of a compound with a carbon-carbon triple bond is ethyne (acetylene), with the molecular formula C_2H_2. Its structure can be represented as:

H-C≡C-H

32. Correct answer: 3

In graphite, each carbon atom is covalently bonded to three other carbon atoms. These bonds form a hexagonal lattice structure where each carbon atom shares three of its valence electrons to form sigma bonds with three neighbouring carbon atoms. This structure is known as a "hexagonal lattice" or "hexagonal network." The fourth valence electron of each carbon atom remains delocalized, creating what is known as a pi-bonding system that extends throughout the structure.

This arrangement gives graphite its unique properties, including it's lubricate and ability to conduct electricity along the planes of carbon atoms.

33. Correct answer: 0

In diamond, each carbon atom forms four covalent bonds with neighbouring carbon atoms in a tetrahedral arrangement, creating a rigid three-dimensional lattice structure. This bonding arrangement leads to the formation of a giant covalent network solid. In diamond, each carbon atom is sp^3 hybridized, meaning it forms sigma bonds with four other carbon atoms.

In sp^3 hybridization, all four valence electrons of carbon participate in bonding, forming four sigma bonds. There are no unpaired electrons because all electrons are involved in bonding pairs within the tetrahedral structure. This configuration gives diamond its exceptional hardness and thermal conductivity, as well as its optical properties like high refractive index and dispersion of light.

In diamond, each carbon atom effectively contributes to the crystal lattice without any unpaired electrons due to the complete bonding in the sp^3 hybrid orbital configuration.

Formation of hydrocarbons

LEVEL-I

1. The correct answer is C.

Alkanes are saturated hydrocarbons with the general formula CnH2n+2, where n is the number of carbon atoms. They contain only single bonds between carbon atoms.

(C) C_3H_8: This fits the alkane formula perfectly. For 3 carbon atoms:

C_nH_{2n+2} = C3H (2*3) +2 = C3H8 , This is propane, a common alkane.

2. The correct answer is B, C.

An unsaturated hydrocarbon contains one or more double or triple bonds between carbon atoms, which results in fewer hydrogen atoms than in a saturated hydrocarbon with the same number of carbon atoms.

- Option (B) C_3H_4: This molecule has 3 carbon atoms and 4 hydrogen atoms. The presence of 2 double bonds (C=C) indicates it is an unsaturated hydrocarbon.
- Option (C) C_5H_8: This molecule has 5 carbon atoms and 8 hydrogen atoms. The presence of 1 double bond (C=C) indicates it is also an unsaturated hydrocarbon.

3. The correct answer is C.

The correct answer is Urea.

Urea was the first organic compound synthesized in a laboratory, a breakthrough achievement by Friedrich Wohler in 1828. This synthesis is considered a pivotal moment in the history of chemistry, as it disproved the prevailing "vitalism" theory, which held that organic compounds could only be produced by living organisms.

Wohler synthesized urea by treating silver cyanate with ammonium chloride. The reaction can be represented as: $AgNCO + NH_4Cl \rightarrow (NH_2)_2CO + AgCl$

4. The correct answer is B.

To form a cyclic hydrocarbon, carbon atoms must be able to bond in a closed ring structure. The minimum number of carbon atoms required to form a stable cyclic hydrocarbon is three, which corresponds to the smallest cyclic hydrocarbon, cyclopropane (C_3H_6).

Cyclopropane consists of a ring of three carbon atoms, each bonded to two hydrogen atoms. Its molecular structure forms a triangle where each carbon atom is bonded to the other two carbons in the ring, fulfilling the requirement for a cyclic structure.

5. The correct answer is B.

To determine which compound has a triple bond, we need to consider the general formula for hydrocarbons and their degree of unsaturation:

(A) C_2H_4: This is ethylene, an alkene with a double bond. Its structure is $H_2C=CH_2$.

(B) C_3H_4: This compound has the correct formula for a hydrocarbon with a triple bond. It can be propyne (also called methyl acetylene), with the structure $HC{\equiv}C-CH_3$.

(C) C_3H_6: This is propene, an alkene with one double bond. Its structure is $H_2C=CH-CH_3$.

(D) All: This is incorrect as only C3H4 has a triple bond.

6. The correct answer is A.

An unsaturated compound in the context of organic chemistry typically refers to a compound that contains double or triple bonds between carbon atoms, allowing for the possibility of further chemical reactions to add more atoms. Among the given compounds, Acetylene (C_2H_2) is unsaturated.

Acetylene, with the chemical formula C_2H_2, consists of two carbon atoms bonded by a triple bond (C≡C) and each carbon atom is also bonded to two hydrogen atoms.

7. The correct answer is D.

To determine the number of sigma bonds in the molecule $CH_2 = C = CHCH_3$, we first need to understand the structure and count the sigma bonds accordingly.

Count the sigma bonds:

- Between C and H in CH_2: 2 sigma bonds (one sigma bond per C-H bond)
- Between C and C in $CH_2 = C$: 1 sigma bond (the single bond between the two carbon atoms)
- Between C and H in $CHCH_3$: 3 sigma bonds (one sigma bond per C-H bond)
- Between C and H in CH_3: 3 sigma bonds (one sigma bond per C-H bond)

Adding these up:

2 (CH_2) + 1 ($CH_2 = C$) + 3 ($CHCH_3$) + 3 (CH_3) = 9 sigma bonds.

8. The correct answer is C.

The number of electrons in the valence shell of carbon is 4. Carbon, with atomic number 6, has an electron configuration of $1s^2\ 2s^2\ 2p^2$. The valence shell of an atom refers to the outermost electron shell, which, for carbon, includes the 2s and 2p orbitals. In these orbital's, there are a total of 4 electrons.

9. The correct answer is B.

Marsh gas is the common name for methane, which has the chemical formula CH_4.

(A) C_2H_2: This is acetylene, an unsaturated hydrocarbon with a triple bond.

(B) CH_4: This is methane, the correct formula for marsh gas.

(C) C_2H_4: This is ethylene, an unsaturated hydrocarbon with a double bond.

(D) C_2H_6: This is ethane, a saturated hydrocarbon.

Methane (CH_4) is the simplest alkane and the main component of natural gas. It's called marsh gas because it's commonly produced in marshes and swamps by the anaerobic decomposition of organic matter.

10. The correct answer is D.

All the compounds listed are inorganic compounds:

(A) Common salt: Also known as sodium chloride (NaCl), it's an ionic compound formed from sodium and chlorine.

(B) Marble: Primarily composed of calcium carbonate ($CaCO_3$), it's a naturally occurring inorganic mineral.

(C) Blue vitriol: This is copper (II) sulphate pentahydrate ($CuSO_4 \cdot 5H_2O$), an inorganic salt of copper.

(D) All: This is correct as all the listed compounds are inorganic.

11. The correct answer is B.

Alkenes are unsaturated aliphatic hydrocarbons containing one carbon-carbon double bond. They have the general formula C_nH_{2n}, where n is the number of carbon atoms. Let's briefly examine each option:

(A) Alkanes: These are saturated hydrocarbons with only single bonds (C_nH_{2n+2}).

(B) Alkenes: These contain one carbon-carbon double bond (C_nH_{2n}).

(C) Alkynes: These contain one carbon-carbon triple bond (C_nH_{2n-2}).

(D) Paraffins: This is another name for alkanes, not for compounds with double bonds.

The structure of the simplest alkene, ethene (C2H4), can be represented as:

$H_2C=CH_2$

12. The correct answer is C.

Organic compounds are primarily composed of carbon and hydrogen atoms, often with other elements like oxygen, nitrogen, sulphur, and halogens. They form the basis of all living organisms and many products derived from them.

Among the options provided:

- (A) Vitamins: Vitamins are organic compounds because they contain carbon atoms in their structure. They are essential nutrients required in small amounts for the proper functioning of the body.

- (B) Oils: Oils are also organic compounds as they are predominantly made up of long hydrocarbon chains derived from plants or animals, containing carbon and hydrogen atoms.

13. The correct answer is C.

Ethene (C_2H_4) is a molecule composed of two carbon atoms and four hydrogen atoms. The structure of ethene consists of a double bond between the two carbon atoms, with each carbon atom also forming single bonds with two hydrogen atoms.

Ethene contains:

- One sigma (σ) bond between each carbon atom and hydrogen atom (4 sigma bonds total for C-H bonds).

- One sigma (σ) bond between the two carbon atoms.

- One pi (π) bond resulting from the overlap of p-orbitals between the two carbon atoms.

14. The correct answer is C.

C_2H_2, also known as acetylene or ethyne, has 3 sigma bonds in its molecular structure. Here's a breakdown:

1. One sigma bond between the two carbon atoms (C-C)
2. Two sigma bonds between each carbon atom and a hydrogen atom (C-H)

The molecular structure of C_2H_2 can be represented as:

H-C≡C-H

In this structure, the triple bond between the carbon atoms consists of one sigma bond and two pi bonds. The total number of bonds in C_2H_2 is 5 (3 sigma + 2 pi).

15. The Correct Answer: (D) butane

Explanation:

L.P.G. (Liquefied Petroleum Gas) is a flammable mixture of hydrocarbon gases used as fuel in heating appliances, cooking equipment, and vehicles.

- It is primarily composed of **butane** (C_4H_{10}) and **propane** (C_3H_8).

- Among these components, **butane** is the major and main constituent, making up about **60% to 65%** of the mixture, while propane constitutes around 35% to 40%.
- Small amounts of other hydrocarbons like ethane and pentane may also be present.

LEVEL-II

Single answer type

1. Saturated hydrocarbons are also known as:

- **Answer: C) Alkanes**
- Explanation: Hydrocarbons are organic compounds composed entirely of carbon and hydrogen. A saturated hydrocarbon is one in which all the carbon-carbon bonds are single covalent bonds. Because each carbon atom forms four single bonds, it is holding the maximum possible number of hydrogen atoms (it is "saturated" with hydrogen). These compounds belong to the homologous series called Alkanes (e.g., methane, ethane, propane).

2. Answer: B and C (Double bonds or Triple bonds)

- Explanation: An unsaturated hydrocarbon contains at least one carbon-carbon double bond (C = C) or triple bond (C ≡ C). Because these multiple bonds exist, the carbon atoms are not holding the maximum possible number of hydrogen atoms. They are capable of undergoing addition reactions to bind more hydrogen atoms.
  - Hydrocarbons with double bonds are called alkenes.
  - Hydrocarbons with triple bonds are called alkynes.

3. Answer: C) Ethyne

- Explanation: The suffix of a hydrocarbon's name tells you the type of bonds it contains:
  - -ane means all single bonds (Alkanes)
  - -ene means it contains a double bond (Alkenes)
  - -yne means it contains a triple bond (Alkynes)

4. General formula of alkenes is:

- Answer: B) C_nH_{2n}
- Explanation: Alkenes contain exactly one carbon-carbon double bond, which removes two hydrogen atoms compared to a fully saturated alkane chain.

ONE OR MORE THAN ONE CORRECT ANSWER QUESTIONS

5. The correct answer is A, C.

A triple bond in alkynes is made up of one sigma bond and two pi bonds.

In alkynes, such as ethyne (C_2H_2), the triple bond between carbon atoms consists of:

1. One sigma (σ) bond
2. Two pi (π) bonds

The sigma bond is formed by the head-on overlap of two sp hybrid orbitals from each carbon atom. The two pi bonds are formed by the side-by-side overlap of two sets of unhybridized p orbitals, one from each carbon atom.

6. The correct answer is A, B, C and D.

1. Clothes: Fabrics like cotton, wool, and synthetic fibers such as polyester all contain carbon. Cotton and wool are natural fibers derived from plants and animals, respectively, which consist primarily of cellulose and keratin, both rich in carbon.

2. Dyes: Many dyes used to colour textiles, food, and other materials are organic compounds containing carbon-based structures. These compounds provide the

colour through their conjugated systems, which absorb light at specific wavelengths.

3. Polymers: Polymers are large molecules composed of repeating units called monomers, many of which are based on carbon backbones. Examples include plastics (polyethylene, PVC), rubber (polyisoprene), and synthetic fibers (nylon, polyester).

4. Medicines: Pharmaceuticals often contain carbon in their structures, essential for the biochemical activity and function of drugs. Carbon is present in various drug classes, including antibiotics, analgesics, and antiviral.

7. The correct answer is D.

The molecular formula C_5H_8 represents a hydrocarbon with 5 carbon atoms and 8 hydrogen atoms. To understand why, let's analyze the molecular formula C_5H_8:

First, calculate the number of bonds carbon can form:

Carbon valency = 4

Total carbon bonds = 4 × 5 = 20

Count the bonds formed with hydrogen:

Hydrogen bonds = 8

Calculate remaining bonds:

Remaining bonds = 20 - 8 = 12

Determine the number of bonds between carbon atoms:

Carbon-carbon bonds = 12 / 2 = 6

In a saturated hydrocarbon (C_nH_{2n+2}), C_5 would have the formula C_5H_{12}. Our compound C_5H_8 has four fewer hydrogen's, indicating the presence of two double bonds or their equivalent.

8. The correct answer is A, B.

The correct answer is both (A) $1s^2\ 2s^2\ 2p^2$ and (B) [He] $2s^2\ 2p^2$. These two representations are equivalent and accurately describe the electronic configuration of carbon.

Carbon is the sixth element in the periodic table, with an atomic number of 6. This means it has 6 protons in its nucleus and 6 electrons surrounding it. The electronic configuration of carbon can be written in two ways:

1. Full notation: $1s^2\ 2s^2\ 2p^2$

This shows all occupied orbitals explicitly.

2. Condensed notation: [He] $2s^2\ 2p^2$

This uses the noble gas core (helium in this case) as shorthand for the completely filled inner shell.

Assertion and reason type

9. Correct Answer:

A) Both A and R are true, and R is the correct explanation of A.

Explanation:

- **Assertion (A) is True:** Saturated hydrocarbons (alkanes) contain only strong, stable carbon-carbon single bonds (σ-bonds) which are difficult to break, making them relatively inert. Unsaturated hydrocarbons (alkenes and alkynes) are significantly more reactive and readily undergo addition reactions at room temperature.
- **Reason (R) is True:** The multiple bonds (double or triple bonds) in unsaturated hydrocarbons consist of a strong σ-bond and one or more weaker π-bonds. The electron cloud of the π-bond lies above and below the plane of the atoms, making it

loosely held and highly susceptible to attack by electrophiles or other chemical reagents.

Since the presence of these weaker, exposed multiple bonds is the exact structural reason *why* unsaturated hydrocarbons are more reactive, Reason (R) is the correct explanation of Assertion (A).

10. Correct Answer:

D) Assertion is false, but Reason is true.

Explanation:

- Assertion (A) is False: The general formula for a cyclic alkane (cycloalkane) is C_nH_{2n} (the same as acyclic alkenes). If a cyclic alkane has 5 carbon atoms ($n = 5$), its formula must be C_5H_{10} (Cyclopentane).

  The given formula, C_5H_8, fits the general formula C_nH_{2n-2}. This means it could represent an alkyne (like pentyne), a diene (like pentadiene), or a cycloalkene (like cyclopentene)—but it cannot be a cyclic alkane.
- Reason (R) is True: Carbon fundamentally belongs to Group 14 and has a valency of 4 (tetravalent), while Hydrogen has a valency of 1 (monovalent).

  Since Assertion (A) is completely incorrect based on chemical formulas but Reason (R) states a correct fundamental fact, the correct choice is D.

Match the following:

11. To match the molecules in Column I with their correct bonding descriptions in Column II, let's analyze each molecule:

(a) C_2H_4 – (P) 5 sigma bonds, (Q) One pi bond

(b) C_2H_2 – (R) 3 sigma bonds, (S) 2 pi bonds

(c) C_3H_6 – (Q) One pi bond, (T) 9 sigma bonds

(d) C_4H_6 – (S) 2 pi bonds, (T) 9 sigma bonds

(a) C_2H_4 (ethylene)

- Ethylene has a double bond between the two carbon atoms and single bonds between carbon and hydrogen.

- Structure: $H_2C=CH_2$

- This means there is one pi bond and five sigma bonds.

- Matches with: (Q) One pi bond and (P) 5 sigma bonds.

(b) C_2H_2 (acetylene)

- Acetylene has a triple bond between the two carbon atoms and single bonds between carbon and hydrogen.

- Structure: HC≡CH

- This means there are two pi bonds and three sigma bonds.

- Matches with: (S) 2 pi bonds and (R) 3 sigma bonds.

(c) C_3H_6 (propene)

- Propene has a double bond between two carbon atoms and single bonds for the rest.

- Structure: $H_2C=CH-CH_3$

- This means there are one pi bond and nine sigma bonds.

- Matches with: (Q) One pi bond and (T) 9 sigma bonds.

(d) C_4H_6 (1, 3-butadiene)

- 1, 3-Butadiene has two double bonds.

- Structure: $H_2C=CH-CH=CH_2$

- This means there are two pi bonds and nine sigma bonds.

- Matches with: (S) 2 pi bonds and (T) 9 sigma bonds.

12.

To match the compounds in Column I with their corresponding formulas in Column II, we need to understand the structure of each type of compound:

(a) Alkane: (S) C_2H_6

(b) Alkene: (Q) C_5H_{10}

(c) Alkyne: (P) C_4H_6

(d) Urea: (R) NH_2-CO-NH_2

(a) Alkane: Alkanes are saturated hydrocarbons with the general formula C_nH_{2n+2}. For example, ethane (C_2H_6) fits this formula.

(b) Alkene: Alkenes are unsaturated hydrocarbons with at least one double bond and have the general formula C_nH_{2n}. For example, pentene (C_5H_{10}) fits this formula.

(c) Alkyne: Alkynes are unsaturated hydrocarbons with at least one triple bond and have the general formula C_nH_{2n-2}. For example, butyne (C_4H_6) fits this formula.

(d) Urea: Urea is an organic compound with the formula NH_2-CO-NH_2.

COMPREHENSION TYPE QUESTIONS

Passage-I

13. The correct answer is A.

A homologous series is a group of organic compounds that share the same functional group and similar chemical properties, but differ in the number of CH_2 units in their molecular structure. Key points about homologous series include:

1. Similar chemical properties: Members of a homologous series react in similar ways due to the presence of the same functional group.

2. Different molecular formulas: Each member differs from the next by a CH2 unit, resulting in different molecular formulas.

3. Same functional group: All members have the same functional group, which defines their chemical behaviour.

14. The correct answer is C.

C_2H_4 (ethene) is not a member of the homologous series of alkanes.

Alkanes are hydrocarbons with the general formula C_nH_{2n+2}, where n is the number of carbon atoms. They are characterized by single bonds between carbon atoms. The members of the alkane series include:

(A) CH4 - Methane

(B) C3H8 - Propane

(D) C2H6 - Ethane

These all fit the general formula for alkanes:

- CH4: n=1, so 2n+2 = 4
- C2H6: n=2, so 2n+2 = 6
- C3H8: n=3, so 2n+2 = 8

However, C_2H_4 (ethene) does not fit this pattern. It has a double bond between carbon atoms and belongs to the alkene series with the general formula C_nH_{2n}.

15. The correct answer is B.

Alkynes are a homologous series of hydrocarbons characterized by the presence of at least one carbon-carbon triple bond. The general formula C_nH_{2n-2} accurately represents this series, where n is the number of carbon atoms in the molecule.

Examples of alkynes:

1. Ethyne (acetylene): C_2H_2 (n=2, so 2n-2 = 2)
2. Propyne: C_3H_4 (n=3, so 2n-2 = 4)

INTEGER TYPE QUESTIONS

16. Correct answer: 8

The number of C-H bonds in C_4H_8 is 8.

To understand this, let's break down the structure of C_4H_8:

1. Carbon atoms: 4
2. Hydrogen atoms: 8
3. Degree of unsaturation: 1 (either one double bond or one ring)

To calculate the number of C-H bonds:

1. First, consider the carbon skeleton. With one degree of unsaturation, we have two possible structures:

a) A linear or branched chain with one double bond

b) A cyclic structure with all single bonds

2. In either case, each carbon atom forms one bond with another carbon atom (either single or as part of a double bond).
3. The remaining bonds of each carbon atom are filled by hydrogen atoms.
4. Since each carbon has 4 bonding capacity, and one is used for C-C bonding, 3 are left for C-H bonds per carbon.
5. However, we only have 8 hydrogen atoms in total.

17. Correct answer: 5

Acetylene, with the chemical formula C_2H_2, is a simple alkyne. The molecule consists of two carbon atoms connected by a triple bond, and each carbon is also bonded to a single hydrogen atom.

In acetylene:

- The triple bond between the two carbon atoms comprises one sigma (σ) bond and two pi (π) bonds.
- Each carbon-hydrogen bond is a sigma (σ) bond.

Structure of acetylene (C_2H_2) is:

H-C≡C-H

- There are two sigma bonds between the carbon atoms and the hydrogen atoms (one for each H-C bond).
- There is one sigma bond and two pi bonds in the triple bond between the carbon atoms.

3 sigma bonds + 2 pi bonds = 5 bonds.

Previous year questions:

1. Correct Answer:C) One C = O **double bond and three** C – H **single bonds**

Explanation:

- **Identifying Compound X:** An organic compound with the molecular formula $C_2H_4O_2$ that has a melting point of 290 K (17°C) and freezes into an ice-like solid during winter is **Ethanoic acid** (commonly known as **acetic acid**, and in its pure form, **glacial acetic acid**).
- **Chemical Structure:** The structural formula of ethanoic acid is CH_3COOH. Let's expand its structure to count the bonds:
- By looking at the structural breakdown, it contains:
  - **Three** C – H **single bonds** (on the methyl group $-CH_3$)
  - One C – C single bond
  - **One** C = O **double bond** * One C – O single bond
  - One O – H single bond

Option C correctly lists the presence of one C = O double bond and three C – H single bonds.

2. Correct Answer:B) Produces carbon dioxide and water on burning in air.

Explanation:

Let's decode the chemical reactions described in the puzzle:

1. **Compound (W)** reacts with sodium metal to evolve H_2 gas, which means (W) is an alcohol. Since the final ester (Z) has 4 carbons, (W) must be a 2-carbon alcohol: **Ethanol (**C_2H_5OH**)**.
2. Heating ethanol (W) at 443 K with excess concentrated H_2SO_4 dehydrates it to form **Compound (X)**, which is **Ethene (**C_2H_4**)**.
3. Heating ethanol (W) with alkaline $KMnO_4$ oxidizes it into **Compound (Y)**, which is **Ethanoic acid (**CH_3COOH**)**.
4. Reacting W (Ethanol) and Y (Ethanoic acid) together forms the sweet-smelling ester **(Z)**, **Ethyl ethanoate (**$CH_3COOC_2H_5$**)**.

The question asks specifically about **Compound (X) — Ethene (**C_2H_4**)**:

- Ethene is a hydrocarbon. Like all hydrocarbons, complete combustion (burning in air) produces **carbon dioxide (**CO_2**)** and **water (**H_2O**)**:

$C_2H_4 + 3O_2 \rightarrow 2CO_2 + 2H_2O$

3. Correct Answer:B) Graphite, Diamond, Allotropes

Explanation:

- **Structure I represents Graphite:** The carbon atoms are arranged in flat, parallel hexagonal layers where each carbon is bonded to three others.
- **Structure II represents Diamond:** The carbon atoms form a rigid, continuous three-dimensional tetrahedral network where each carbon is bonded to four others.
- **Relationship:** Since both diamond and graphite are different physical structural forms of the exact same pure element (Carbon), they are called **Allotropes**.

4. Correct Answer:D. C_5H_{10}

Explanation:

A homologous series is a group of organic compounds that share the same functional group and general chemical formula, with each successive member differing by a fixed $-CH_2-$ unit.

Let's analyze the molecular formulas of the choices:

- A. C_2H_6 (Ethane): Fits the general formula C_nH_{2n+2} → Alkane
- B. C_4H_{10} (Butane): Fits the general formula C_nH_{2n+2} → Alkane
- C. C_3H_8 (Propane): Fits the general formula C_nH_{2n+2} → Alkane
- D. C_5H_{10} (Pentene): Fits the general formula C_nH_{2n} → Alkene

Since compounds A, B, and C all belong to the alkane homologous series, C_5H_{10} is the odd one out because it belongs to the alkene series.

5. Correct Answer:D. III only

Explanation:

First, let's write down the molecular formula and family type for each given hydrocarbon:

- P ($CH_3CH_2CH_2CH_2CH_3$): Total atoms = C_5H_{12} → Alkane (General Formula: C_nH_{2n+2})
- Q (CH_3-C ≡ C-CH_2CH_3): Total atoms = C_5H_8 → Alkyne (General Formula: C_nH_{2n-2})
- R (CH_3CH_2-CH = CH-CH_3): Total atoms = C_5H_{10} → Alkene (General Formula: C_nH_{2n})

5. Chemical Reactions and Equations

LEVEL – I

1. Correct option: (A) Balanced equation

A chemical equation in which the number of atoms of each element is the same on the side of reactants and products is called as balanced chemical equation.

2. Correct option: (A) Created, destroyed

In a chemical reaction the atoms are neither created nor destroyed. It changes its chemical configuration.

3. Correct option: (B)

Here equation $3CuSO_4 + KI \rightarrow Cu_2I_2 + 2I_2 + 2K_2SO_4$ is not balanced both in reactants side and products side. Cu, potassium, Sulfur, Oxygen are not balanced.

4. Correct option: (C)

Here equation $2FeS + 3O_2 \rightarrow 2FeO + 2SO_2$ is balanced. The number of atoms on both reactants and product side are equal.

5. Correct option: (D) Both A and B

Here equations $(NH_4)PO_4 \rightarrow 3NH_3 + H_2O + HPO_3$, $N_2 + 3H_2 \rightarrow 4NH_3$ both are unbalanced equations due to difference in number of atoms on both reactants and product side.

6. Correct option: (A)

$N_2 + 3H_2 \rightarrow 2NH_3$ is a balanced equation in this reaction number of atoms on both reactants and product side are equal.

7. Correct option: (A)

The balanced equation for the decomposition of $2KNO_3$ is:

$2KNO_3 \rightarrow 2KNO_2 + O_2$

8. Correct option: (A)

The law of conservation of mass in chemical reactions states that the total mass of the reactants must be equal to the total mass of the products. In this case, it can be representing as: n + m = p + q

9. Correct option: (B)

The balanced equation of reaction between Cu and $AgNO_3$ is

$Cu + 2AgNO_3 \rightarrow Cu(NO_3)_2 + 2Ag$

10. Correct option: (B)

The balanced equation for this reaction is $N_2 + 3H_2 \rightarrow 2NH_3$.

11. Correct option: (B)

$NH_4Cl(s) \underline{Heat cool} NH_3(g) + HCl(g)$ is the final equation of the reaction.

12. Correct option: (B)

Given chemical reaction is $FeS + HCl \rightarrow H_2S + x$.

$FeS + 2HCl \rightarrow H_2S + FeCl_2$ is the balanced chemical reaction. So x value is $FeCl_2$.

13. Correct option: (A)

A balanced chemical equation provides information about the substances involved in a chemical reaction, the symbols and formulas of all substances, and the

number of atoms or molecules of all substances involved in the reaction. It does not provide any information regarding speed or rate of the chemical reaction.

14. Correct option: (B) Two moles of phosphine

The balanced chemical equation for the reaction between one mole of calcium phosphide $\left[Ca_3P_2\right]$ and an excess of water is as follows:

$Ca_3P_2 + 6H_2O \rightarrow 2PH_3 + 3Ca\left(OH\right)_2$

Above equation explains that one mole of calcium phosphide reacts with six moles of water to produce two moles of phosphine (PH_3 PH3) and three moles of calcium hydroxide ($Ca\left(OH\right)_2$).

15. Correct option: (D) 0.1

$3BaCl_2 + 2Na_3PO_4 \rightarrow Ba_3\left(PO_4\right)_2 + 6NaCl$

Here 3 moles of $BaCl_2$ react with 2 moles of Na_3PO_4 to yield 1 mole of $Ba_3\left(PO_4\right)_2$.

Ratio of $BaCl_2$: Na_3PO_4 =3:2.

0.5 moles of $BaCl_2$ requires $\frac{2}{3}$ x 0.5 mole (0.333 mole) Na_3PO_4.

Only 0.2 moles Na_3PO_4. So Na_3PO_4 it is limiting and $BaCl_2$ is excess in quantity.

2 moles of Na_3PO_4 will yield 1 mole $Ba_3\left(PO_4\right)_2$

0.2 moles of Na_3PO_4 yield 0.1 mole of $Ba_3\left(PO_4\right)_2$.

16. Correct option: (A)

Balanced equation for the combustion of octane is

$2C_8H_{18} + 25O_2 \rightarrow 16CO_2 + 18H_2O$

Moles of octane equal to a mass of 570 grams can be calculated using its molar mass:

$$n_{C_8H_{18}} = \frac{mass}{molarmass} n_{C_8H_{18}} = \frac{m_{n_{C_8H_{18}}}}{M_{n_{C_8H_{18}}}} = \frac{570g}{114\frac{g}{mol}} = 5mol$$

From the above equation, 2 moles of octane react with 25 mol of oxygen. So, 5 moles of octane reacted with 62.5 mol of oxygen.

$$n_{O_2} = \frac{25molesofO_2}{2molC_8H_{18}} \times 5molC_8H_{18}$$

$$n_{O_2} = 62.5miolO_2$$

Mass of oxygen can be given by below formula

$$m_{O_2} = moles \times MolarmassO_2 = n_{O_2} \times M_{O_2} \times m_{O_2}$$

$$m_{O_2} = 62.5mol \times 32\frac{g}{molm_{O_2}} = 2000g$$

17. Correct option: (B)

Given $CaCO_3$ is 95% pure and mass is 200kg.

Reaction can be given as $CaCO_3\left(s\right) \rightarrow CaO\left(s\right) + CO_2 \uparrow$

Amount of pure limestone is given by $\frac{95}{100} \times 200 = 190kg = 190000grams$

100 kg $CaCO_3$ gives 56 kg CaO.

190 $CaCO_3$ gives ($\frac{56 \times 190}{100}$) = 106.4 kg of CaO.

18. Correct option: (A)

Balanced equation of HCl reacted with MgO.

$$\underset{1mol}{MgO} + \underset{2mol=(2\times36.5)=73grams}{2HCl} \rightarrow \underset{1mol=(24+71)95g}{MgCl_2} + \underset{1mol}{H_2O}$$

73 g of HC1 produce 95g $MgCl_2$.

1 g of HCI produces $\frac{95}{73} grams$ of $MgCl_2$.

17 g of HCl will produce $\frac{95}{73} \times 17 grams = 22.12g$

19. Correct option: (A)

Balanced chemical equation for the given statement is

$3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$

$$\frac{MoleofFe}{MoleofH_2O} = \frac{3}{4}$$

$$Moleofsteam = \frac{mass}{molarmass} = \frac{18}{18}$$

$$MoleofFe = \frac{18}{18} \times \frac{3}{4} = 0.75$$

$massofFe = mole \times molarmass$

$massofFe = 0.75 \times 56 = 42$

$massofFe = 42g$

20. Correct option: (D)

Balanced chemical equation can be given below.

$2H_2O \rightarrow 2H_2 + O_2$

From the equation we can observe that 2 moles of water → 1 mole of oxygen.

$36gH_2O = 32gO_2$

$$72gH_2O = \frac{72 \times 32}{36} = 64gO_2$$

Level-II

1. Correct option: (D)

Balanced chemical equation for the given statement is given as

$$\underset{100g}{CaCO_3}(s) \rightarrow \underset{56g}{CaO}(s) + CO_2 \uparrow$$

1 mol of CaO is obtained by the decomposition of 1 mol of $CaCO_3$.

56g of CaO will be obtained from 100g of $CaCO_3$.

$\frac{28 \times 100}{56} = 50 grams$ Of $CaCO_3$ is obtained from 28grams of residue.

2. Correct option: (D)

The atomic number of carbon is 6 and atomic mass of carbon is 12.

Given weight of carbon in carbon dioxide is 4.4grams.

$$\frac{12\times 4.4}{44}=1.2$$

Now, calculate the percentage of carbon in the organic compound using the mass of carbon and the mass of the organic compound.

$$Percentage\ of\ carbon\ =\ \left(\frac{Mass\ of\ carbon}{Mass\ of\ organic\ compound}\right)\times 100$$

$$Percentage\ of\ carbon\ =\ \left(\frac{1.2\ g}{1.6g}\right)\times 100\ =\ 75\%$$

3. Correct Option: A) 16 g 11.2 L

Chemical Equation:

$2H_2O_2 \rightarrow 2H_2O + O_2$

- Molar Mass of H_2O_2: $2(1) + 2(16) = 34$ g/mol
- Moles of H_2O_2: $\frac{34\ g}{34\ g/mol} = 1$ mol
- Stoichiometry: According to the balanced equation, 2 moles of H_2O_2 yield 1 mole of O_2. Therefore, 1 mole of H_2O_2 will yield 0.5 mole of O_2.
- Mass of O_2: $0.5\ mol \times 32\ g/mol = 16\ g$
- Volume of O_2 at NTP: $0.5\ mol \times 22.4\ L/mol = 11.2\ L$

4. Correct Option: A) 81.7

Chemical Equation:

$2KClO_3 \rightarrow 2KCl + 3O_2$

- Moles of O_2 produced: $\frac{2.24\ L}{22.4\ L/mol} = 0.1$ mol
- Stoichiometry: 3 moles of O_2 come from 2 moles of $KClO_3$.
  Thus, 0.1 mole of O_2 comes from $\frac{2}{3}\times 0.1 = \frac{0.2}{3}$ moles of $KClO_3$.
- Mass of pure $KClO_3$: $\frac{0.2}{3}\ mol \times 122.5\ g/mol \approx 8.17\ g$
- Percentage Purity: $\frac{8.17\ g}{10\ g}\times 100\% = 81.7\%$

5. Correct Option: D) 7 g

- Chemical Equation:
  $C_2H_5Cl \rightarrow C_2H_4$ (alkene) + HCl
- Molar Mass of C_2H_5Cl: $24 + 5 + 35.5 = 64.5$ g/mol
- Moles of Reactant: $\frac{32.25\ g}{64.5\ g/mol} = 0.5$ mol
- Theoretical Yield of Alkene (C_2H_4): $0.5\ mol \times 28\ g/mol = 14\ g$
- Actual Yield (at 50% efficiency): $14\ g \times 0.50 = 7\ g$

6. Correct Option: D) 5 ml of O2

- Chemical Equation:
  $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$
- By Gay-Lussac's Law of combining volumes, 2 volumes of H_2 react with 1 volume of O_2.
- 30 ml of H_2 requires exactly $\frac{30}{2} = 15$ ml of O_2.
- Since 20 ml of O_2 is provided, H_2 is the limiting reagent and gets completely consumed.
- Remaining O_2: $20\ ml - 15\ ml = 5$ ml

7. Correct Option: A) 2,2

- Balancing the Equation:
  $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$

- Checking the atoms on both sides confirms 6 C, 12 H, and 6 O are perfectly balanced. The coefficients for the products (C_2H_5OH and CO_2) are 2 and 2.

8. Correct Option: B) Skeleton equation

A chemical equation that is written without balancing the number of atoms on both sides is known as a skeleton equation.

9. Correct Option: A) 3,7

Let the equation be balanced using the right side as a reference:

On the right side, there are 3 atoms of Mg. Thus, the coefficient of Mg on the left must be 3.

- Counting chlorate/chlorine atoms on the right side: $3 \times 2 = 6$ (from $Mg(ClO_3)_2$) + 1 (from HCl) = 7 Cl atoms. Thus, the coefficient of $HClO_3$ must be 7.
- Verifying with hydrogen and oxygen balances perfectly with 3 and 7.

10. Correct Option: D) 25

Problem: $FeS_2 + O_2 \rightarrow Fe_2O_3 + SO_2$, the sum of coefficients of reactants and products is:

- Balanced Chemical Equation:
  $4FeS_2 + 11O_2 \rightarrow 2Fe_2O_3 + 8SO_2$
- Sum of Coefficients: $4 + 11 + 2 + 8 = 25$

11. Correct Option: B) 28 g

Chemical Equation:

$CaCO_3 \rightarrow CaO + CO_2$

- Molar Masses: $CaCO_3$ = 100 g/mol, CaO = 56 g/mol
- Calculation: 100 g of $CaCO_3$ produces 56 g of CaO.

Therefore, 50 g of $CaCO_3$ will produce exactly half: $\frac{56}{2} = 28$ g.

12. Correct Option: C) 11.2 L

- Chemical Equation:
  $2H_2O \rightarrow 2H_2 + O_2$
- Moles of H_2O: $\frac{18\text{ g}}{18\text{ g/mol}} = 1$ mol
- Stoichiometry: 2 moles of H_2O yield 1 mole of O_2. Therefore, 1 mole of H_2O yields 0.5 mole of O_2.
- Volume at NTP: $0.5\text{ mol} \times 22.4\text{ L/mol} = 11.2\text{ L}$

13. Correct Option: B) 3

- Moles of H_2O: $\frac{108\text{ g}}{18\text{ g/mol}} = 6$ moles
- Since the ratio of $H_2O : O_2$ is 2:1, the number of moles of O_2 produced is $\frac{6}{2} =$ 3 moles.

14. Correct Option: A) 0.5 mol

- Chemical Equation:
  $BaCl_2 + H_2SO_4 \rightarrow BaSO_4 + 2HCl$
- The stoichiometric reaction ratio is 1:1.
- Since we only have 0.5 mole of $BaCl_2$ compared to 1 mole of H_2SO_4, $BaCl_2$ is the limiting reactant.
- The amount of product formed depends completely on the limiting reactant: 0.5 mole of $BaCl_2$ yields 0.5 mole of $BaSO_4$.

15. Correct Option: A,B,C

Explanation: A balanced chemical equation gives qualitative information (what reactants and products are involved) and quantitative information (the

stoichiometric numbers showing the relative molecules, moles, and masses/weights of the species involved). However, it does not provide physical traits like the color of the chemical substances. Therefore, the question likely intended to ask which information is not conveyed, making D the incorrect fact. If it is a multiple-correct question asking what it *does* convey, A, B, and C are all valid pieces of information derived from it.

16. **Correct Option:** (B) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

Explanation: Assertion is true: By definition, a chemical reaction reorganizes atoms to form entirely new chemical substances with distinct properties.

- Reason is true: A chemical equation uses chemical formulas and symbols to represent a real-world chemical reaction on paper.
- Connection: While both statements are scientifically true, the fact that an equation is a symbolic tool (Reason) does not explain *why* a chemical reaction must form new substances (Assertion).

17. **a) → S; b) → R; c) → P; d) → Q**

18. Correct Option: A) 4.48 L

Chemical Equation:

$CaCO_3 \rightarrow CaO + CO_2$

Molar Mass of $CaCO_3$: $40 + 12 + (3 \times 16) = 100$ g/mol

Moles of $CaCO_3$: $\frac{20 \text{ g}}{100 \text{ g/mol}} = 0.2$ mol

Stoichiometry: 1 mole of $CaCO_3$ produces 1 mole of CO_2. Thus, 0.2 mol yields 0.2 mol of CO_2.

Volume at NTP: $0.2 \text{ mol} \times 22.4 \text{ L/mol} = 4.48$ L

19. **Correct Option: B) 0.224 L**

Chemical Equation:

$2NaNO_3 \rightarrow 2NaNO_2 + O_2$

Molar Mass of $NaNO_3$: $23 + 14 + (3 \times 16) = 85$ g/mol

Moles of $NaNO_3$: $\frac{1.70 \text{ g}}{85 \text{ g/mol}} = 0.02$ mol

Stoichiometry: 2 moles of $NaNO_3$ yield 1 mole of O_2.

Therefore, 0.02 mol yields $\frac{0.02}{2} = 0.01$ mol of O_2.

Volume at NTP: $0.01 \text{ mol} \times 22.4 \text{ L/mol} = 0.224$ L

20. **Correct Option: A) 8.7 g**

Chemical Equation:

$2KClO_3 \rightarrow 2KCl + 3O_2$

Moles of O_2 needed: $\frac{2.4 \text{ L}}{22.4 \text{ L/mol}} \approx 0.1071$ mol

Stoichiometry: 3 moles of O_2 require 2 moles of $KClO_3$.

Moles of $KClO_3 = \frac{2}{3} \times 0.1071 \approx 0.0714$ mol.

Mass of $KClO_3$: $0.0714 \text{ mol} \times 122.5 \text{ g/mol} \approx 8.75$ g

The value rounds nearest to 8.7 g.

Type of Chemical Reactions

Level – I Questions

1. **Correct Option: C) Chemical synthesis**

Explanation: When a substance is prepared by combining its direct constituent elements (e.g., $C + O_2 \rightarrow CO_2$), it is specifically called a chemical synthesis reaction.

2. **Correct Answer: A) Chemical combination**

- **Explanation:** A combination reaction occurs when two or more reactants combine to form a single, new product. Here, calcium oxide (CaO) and water (H_2O) combine to form a single product, calcium hydroxide ($Ca(OH)_2$).

3. **Correct Answer: A) Chemical combination**

- **Explanation:** This is the standard definition of a **combination reaction**. The key indicator is that there is *only one* product formed regardless of the number of reactants.

4. $2Na + Cl_2 \rightarrow 2NaCl$

Correct Answer: A) Combination reaction

- **Explanation:** two elements (sodium and chlorine) react together to form a single compound (sodium chloride).

5. $2H_2O(l) \xrightarrow{\text{Heat}} 2H_2(g) + O_2$

Correct Answer: B) Chemical decomposition

- **Explanation:** A decomposition reaction is the opposite of a combination reaction. It involves a single reactant breaking down into two or more simpler products, usually triggered by energy sources like heat, light, or electricity.

6. $AgNO_3 + NaCl \rightarrow X + NaNO_3$

Correct Answer: B) AgCl - white precipitate

- **Explanation:** This is a double displacement (precipitation) reaction. Silver ions (Ag^+) swap partners with sodium ions (Na^+). The reaction yields sodium nitrate and silver chloride (AgCl), which is a well-known insoluble **white precipitate**.

7. **Correct Answer: B) $Ca + Cl_2 \rightarrow CaCl_2$**

- **Explanation:** * Options A, C, and D are all single displacement reactions where a more reactive element displaces a less reactive one from its compound.
  - Option B is a **combination reaction** because calcium and chlorine gas are combining to form a single product.

8. $Zn + H_2SO_4$ (dil.) $\rightarrow ZnSO_4 + H_2 \uparrow$

Correct Answer: B) Single displacement reaction

- **Explanation:** Zinc (Zn) is more reactive than hydrogen (H). Therefore, it displaces hydrogen from the dilute sulfuric acid to form zinc sulfate and releases hydrogen gas.

9. **Correct Answer: A) Na, Fe, Cu, Ag**

- **Explanation:** According to the metal reactivity series, the activity decreases as you go down from alkali metals to noble metals:
  Sodium (Na) > Iron (Fe) > Copper (Cu) > Silver (Ag)

10. **Correct Answer:** D) $Ag + HNO_3 \rightarrow AgNO_3 + H_2$

- **Explanation:** Silver (Ag) lies below hydrogen (H) in the reactivity series, meaning it is less reactive than hydrogen and **cannot displace hydrogen gas** from acids.
  (Note: While silver does dissolve in concentrated nitric acid, it produces toxic nitrogen oxide gases (NO_x) and water, never hydrogen gas).

11. **Correct Answer: C)** $Cu + 2AgNO_3 \rightarrow Cu(NO_3)_2 + 2Ag$

- **Explanation:** A single displacement reaction occurs only if a free metal is **more reactive** than the metal combined in the compound. Looking at the activity series:

12. **Correct Answer: D) Pt**

- **Explanation:** Platinum (Pt) is a noble metal positioned at the very bottom of the reactivity series. Since it is the least reactive element among the choices (Zn > Fe >

Sn > Pt), it lacks the chemical activity to displace any of the other three metals from their respective compounds.

13. Correct Answer: D) Both A and B

- **Explanation:** 1. It is a **double displacement reaction** because the ions (Ag^+ and Na^+) mutually exchange their partners.
  2. It is a **precipitation reaction** because one of the products formed (AgCl) is an insoluble solid that falls out of solution as a white precipitate (indicated by the downward arrow ↓).

14. Correct Answer: B) 1 and 2

- **Explanation:** * It is a **neutralisation reaction** because it involves an acid (H_2SO_4) reacting with a base (NaOH) to yield salt (Na_2SO_4) and water (H_2O).
  - It is also inherently a type of **double displacement reaction**, as the hydrogen ions and sodium ions swap anion partners.

15. Correct Answer: D) Chemical double displacement

- **Explanation:** By definition, a double displacement reaction involves two ionic compounds exchanging their ionic parts (cations/metallic parts and anions/non-metallic parts) to form two entirely new compounds.

16. Correct Answer: D) $2AgBr \rightarrow 2Ag + Br_2$

- **Explanation:** A **photochemical decomposition** reaction breaks down a compound using light energy. Silver bromide (AgBr) decomposes into silver metal and bromine gas when exposed to sunlight. (This property makes it widely utilized in traditional black-and-white photography). Options A and B require thermal energy (heat), while option C is a single displacement reaction.

17. Correct Option: C) K

Explanation: Calcium can displace metals lower than itself (Cu, Pt, Pb). However, Potassium (K) is higher than Calcium in the reactivity series, so Calcium cannot displace it.

18. Correct Option: C) Pb

Explanation: Iron (Fe) is more reactive than Lead (Pb), so it can displace lead from its compounds. It cannot displace Na, Zn, or Mg because they are more reactive than iron.

19. Correct Answer: D) $Pb(NO_3)_2 + 2KI \rightarrow PbI_2 + 2KNO_3$

Explanation: Look for the choice with two compounds exchanging partner ions: $AB + CD \rightarrow AD + CB$.

- Correct Option: The corresponding option showing icon/partner swapping.

20. Correct Answer: D) Zn > Fe > Pb

Explanation: The standard sequence follows: K > Na > Ca > Mg > Al > Zn > Fe > Pb > H > Cu > Hg > Ag > Au > Pt.

Level – II

1. Correct Option: B)

Explanation: Let's look at statement B: "The reactions in which one element displaces another element in a compound are known as displacement reactions." This is the precise textbook definition of a single displacement reaction.

2. Correct Option: C) Iron

Explanation: Finely divided Iron (Fe) is used as the catalyst in the Haber process for industrial ammonia manufacturing, with Molybdenum acting as a promoter.

3. **Correct Answer: A)** $2H_2O \rightarrow 2H_2 + O_2(g)$

Explanation: The electrolytic decomposition of water ($2H_2O \xrightarrow{\text{electricity}} 2H_2 + O_2$) is driven by electrical energy.

4. Correct Option: C) Photochemical reactions

Explanation: Reactions driven by photons or light absorption are termed photochemical reactions.

5. Correct Option: A) CuO

Explanation: The substance providing oxygen or getting reduced is the oxidizing agent. Here, CuO loses oxygen to become Cu (reduction), so CuO is the oxidizing agent.

6. **Correct Answer: D)** $CaCO_3 \longrightarrow CaO + CO_2$

Explanation: Decomposition reactions like the thermal cracking of calcium carbonate ($CaCO_3 \rightarrow CaO + CO_2$) or the combination of nitrogen and oxygen ($N_2 + O_2 \rightarrow 2NO$) require the continuous absorption of heat.

7. Correct Option: B) Rusting of iron

Explanation: The slow degradation of metal by action of air, moisture, or chemicals on their surface—most commonly seen as the rusting of iron—is corrosion.

One or More Than One Correct Answer Questions

8. **Correct Answers: A, B, and C**

- **Explanation of correct options:**
  - **A is correct:** Mechanistically, corrosion involves a redox process where the metal atoms lose electrons and undergo **oxidation** to form metal oxides or other salts.
  - **B is correct:** Iron requires both oxygen (from air) and water/moisture simultaneously to rust. Eliminating either one stops the process.
  - **C is correct:** Galvanisation is the process of applying a protective zinc coating to iron or steel. Zinc corrodes preferentially to protect the underlying iron, successfully preventing corrosion.

9. **Correct Answers: A, B, and D**

- **Explanation:** A **precipitation reaction** occurs when two aqueous solutions react to form an insoluble solid compound, known as a precipitate. In chemical equations, this insoluble solid is explicitly denoted by either a solid state symbol (s) or a downward arrow ↓.
  - **A is a precipitation reaction:** Forms insoluble silver chloride ($AgCl(s)$ ↓).
  - **B is a precipitation reaction:** Forms insoluble barium sulfate ($BaSO_4(s)$ ↓).
  - **D is a precipitation reaction:** Forms insoluble silver bromide ($AgBr(s)$ ↓).

Assertion and Reason Type

10. Energetics of Exothermic Reactions

- Problem: Assertion (A): Exothermic reactions release heat energy during the reaction.

  Reason (R): The energy released during bond formation in products is greater than the energy required to break bonds in reactants.
- Explanation: The net energy output is positive (released as heat) precisely because product bond formation releases more energy than what is consumed to break the original reactant bonds. Thus, the reason correctly justifies the assertion.
- Correct Option: (A) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

11. Conditions for Neutralisation

- Problem: Assertion (A): In a neutralisation reaction, an acid reacts with a base to form salt and water.
  Reason (R): Neutralisation reactions occur only in the absence of water.
- Explanation: The Assertion is true. However, the Reason is completely false; neutralization reactions typically occur in aqueous solutions where acids and bases liberate H^+ and OH^- ions.
- Correct Option: (C) Assertion is true, but Reason is false.

Matrix Matching Type

a) → R; b) →Q; c) → Sd) → P

Comprehension Type

Passage-I: Redox Principles

13. Correct Option: C) Gain of oxygen

Explanation: Classical definition: Oxidation is the gain of oxygen or loss of hydrogen.

14. Correct Option: B) Reducing agent

Explanation: The substance that reduces another species by getting oxidized itself is known as a reducing agent.

15. Correct Option: B) Loss of oxygen / C) Gain of hydrogen

Explanation: In terms of classical definitions, reduction involves the loss of oxygen or the gain of hydrogen. (Options B and C are both correct classic interpretations here).

Passage-II: Rancidity

16. Correct Option: B) Spoilage of food due to oxidation

Explanation: Rancidity is the aerial spoilage of food containing fats and oils due to oxidation, creating bad smells and tastes.

17. Correct Option: C) Adding antioxidants

Explanation: Adding antioxidants (like BHA or BHT) prevents fats and oils from reacting with oxygen.

18. Correct Option: C) To prevent oxidation

Explanation: Nitrogen is an inert gas used to flush out oxygen from food packets, thereby preventing oxidation and keeping the food fresh.

PREVIOUS YEAR QUESTIONS

1. Step 1: Identify Oxidation and Reduction

- Reduction: Manganese (Mn) goes from an oxidation state of +4 in MnO_2 to 0 in Mn. Thus, MnO_2 is the substance being reduced.
- Oxidation**:** Aluminum (Al) goes from an oxidation state of 0 to +3 in Al_2O_3. Thus, Al is the substance being oxidized.
  Step 2: Balance the Equation
  We can balance this using the hit-and-trial method or by balancing atoms:

1. Look at Oxygen: There are 2 on the left and 3 on the right. To balance, we use a common multiple of 6.
2. Place a 3 before MnO_2 and a 2 before Al_2O_3.
3. Now balance the metals:

- 3 Mn on the left → place a 3 before Mn.
- 4 Al on the right (2 X 2) place a 4 before Al.

Balanced Equation:
$3MnO_2 + 4Al \rightarrow 3Mn + 2Al_2O_3$
Step 3: Determine the Ratio

- Coefficient of substance reduced (MnO_2) = 3
- Coefficient of substance oxidized (Al) = 4

Ratio = 3:4

So, answer is A

2. **C. Correct:** Sodium bicarbonate is indeed used as an antacid (to neutralize stomach acid) and is the primary component in soda-acid fire extinguishers.

3. **Correct Option: B**

Analysis: When an electric current is passed through water, it decomposes into its constituent elements: hydrogen gas and oxygen gas.

Chemical Equation**:** $2H_2O(l) \xrightarrow{electricity} 2H_2(g) + O_2(g)$

Conclusion**:** Since a single compound (water) splits into two simpler substances, it is a decomposition reaction.

4. **Correct Option: A**

Analysis**:** A classic double displacement reaction that produces a characteristic **y**ellow precipitate involves Lead (II) nitrate and Potassium iodide.

5. **Correct Option: C**

H_2 is being oxidized, not reduced, because it gains oxygen to form H_2O.

6. **Correct Option: D) 2 and 4**

Pb: 2 on left, 2 on right.

N: 4 on left, 4 on right.

O: 12 on left (2 X 3 X 2), 12 on right (2 + 8 + 2).

7. **Correct Option: A) (ii), (iii) and (iv) only**

(ii) Decomposition of vegetable matter into compost – Endothermic reaction: Incorrect**.** This is a biological decomposition process that releases energy (heat), making it exothermic.

(iii) Reaction of zinc with copper sulphate – Decomposition reaction: Incorrect. This is a Single Displacement reaction (Zn + CuSO4 → ZnSO4 + Cu).

(iv) Reaction of barium chloride with sodium sulphate – Single displacement reaction: Incorrect. This is a Double Displacement (Precipitation) reaction (BaCl2+ Na2SO4→ BaSO4 + 2NaCl).

8. **Correct Option: C) 2, 2 and 3**

Place a 2 before KClO3 (2 X 3 = 6 Oxygens).

Place a 3 before O2 (3 X 2 = 6 Oxygens).

2KClO3 → 2KCl + 3O2 ;x=2, y=2, and z=3

9. $NaOH_{(aq)} + HCl_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(l)}$

Sodium (Na): Remains +1 on both sides.

- Oxygen (O): Remains -2 on both sides.
- Hydrogen (H): Remains +1 on both sides.
- Chlorine (Cl): Remains -1 on both sides.
- Verdict: This is an acid-base neutralization reaction. There is no transfer of electrons or change in oxidation states. This is NOT a redox reaction.

10. C) Oxidising agent, Oxidation, reduction, Reducing agent

Zinc Oxide (ZnO) to Zinc (Zn): ZnO loses oxygen to become Zn. The removal of oxygen is reduction. Because ZnO provides oxygen to the other reactant, it acts as the oxidising agent.

Carbon (C) to Carbon Monoxide (CO): Carbon gains oxygen to become CO. The gain of oxygen is oxidation. Because Carbon removes oxygen from ZnO, it acts as the reducing agent.

6. Stoichiometry

- **Balancing Chemical Equations**
- **Quantitative Relations in Chemical Reactions**
- **Calculations Based on Mole-Mole Relationship**
- **Calculations Based on Mass-Mass Relationship**
- **Calculations Based on Mass-Volume Relationship**
- **Calculations Based on Volume-Volume Relationship**

LEVEL – I

1. Correct option: (B)

The balanced chemical equation for the reaction of one mole of calcium phosphide (Ca_3P_2) with excess water is as follows:

$$Ca_3P_2 + 6H_2O \rightarrow 3Ca(OH)_2 + 2PH_3$$

2. Correct option: (B)

A balanced equation is:

$$Mg(OH)_2 + 2HCl \rightarrow MgCl_2 + 2H_2O$$

Number of moles of stomach acid neutralized.

=2×number of moles of $Mg(OH)_2$

$$= 2 \times \frac{1}{58.33} = 0.0343$$

3. Correct option: (C)

Number of moles of CO_2 =Number of moles of $CaCO_3$

$$= \frac{Mass}{Molarmass} = \frac{50}{100} = \frac{1}{2}$$

4. Correct option: (D)

$$3BaCl_2 + 2Na_3PO_4 \rightarrow Ba_3(PO_4)_2 + 6NaCl$$

The coefficients of the balanced equation tell you the ratio of the reactants and the products.

3 moles of $BaCl_2$ react with 2 moles of Na_3PO_4 to yield 1 mole of $Ba_3(PO_4)_2$.

Ratio of $BaCl_2 : Na_3PO_4 = 3:2$

Hence 0.5 mole of $BaCl_2$ require $= = \frac{2}{3} \times 0.5 = 0.33$ moles of Na_3PO_4

Since, available Na_3PO_4 (0.2 *mole*) *is less than required mole* (0.33), *it is the limiting reactant and would determine the amount of product* $Ba_3(PO_4)_2$

$$2C_8H_{18} + 25O_2 \rightarrow 16CO_2 + 18H_2O$$

For burning $2 \times 114g$ of octane, oxygen required = $25 \times 32g$

For burning570 g of octane, oxygen required $\frac{25 \times 32}{2 \times 114} \times 570g = 2000g$

5. Correct option: (A)

6. Correct option: (B)

Amount of pure $CaCO_3 = \frac{95}{100} \times 200 = 190kg = 190000g$

$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$

1 mole of $CaCO_3$ = 1mole of CaO

100grams of $CaCO_3$ =56grams of CaO

190000grams of $CaCO_3$ gives $\frac{56}{100} \times 190000g = 106400g = 106.4kg$

7. Correct option: (A)

Balanced chemical equation is

$MgO + 2HCl \rightarrow MgCl_2 + H_2O$

$MgO(1\,\text{mole})$

$2HCl(2\,\text{mole}) = (2 \times 36.5)\,\text{g} = 73\,\text{g}$

$MgCl_2(1\,\text{mole}) = (24 + 71)\,\text{g} = 95\,\text{g}$

$H_2O(1\,\text{mole})$

73 g of HCl produce $MgCl_2$ =95g

1g of HCl produce $\frac{95}{73}g$ of $MgCl_2$

17g of HCl produce $\frac{95}{73} \times 17g = 22.12g$ of $MgCl_2$

8. Correct option: (A)

The balanced equation can be written as

$3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$

Mole ratio of reaction suggest that $\frac{MoleofFe}{MoleofH_2O} = \frac{3}{4}$

$Mole\ of\ Fe = \frac{18}{18} \times \frac{3}{4}$

$Weight\ of\ Fe = \frac{3}{4} \times 56 = 42g$

9. Correct option: (D)

$2H_2O \rightarrow 2H_2 + O_2$

From the above equation,

2 moles of water gives 1 mole of O_2

So 36grams of water gives 32grams of oxygen.

72grams of water= $\frac{72 \times 32}{36} = 64 gramsofO_2$

10. Correct option: (D)

The balanced chemical equation is $CaCO_3 \rightarrow CaO + CO_2$

1 mol of CaO will be obtained by the decomposition of 1 mol of $CaCO_3$

56g of CaO will be obtained from 100g of $CaCO_3$

28grams of residue will be formed from $\frac{28 \times 100}{56} = 50g\ ofCaCO_3$

11. Correct option: (C)

$4Al + 3O_2 \rightarrow 2Al_2O_3$

4×27 grams of aluminum reacts with 3×32 grams of oxygen.
108 grams of aluminum reacts with 96 grams of oxygen.
13.5 grams of alluminium reacts with X grams of oxygen.

$X = \frac{13.5 \times 96}{108} = 12\,grams.$

12. Correct option: (D)

The atomic mass of carbon is 12.

Weight of carbon in the 4.4g CO_2 is $= \frac{12 \times 4.4}{44} = 1.2\,grams.$

So the mass of carbon in the compound was 1.2 grams.
The % mass of carbon in the compound of the total weight 1.6g is

$\frac{1.2}{1.6} \times 100 = 75\ \%$

The percentage of carbon in the compound = 75%.

13. Correct option: (D)

$CaCO_3 \underrightarrow{\Delta} CaO + CO_2 \uparrow$

The molar mass of $CaCO_3$ =100g/mole

10 g of 90% pure lime = $= \frac{10g}{100g/mol} \times \frac{90}{100} = 0.09\,moles\,CaCO_3$

0.09 moles $CaCO_3$ on heating gives 0.09 moles CO_2

At STP, 1 mole of CO_2 = 22.4L.

At STP, $0.09\ mole\,CO_2 = 0.09 \times 22.4 = 2.016\,L.$

Hence, the volume of CO_2 liberated at STP when 10 g of 90% pure lime is heated completely is 2.016 L.

14. Correct option: (A)

$2KClO_3 \rightarrow 2KCl + 3O_2$

$2 \times 22.4L$ (atNTP) $\times 22.4L(atNTP)$

$2 \times 22.4L\,of\ KClO_3$ generates $3 \times 22.4L\,of\ O_2\,at\ NTP$

$2.4L\,of\ O_2\ needs\ \frac{2 \times 22.4 \times 2.4}{3 \times 22.4} = 1.6L\,of\ KClO_3$

15. Correct option: (A)

$2H_2O_2 \rightarrow 2H_2O + O_2$

$\frac{number\,of\,moles\,of\,H_2O_2}{2} = number\,of\,moles\,of\,O_2$

$n_{H_2O_2} = \frac{34}{34} = 1$

$n_{O_2} = \frac{1}{2} = \frac{w}{m_{O_2}} = \frac{w}{32}$

$w = 16g$

$Volume = n \times 22.4lt$

$Volume = \frac{1}{2} \times 22.4lt = 11.2liters$

16. Correct option: (A)

$KClO_3 \rightarrow 2KCl + 3O_2$

3 moles of oxygen yielded from $= \dfrac{2\times\left(\dfrac{22.4}{22.4}\right)}{3}$

$= 0.0667$ moles of $KClO_3$

Amount of $KClO_3 = 0.0667 \times M_{KClO_3} = 8.1667g$

Percentage of purity of sample is $= \dfrac{8.1667}{10}\times 100 = 81.667\%$

17. Correct option: (B)

Balancing the reaction $CaCO_3 \rightarrow CaO + CO_2(g)\uparrow$

1 mole of $CaCO_3$ gives of 1 mole of CO_2

From Avagadro's Law, 1 mole of CO_2 at NTP occupies 22.4L.

1 mole of $CaCO_3$ = 22.4L of CO_2

$100g \ = \ 22.4L \ of \ CO_2$

18. Correct option: (A)

19. Correct option: (A)

20. Correct options: (A, D)

When 30ml of H_2 will react with 15ml of O_2 to form water it can be represented in a balanced chemical equation:

$H_2(g) + \dfrac{1}{2}O_2(g) \rightarrow H_2O(g)$

$H_2(g) \rightarrow (30\,\text{ml} \rightarrow 0ml)$

$\dfrac{1}{2}O_2(g) \rightarrow (20ml \rightarrow 20-15ml)$

$H_2O(g) \rightarrow (0ml \rightarrow 30ml)$

In the end of the reaction 30ml of water and 5ml of oxygen and 10ml of hydrogen will be left.

LEVEL - II

1. Correct option: (B)

$2NaNO_3 \rightarrow 2NaNO_2 + O_2\uparrow$

Molecular weight of $NaNO_3 = 85\dfrac{g}{mol}$

Number of moles of $NaNO_3 = \dfrac{1.7}{85} = 0.02moles$

2 moles of $NaNO_3$ produces 0.01 moles of oxygen gas on heating.

At N.T.P

1 mole of oxygen gas has 22.4L of volume.

1.01 moles of oxygen gas has a volume.

$22.4\times 0.01 = 0.224L$

2. Correct option: (B)

$CaCO_3 \rightarrow CaO + CO_2 \uparrow$

Molecular weight of limestone= 100g/mole.

No. of moles of limestone= $\frac{50}{100} = 0.5 moles$

From the given equation

1mole of limstone forms 1 mole of quicklime.

So, 0.5mole of limstone forms 0.5 mole of quicklime.

Molecular weight of Quicklime =- 56 g/mole.

Then weight of quicklime produced $0.5 \times 56 = 28 grams$

3. Correct option: (C)

Balanced equation for electrolysis of water is $2H_2O \rightarrow O_2 + 2H_2$

Molecular weight of water=18g/mole.

Weight of water=18grams.

Number of moles= $\frac{18}{18} = 1 mole$

1mole of water produce 0.5 mole of oxygen.

At NTP, 0.5 moles of oxygen produces 11.2L.

4. Correct option: (B)

Electrolytic decomposition reaction of water is $2H_2O \rightarrow 2H_2 + O_2$

From the above reaction 2 moles of water of 1 mole of oxygen.

108 grams of water will produce $= \frac{6 \times 1}{2} = 3 moles$ of oxygen.

5. Correct option: (A)

$BaCl_2 + H_2SO_4 \rightarrow BaSO_4 + 2HCl$

1 mole of Barium chloride reacts with 1 mole of H_2SO_4 to produce 1 mole of $BaSO_4$.

Here, Barium chloride acts as a limiting reagent. So, 0.5 mole of $BaCl_2$ produce 0.5 mole of $BaSO_4$.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

6. Correct option: (C,D)

$2KClO_3 \rightarrow 2KCl + 3O_2$

Moles of $O_2 = \frac{1.5}{32} = 4.68 \times 10^{-2} mol$

From balanced equation, 3 moles of oxygen = 2 moles of KCl.

$4.68 \times 10^{-2} mol$ of oxygen $= \frac{2 \times 4.68 \times 10^{-2}}{3} = 3.12 \times 10^{-2} mol$

Mass of KCl= moles(molar mass) $= 2.324$

7. Correct option: (A)

The balanced chemical equation is $CaCO_3 \rightarrow CaO + CO_2$

100g of $CaCO_3$ = 56g of CaO.

90grams of $CaCO_3 = \frac{56}{100} \times 90g$ CaO $= 50.4g$

8. Correct options: (C, D)

22.4 L of a gas at STP = 1 mole.

11.2 L of a gas at STP = 0.5 mole.

11.2 L of a gas at STP will weigh 14 g.

0.5 moles of a gas at STP will weigh 14 g.

1 mole of a gas at STP will weight $\frac{1}{0.5}\times 14 = 28 grams$

$Molecular\ weight\ of\ N_2O = 2(14)+16 = 44\,g/mol$

$Molecular\ weight\ of\ NO_2 = 14+2(16) = 46\,g/mol$

$Molecular\ weight\ of\ N_2 = 2(14) = 28\,g/mol$

$Molecular\ weight\ of\ CO = 12+16 = 28\,g/mol$

$11.2\ L\ of\ N_2\ and\ CO\ at\ STP\ will\ weigh\ 14\ g.$

9. Correct option: (B)

$NaOH + CO_2 \rightarrow NaHCO_3$

According to the given equation 1 mole of CO_2 produces 84 grams of $NaOH$ So 2 mole of CO_2 produces 168 grams of $NaOH$.

10. Correct option: (B)

$MnO_2 + HCl \rightarrow Cl_2 + MnCl_2 + H_2O$

$2KI + Cl_2 \rightarrow KCl + I_2$

molecular weight of MnO_2 = 87 g

Number of moles of MnO_2 = 0.1 mole.

From ist equation we have 1 mole of MnO_2 produce 1 mole of Cl_2 and from 2nd equation 1 mole of Cl_2 is producing

amount of I_2 produces = moles (molecular mass)= 0.1(254)= 25.4 g

COMPREHENSION TYPE

11. Correct option: (B)

Here, hydrogen is 100 g. So, chlorine is the limiting reagent and hydrogen is in excess.

12. Correct option: (A)

According to the reaction, one mole or 70 g chlorine reacts with 1 mole (or 2 g) hydrogen to produce 2 moles ($2\times 36.5 = 73 gHCl$)

100 g of chlorine will react with $\frac{2}{70}\times 100 = 2.86\,g$ of hydrogen. Here, hydrogen is 100 g so chlorine acts as a limiting reagent.

Hydrogen is in excess by $= 100 - 2.86 = 97.14 g$

70 g of chlorine produces 73 g HCl. 100 g of chlorine produces $\frac{73}{70}\times 100 = 104.28\,g\ HCl$

13. Correct option: (B)

100 g of chlorine will react with $\frac{2}{70}\times 100 = 2.86\,g$ of hydrogen. Here, hydrogen is 100 g so chlorine acts as a limiting reagent.

Hydrogen is in excess by $= 100 - 2.86 = 97.14 g$

4. Correct option: (D)

$100\ g\ of\ H_2 = 50\ mole$

$100\ g\ of\ Cl_2 = 1.4\ mole$

According to the reaction 1 mole of H_2 is reacting with 1 mole of Cl_2 So, Cl_2 is the limiting reagent.

moles of HCl formed $= 2\times1.4\ mole$

Amount of HCl $=104.28\,g$

Since, the reaction is giving 90 % yield, amount of HCl formed is $=104.8\times\frac{90}{100}$

MATRIX MATCHING TYPE

15. Correct Matching is: a-P; b-S; c-Q; d-R

a) Molar mass of gas - P) 12 gms

b) Avogadro number - S) 6.023×10^{23}

c) 1 Mole of any gas occupies - Q) $22400cm^3$

d) 1 gram atom of carbon - R) 2 × vapor density in volume

16. Correct Matching is: a-P; b-R; c-T; d- Q

a) 32 g of SO_2 - P) 2g

b) 22.4 lit Hydrogen at STP - R) 1 mole

c) 0.5 mole of CH_4 - T) 8 g

d) 32g Methane - Q) 44.8 lit at STP

INTEGER TYPE

17. Correct Answer is: 5

The balanced chemical equation is

$3Br_2 + 3Na_2CO_3 \rightarrow 5NaBr + NaBrO_3 + 3CO_2$

From the above eqation we can observe that 5 molecules of $NaBr$ are obtained.

18. Correct Answer is:

Balanced chemical equation can be written as

$CaCl_2 + 2Na3PO_4 \rightarrow 6NaCl + Ca_3(PO_4)_2$

Total number of moles of $CaCl_2$ = $\frac{89.3}{110.98}\frac{g}{\frac{g}{mole}} = 0.805mole$

The ratio between $CaCl_2$ and $Ca_3(PO_4)_2$ is 3:1

Moles of $Ca_3(PO_4)_2 = \frac{0.805}{3} = 0.268$

Mass of $Ca_3(PO_4)_2 = 0.268mol\times310.17\frac{g}{mol} = 83.1g$

19. Correct Answer is:

The balanced chemical equation can be written as

$2H_2O_2 \rightarrow 2H_2O + O_2$

$2molofH_2O_2 \rightarrow 1molofO_2$

$2\times34gmH_2O_2 \rightarrow 32gm$

$\frac{136}{68} = 2mol\ of\ H_2O_2\ gives \Rightarrow 1mol$ of oxygen.

20. Correct Answer is:

$22.4\ L\ of\ gas\ =\ 1\ mole$

$30\ L\ of\ gas\ =\frac{30}{22.4}=\ 1.33\ mole$

According to equation, we have

3 mole of O_2 is producing 2 moles of O_3

so, 1.33 moles of O_2 produce $O_3=\frac{2}{3}\times 1.33=0.886\ mole$ and only 10% is converted to ozone.

The amount of ozone produced is equal to $=\ 0.886\times 0.1=0.0886\ mole$

Volume of ozone produced is equal to $=\ 0.0886\times 22.4\ L=1.984\ L$

Oxidation Number

LEVEL – I

1. Correct option: (D)

Potassium (K) typically has an oxidation number of +1.

Cyanide ion (CN^-) has a charge of -1.

overall charge of the complex is 0 because $K_4Fe(CN)_6$ is a neutral compound.

Let x be the oxidation number of iron (Fe).

Oxidation numbers being equal to the overall charge:

$4(1)+x+6(-1)=0$

$4+x-6=0$

$x-2=0$

$x=+2$

2. Correct option: (A)

$In\ HClO_4,$

$1+x+4(-2)=0$

$x=+7$

$In\ HClO_3,$

$1+x+3(-2)=0$

$x=+5$

$In\ HClO_2,$

$1+x+2(-2)=0$

$x=+3$

$In\ HClO,$

$1+x+1(-2)=0$

$x=+1$

3. Correct option: (A)

NO in brown ring complex has +1 oxidation state.

Let oxidation state of Fe be x.

Overall charge on the coordination sphere, $[Fe(H_2O)_5NO^+]$ is +2

So sum of oxidation numbers of all elements must be equal to +2.

$x+1=+2$

$x=+1$

Correct option: (C)

In $[Fe(H_2O)_6Cl_3]$ oxidation state of iron is $x+6(0)+3(-1)=0$

$x=+3$

5. Correct option: (C)

The oxidation state of phosphorus can vary, but the common range is from -3 to +5.

6. Correct option: (B)

CH_3Cl : In this compound, carbon has an oxidation state of -3.

CCl_4 : In this compound, carbon has an oxidation state of +4.

$CHCl_3$: In this compound, carbon has an oxidation state of +3.

CH_2Cl_2 : In this compound, carbon has an oxidation state of +2.

Among the options, CCl_4 (carbon tetrachloride) has the highest oxidation state for carbon, which is +4.

7. Correct option: (C)

$HClO_4$ (perchloric acid): Chlorine has an oxidation state of +7.

$HClO_2$ (chlorous acid): Chlorine has an oxidation state of +3.

$HClO_3$ (chloric acid): Chlorine has an oxidation state of +5.

HCl (hydrochloric acid): Chlorine has an oxidation state of -1.

8. Correct optio: (B)

Phosphorous acid (H_3PO_3): In this compound, phosphorus has an oxidation state of +3.

9. Correct option: (B)

A) K_2MnO_4 : Manganese has an oxidation state of +6.

B) $KMnO_4$: Manganese has an oxidation state of +7.

C) MnO_2 : Manganese has an oxidation state of +4.

D) Mn_2O_3 : There seems to be a typo in the compound, and it should be Mn_2O_3.

In Mn_2O_3 , manganese has an oxidation state of +3.

10. Correct option: (A)

Oxidation number of phosporous in PO_4^{-3} , $P_4O_{10}, P_2O_7^{-4}$ is +5.

11. Correct option: (B)

Let the Oxidation number of Cr in $K_2Cr_2O_7$ be x.

We know that,oxidation number . of K=+1

Oxidation number of O = −2

$2(+1)+2(x)+7(-2)=0$

$2+2x-14=0$

$2x=14-2=+12$

$x=+122=+6$

12. Correct option: (B)

$3Br_2+6NaOH \rightarrow 5NaBr+NaBrO_3+3H_2O$

Oxidation number of bromine changes from 0 in Br_2 to +5 in $NaBrO_3$.

13. Correct option: (A)

Oxidation is defined as loss of electrons.

14. Correct option: (B)

A reducing agent is a substance that can donate electrons.

15. Correct option: (B)

Magnesium reacts with acids to produce hydrogen gas and magnesium salts. In such reactions, magnesium undergoes Oxidation.

During the reaction, magnesium donates electrons to form magnesium ions, which is characteristic of oxidation. The oxidation state of magnesium increases, indicating the loss of electrons.

16. Correct option: (B)

In the given reaction $H_2O(g)+C(glowing)\rightarrow CO+H_2$ carbon (C) undergoes oxidation, and water (H_2O) acts as a reducing agent.

17. Correct option: (C)

In a reduction reaction, a species gains electrons, leading to a decrease in oxidation number or an increase in valency.

18. Correct option: (C)

$Cl_2+OH_\rightarrow Cl^-+ClO^{-4}+H_2O$

Here in this reaction $\begin{matrix} Cl\rightarrow(0\rightarrow-1) \\ Cl_2\rightarrow ClO_4^{-1}(0\rightarrow+7)\end{matrix}$

LEVEL - II

1. Correct option: (D)

The oxidation states of carbon atoms in diamond and graphite are both considered to be zero. In these allotropes of carbon, the carbon-carbon bonds are purely covalent, and carbon does not gain or lose electrons. Therefore, the correct answer is Zero, Zero.

2. Correct option: (B)

S_8 : The sulfur atoms in elemental form (like S_8) have an oxidation number of 0.

S_2F_2: In S_2F_2, the oxidation state of sulfur is +1. Fluorine is more electronegative, so sulfur is assigned a positive oxidation state.

H_2S: In H_2S, hydrogen is usually assigned a +1 oxidation state, and sulfur is assigned a -2 oxidation state.

3. Correct option: (A)

Caro's acid (H_2SO_4), Marshall's acid ($H_2S_2O_8$):

The oxidation state in both the acids is +6 each because oxidation state cannot be greater than the number of valence electrons.

4. Correct option: (D)

In $[Fe(CN)_6]^{4-}$ Oxidation state of Fe $\begin{matrix} =x+6(-1)=-4 \\ x=+2\end{matrix}$

In $[Fe(CN)_6]^{3-}$ Oxidation state of Fe $\begin{matrix} x+6(-1)=-3 \\ x=+3\end{matrix}$

In $[Cr(CN)_6]^{3-}$ Oxidation state of Cr $\begin{matrix} x+6(-1)=-3 \\ x=+3\end{matrix}$

In CrO_2Cl_2 Oxidation state of Cr $\begin{matrix} x+2(-2)+2(-1)=0 \\ x=+6\end{matrix}$

5. Correct option: (C)

$H-O-Ti-O-O-Ti-O-H$, $p(OH)_2=O-O-O-p(OH)_2=O$

Both H_2TiO_4 and $H_4P_2O_8$ contain peroxy linkages.

6. Correct option: (B)

Among the given options $KMnO_4$ (permanganate) is a strong oxidizing agent.

7. Correct option: (A)

In bleaching powder,($CaOCl_2$)two Cl atoms are in different oxidation states i.e., one Cl^- having oxidation number of –1 and OCl^- having oxidation number of +1.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

8. Correct options: (B, C)

$FeCr_2O_4 \Leftrightarrow \left(Fe^{+2}\left(Cr_2O_4\right)^{-2}\right)$ oxidation state of Chromium is +3

$KCrO_3Cl \Leftrightarrow \left[K^+\left(CrO_3Cl\right)\right]^-$ oxidation state of Chromium is +6.

CrO_5 oxidation state of Chromium is +6. (Two peroxo linkage).

[Cr(OH)4]⊖

$[Cr(OH)_4]^-$ oxidation state of Chromium is +3.

9. Correct options: (A, B, C, D)

Lets consider the oxidation number of carbon as x.

$HCHO$

$x+2(1)+(-2)=0$

$x=0$

CH_2Cl_2

$x+2(+1)+2(-1)=0$

$x=0$

$C_6H_{12}O_6$

$x+6(-2)+12(+1)=0$

$x=0$

$C_{12}H_{22}O_{11}$

$12(x)+22(+1)+11(-2)=0$

$x=0$

10. Correct option: (D)

The reduction of the metal center in aqueous permanganate ion (MnO_4^-) involves 5 electrons. The balanced half-reaction in an acidic medium is as follows:

$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$

11. Correct options: (A, B, D)

The balanced equation is $ClO^{-3} + 6I^- + 6H_2SO_4 \rightarrow 3I_2 + Cl^- + 6HSO_4^- + 3H_2O$

Sulphur (S) is oxidized in this reaction. It goes from a +4-oxidation state in ClO_2 to a +6-oxidation state in HSO_4^-. So, statement C is incorrect.

Comprehension:

12. Correct options: (A, C)

Let x is the oxidation state of carbon.

$C_6H_{12}O_6$:

$x\times 6+12(+1)+6(-2)=0$

$6x+12-12=0$

$6x=0$

$x=0$

Oxidation state of C is 0.

$HCOOH$:

$x+2(+1)+2(-2)=0$

$x+2-4=0$

$x=2$

Oxidation state of carbon is 2.

$HCHO$:

$x+2(+1)+1(-2)=0$

$x+2-2=0$

$x=0$

Oxidation state of carbon is 0.

CH_4 :

$x+4(+1)=0$

$x+4=0$

$x=-4$

Oxidation state of carbon is -4.

13. Correct option: (C)

Oxidation number of carbon in carbon monoxide $\frac{(2\times1-0)}{1}=+2$

Oxidation number of carbon in carbon dioxide $=\frac{(2\times2-0)}{1}=+4$

Oxidation number of carbon in carbon dioxide $=\frac{(2\times2-0)}{3}=\frac{4}{3}$

oxidation state of carbon is fractional in carbon suboxide.

14. Correct option: (D)

A) CH_4 (methane): Carbon has an oxidation state of -4 in CH_4 .

B) CH_3OH (methanol): Carbon has an oxidation state of -2 in CH_3OH .

C) CH_2O (formaldehyde): Carbon has an oxidation state of 0 in CH_2O .

D) $HCOOH$ (formic acid): Carbon has an oxidation state of +2 in $HCOOH$.

15. Correct Matching is: a-R; b-S; c-Q; d-P

a) S - R) 0

b) H_2S - S) – 2

c) S_2Cl_2 - Q) +1

d) $H_2S_2O_2$ - P) +6

16. Correct Matching is: a-S; b-R; c-P; d- Q

a) Sulphurous acid (H_2SO_3) S) +4.

b) Peroxy monosulfuric acid (H_2SO_5) +6.

c) Dithionic acid ($H_2S_2O_6$) +5.

d) Dithionous acid ($H_2S_2O_4$) +3.

INTEGER TYPE

17. Correct Answer is: 6

In the alkaline oxidative fusion of MnO_2, manganese (Mn) undergoes oxidation to form the product. The balanced chemical reaction is typically represented as:

$MnO_2 + 2NaOH + O_2 \rightarrow Na_2MnO_4 + H_2O$

In the product Na_2MnO_4, manganese is in the +6 oxidation state.

18. Correct Answer is: 2

Chromium pentoxide, CrO_5, has a structure where oxygen atoms form peroxy (O-O) links. In CrO_5, there are two peroxy links. Each peroxy link involves two oxygen atoms bonded together. So, in CrO_5, you have a total of four oxygen atoms involved in two peroxy links.

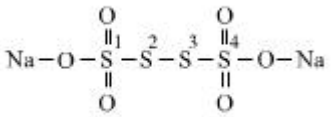
19. Correct Answer is: 6

The balanced chemical equation for this conversion is typically represented as:

$Cr_2O7_2^- + 6e^- + 14H^+ \rightarrow 2Cr^{+3} + 7H_2O$

In this equation, 6 moles of electrons ($6e^-$) are involved in the conversion of 1 mole of $Cr_2O7_2^-$ into Cr^{+3} ions.

20. Correct Answer is: 2



Out of the four Sulphur atoms in $Na_2S_4O_6$, two (marked 2 and 3) are bonded with Sulphur atoms. The oxidation state of these two Sulphur atoms is zero.

Redox Reactions

LEVEL - I

1. Correct option: (A)

The oxidation number of iron changes from +3 to +2. So, $FeCl_3$ is reduced and acts as oxidizing agent.

The oxidation number of Sulphur changes from -2 to 0. So, H_2S is oxidized and acts as reducing agent.

2. Correct option: (C)

$H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + H_2O$

Above reaction is a acid-base neutralization reaction.

$3O_2 \rightarrow 2O_3$ changes in one allotrope of oxygen to another. Thus, it's an allotropic formation reaction.

N has oxidation number 0 and in NO it is +2. Thus N_2 is reduced and similarly, oxidation of oxygen changes from 0 to -2 and thus oxygen is oxidized.

$H_2O(l) \rightarrow H_2O(g)$ Only Physical state changes.

The formation of oxides of nitrogen from nitrogen and oxygen by lightning is the redox reaction where oxidation, as well as reduction, takes place, simultaneously.

3. Correct option: (A)

$CaCO_3 \rightarrow CaO + CO_2$

$\overset{+2}{Ca}\overset{+4}{C}\overset{-2}{O_3} \rightarrow \overset{+2}{Ca}\overset{-2}{O} + \overset{+4}{C}\overset{-2}{O_2}$

Above reaction is a decomposition reaction. There is no change in the oxidation number of any species so it is not a redox reaction.

4. Correct option: (C)

A redox (reduction-oxidation) reaction is a chemical reaction in which there is a transfer of electrons between reactants. In a redox reaction, one substance undergoes oxidation (loses electrons) while another substance undergoes reduction (gains electrons).

5. Correct option: (A)

Oxidation number of Chromium in CrO_2^{-4}: $x + 4 \times (-2) = -2$, $x = 6$

Oxidation number of Chromium in CrO_7^{-2}: $2x + 7 \times (-2) = -2$, $x = 6$

6. Correct option: (C)

Nitrogen oxides form nitrogen and oxygen by lightning - This reaction involves the decomposition of nitrogen oxides, and the formation of nitrogen and oxygen. It does involve a transfer of electrons and is a redox reaction.

7. Correct option: (C)

$CaCO_3 \rightarrow CaO + CO_2$ this equation is not a combination reaction it is a decomposition reaction. Here Calcium carbonate decomposed into CaO and CO_2.

8. Correct option: (D)

A decomposition reaction is a type of chemical reaction in which a single compound breaks down into two or more simpler substances. Let's analyze each option:

A) $2H_2O \rightarrow 2H_2 + O_2$ - This represents the decomposition of water into hydrogen and oxygen.

B) $2NaH \rightarrow 2Na + H_2$ - This represents the decomposition of sodium hydride into sodium and hydrogen.

C) $2KClO_3 \rightarrow 2KCl + 3O_2$ - This represents the decomposition of potassium chlorate into potassium chloride and oxygen.

9. Correct options: (C)

The given reaction $3ClO^-(aq) \rightarrow ClO_3^- + 5Cl^-(aq)$ represents a disproportionation reaction. In disproportionation reactions, a single species undergoes both oxidation and reduction within the same chemical reaction.

10. Correct option: (C)

$2H_2O_2 \rightarrow 2HO + O_2$ this reaction is a disproportionation reaction. “In a disproportionation reaction an element in one oxidation state is simultaneously oxidized and reduced”.

The oxidation state of the oxygen present in peroxide is –1.

The oxidation state of oxygen present in the product is 0.

The oxidation state of oxygen present in water is –2.

11. Correct option: (A)

The reactivity of metals is often associated with their position in the reactivity series. The reactivity series is a list of metals arranged in order of their decreasing reactivity. Among the options provided:

A) Al (aluminum)

B) Ni (nickel)

C) Pb (lead)

D) Cu (copper)

Aluminum (Al) is the most reactive metal among the given options. The reactivity series generally follows the trend that metals at the top are more reactive than those at the bottom.

12. Correct option: (C)

$2Na + 2H_2O \rightarrow 2NaOH + H_2$

In the reactant, each sodium atom has an oxidation state of 0 because it is in its elemental form. In the product (NaOH), each sodium atom has an oxidation state of +1because it is bonded to a more electronegative element (oxygen) and has donated one electron.

The change in oxidation number for sodium is from 0 to +1.

13. Correct option: (B)

The oxidation number is a measure of the number of electrons that an atom gains or loses in order to form a chemical bond with another atom. In the case of the M^{3+} ion losing 3 electrons, it means that the ion has undergone oxidation, and its oxidation number is equal to the charge on the ion.

The M^{3+} ion has a charge of 3+, so its oxidation number is +3.

14. Correct option: (C)

$M + water \rightarrow oxide or hydroxide + H_2$

A) Iron (Fe) - Reacts with water to form iron oxide and hydrogen gas.

B) Sodium (Na) - Reacts with water to form sodium hydroxide and hydrogen gas.

C) Mercury (Hg) - Does not react with water under normal conditions; it is not an alkali or alkaline earth metal.

D) Magnesium (Mg) - Reacts with water to form magnesium hydroxide and hydrogen gas.

15. Correct option: (C)

When $KMnO_4$ acts as an oxidizing agent and ultimately forms

$MnO_4^{-2}, MnO_2, Mn_2O_3, Mn^{+2}$ then the number of electrons transferred in each case are 1,3,4,5 respectively.

The oxidation number of Mn in $MnO_4^{-2}, MnO_2, Mn_2O_3, Mn^{+2}$ is 7, 6, 4, 3 and 2 respectively. The number of electrons transferred corresponds to the change in the oxidation number.

16. Correct option: (D)

$MnO_4^{-2} \rightarrow Mn^{+2} + xH_2O$

The oxidation state of manganese changes from $+7$ in $\mathrm{MnO_4^{-2}} to + 2Mn^{+2}$

To find the number of electrons (n) involved in this change, you can use the formula:

$n = oxidation\ state\ of\ oxidized\ element - oxidation\ state\ of\ reduced\ element$

$n = (+7) - (+2) = 5$

Each electron is associated with a water molecule, so x=5.

17. Correct option: (C)

The balanced reaction written as $3Cl_2 + 6OH^- \rightarrow ClO_3^- + 5Cl^- + 3H_2O$

Hence, X=3, Y=5.

18. Correct option: (B)

$NO_2^- + H_2O \rightarrow NO_3^- + 2H^+ + 2e^-$

In this reaction, n stands for the number of electrons transferred. Looking at the coefficients of the electrons (e^-) in the balanced equation, we see that n=2.

19. Correct option: (A)

$2MnO_4^- + 5C_2O_4^{-2} + 16H^+ \rightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$

The above reaction is the balanced chemical reaction. Thus the value of x, y and z is 2,5, and 16.

20. Correct option: (B)

Given equation is $MnO_4^- + Fe^{2+} + H^+ \rightarrow Mn^{2+} + Fe^{3+} + H_2O$

Splitting into two half reactions,

$MnO_4^- + H^+ \rightarrow Mn^{2+} + H_2O; (\text{Re}\,ductionhalfreaction)$

$Fe^{2+} \rightarrow Fe^{3+} (Oxidationhalfreaction)$

Adding hydrogen ions to the side deficient in hydrogen,

$MnO_4^- + 8H^+ \rightarrow Mn^{2+} + 4H_2O$

Adding electrons to the sides deficient in electrons,

$MnO_4^- + 8H^+ + 5e \rightarrow Mn^{2+} + 4H_2O$

$Fe^{2+} \rightarrow Fe^{3+} + e$

Balancing electrons in both half reactions,

$MnO_4^- + 8H^+ + 5e \rightarrow Mn^{2+} + 4H_2O$

$5Fe^{2+} \rightarrow 5Fe^{3+} + 5e$

Adding both the half reactions,

$MnO_4^- + 8H^+ + 5Fe^{2+} \rightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$

21. Answer: D) Electron transfer reaction

Explanation:

A **redox (reduction–oxidation) reaction** is a chemical reaction in which **electrons are transferred** from one substance to another.

- **Oxidation** = Loss of electrons.
- **Reduction** = Gain of electrons.

Both oxidation and reduction occur simultaneously.

Example:

$Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

- Zn loses electrons (oxidation).
- Cu^{2+} gains electrons (reduction).

Hence, a redox reaction is an **electron transfer reaction**.

22. Answer: C) Dissolving a salt in water

Explanation:

Let's examine each option:

- **C) Dissolving a salt in water Not a redox reaction**
  The salt simply dissociates into ions. There is **no change in oxidation state** and **no electron transfer**.
  Example:
  $NaCl \rightarrow Na^+ + Cl^-$
  Therefore, the correct answer is **C) Dissolving a salt in water**

LEVEL – II

1. Correct option: (A)

The balanced chemical reaction can be represented as:

$aFeS_2 + bO_2 \rightarrow cFe_2O_3 + dSO_2 \approx 4FeS_2 + 11O_2 \rightarrow 2Fe_2O_3 + 8SO_2$

$(a+b+c+d) = 4+11+2+8$

$(a+b+c+d) = 25$

2. Correct option: (D)

$2H_2O_2 \rightarrow 2H_2O + O_2$ -his is a disproportionation reaction because hydrogen peroxide (H_2O_2) is both oxidized and reduced in the reaction.

3. Correct option: (B)

copper (Cu) does not react with acids to produce hydrogen gas.

4. Correct option: (C)

The reactivity of non-metals generally increases from left to right across a period and decreases down a group on the periodic table. Among the given options, fluorine (F) is the most reactive non-metal.

5. Correct option: (C)

In a displacement reaction, one element displaces another from a compound. Displacement reactions involve the transfer of electrons, making them redox reactions.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

6. Correct option: (D)

The oxidation state of iodine in HI is -1 as the iodine is more electronegative than hydrogen.

The oxidation state of iodine in I_2 is zero.

In HIO_4 oxidation state of iodine is +7.

In ICl the oxidation state of iodine is +1 since chlorine is more electronegative than iodine.

Increasing oxidation number of iodine is: $HI < I_2 < ICl < HIO_4$

7. Correct options: (A, B, D)

SO_2: Sulphur is present in +4 oxidation state so it act as both oxidizing as well as reducing agent.

H_2O_2: Oxygen is present in -1 oxidation state so it can act as both oxidizing as well as reducing agent.

HNO_2: Oxidation number of nitrogen is +3. So, it can easily increase or decrease its oxidation state. So, it acts both as an oxidizing and reducing agent.

8. Correct options: (B, C, D)

Disproportionation is a specific type of redox reaction in which a species is simultaneously reduced and oxidized to form two different products.

$$3H_2O + \overset{0}{P_4} + 3\overset{-}{O}H \rightarrow 3\overset{-3}{P}H_3 + 3H_2\overset{+1}{P}O_2^-$$

$$\overset{0}{S_8} + 12\overset{-}{O}H \rightarrow 4S^{-2} + 2S_2^{+2}O_3^{-2} + 6H_2O$$

$$\overset{0}{Br_2} + 2\overset{-}{O}H \rightarrow \overset{-1}{B}rO^- + \overset{-1}{B}r + H_2O$$

9. Correct options: (B, D)

Copper in Cu_2O is reduced as its oxidation number decreases from +1 to 0.

Copper in Cu_2S is reduced as its oxidation number decreases from +1 to 0.

Both Cu_2O and Cu_2S are reduced.

10. Correct options: (A, B)

COMPREHENSION TYPE

11. Correct option: (B)

$FeS_2 \rightarrow Fe^{3+} + SO_2$ in this reaction n factor for Fe $\begin{matrix} Fe(+2 \rightarrow +3) \\ S(-1 \rightarrow +4) \end{matrix}$

So net change is $= 1 + (5 \times 2) = 11$

12. Correct option: (D)

$Br_2 \rightarrow 2BrO_3^- + 10e^-$

$Br_2\ (2x = 0)$

$2BrO_3^-\ (2x - 12 = -2)$

$Br_2 \rightarrow 2BrO_3^- + 10e^-$

$2e^- + Br_2 \rightarrow 2Br^-] \times 5$

$- - - - - - - - - - - - - -$

$6Br_2 \rightarrow 2BrO_3^- + 10Br^-$

13. Correct option: (B)

Given equation is $As_2S_3 \rightarrow As^{5+} + SO_4^{-2}$ it can be written into two equations

$As_3^{+2} \rightarrow 2As^{5+} + 4e^-$

$As_3^{+2}\ (2x = 6)$

$2As^{5+} + 4e^-\ (2x = 10)$

$S_3^{-2} \rightarrow 3SO_4^{-2} + 24e^-$

$S_3^{-2}\ (3x = -6)$

$$3SO_4^{-2} + 24e^- \begin{pmatrix} 3x - 34 = -6 \\ 3x = 34 - 6 \\ 3x = 18 \end{pmatrix}$$

n-factor is 28.

14. Correct Matching is: a-R; b-S; c-Q; d- P

a) H_2S: Sulfide ion (S2-), so the oxidation number of S is R) -2.

b) H_2SO_4: Sulfuric acid, H_2SO_4. The oxidation numbers are: H (+1), S (+6), O (-2). Therefore, the oxidation number of S is S) 6.

c) SO_2 : Sulfur dioxide. The oxidation numbers are: S (+4), O (-2). So, the oxidation number of S is Q) 4.

d) S_8 : Elemental sulfur (S_8). Each S atom has an oxidation number of 0.

15. Correct Matching is: a- Q; b-R; c-P; d- S

a) $Mn(OH)_2 + H_2O_2 \rightarrow MnO_2$

The oxidation state of manganese (Mn) changes from +2 in $Mn(OH)_2$ to +4 in MnO_2

Number of electrons lost: $4-2=2$

Answer: Q corresponds to 2.

b) $AlCl_3 + 3K \rightarrow Al + 3KCl$

The oxidation state of aluminum (Al) changes from +3 in $AlCl_3$ to 0 in Al. Number of electrons lost:3

Answer: R corresponds to 3.

c) $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$

$Valancy\ of\ Fe in\ Fe_3O_4\ is +2,$

$lost\ electrons = 2\times 3 = +6$

d) $H_2S + NO_3^- \rightarrow S + NO$

Number of electrons lost: 6.

16. Correct Answer is:

$IO_3^- + aI^- + bH^+ \rightarrow cH_2O + dI_2$

The balanced equation can be written as

$IO_3^- + 5I^- + 6H^+ \rightarrow 3H_2O + 3I_2$

$a=5; b=6; c=3; d=3$

17. Correct Answer is: 14

18. Correct Answer is:

Given equation is $Cl_2 + 6OH^- \rightarrow ClO_3^- + Cl^- + 3H_2O$

Step-1: $Cl_2 + 6OH^- \rightarrow ClO_3^- + 3H_2O$

$Cl_2 \rightarrow Cl^-$

Step-2: Balancing the above reaction $Cl_2 + 12OH^- \rightarrow 2ClO_3^- + 6H_2O$

$Cl_2 \rightarrow 2Cl^-$

Step-3: Adding electron to sides deficient in electrons,

$Cl_2 + 12OH^- \rightarrow 2ClO_3^- + 6H_2O + 10e^-$

$Cl_2 + 2e^- \rightarrow 2Cl^-$

Step-4: Balancing electron in both half reactions,

$Cl_2 + 12OH^- \rightarrow 2ClO_3^- + 6H_2O + 10e^-$

$5\left[Cl_2 + 2e^- \rightarrow 2Cl^-\right]$

Step-5: Adding both

$Cl_2 + 12OH^- \rightarrow 2ClO_3^- + 6H_2O + 10Cl^-$

Step-6: Dividing by 2,

$3Cl_2 + 6OH^- \rightarrow ClO_3^- + 3H_2O + 5Cl^-$

X, y values are 3, 5.

19. Correct Answer is: 6

The oxidation number of hydrogen (H) is +1 (there are two hydrogen atoms).
The oxidation number of oxygen (O) is -2 (there are four oxygen atoms, and the overall charge of oxygen is -8).
Let the oxidation number of sulfur (S) be x.

$2(+1)+x+4(-2)=0$

$2+x-8=0$

$x-6=0$

$x=+6$

20. Correct Answer is: 8

Answer: Balanced chemical equation can be written as

$4Zn+10HNO_3 \rightarrow 4Zn(NO_3)_2+3H_2O+NH_4NO_3$

$sum\ of\ the\ co-efficients\ is\ (a+b+c)=4+3+1=8$

Equivalent Weight

LEVEL – I

1. Correct option: (D)

The molecular weight of orthophosphoric acid (H_3PO_4) is represented by M.

Orthophosphoric acid (H_3PO_4) has three acidic hydrogens, each of which can potentially ionize. The equivalent weight is defined as the molecular weight divided by the number of acidic hydrogens that can be replaced.

In H_3PO_4, there are three acidic hydrogens, so the equivalent weight (E) is given by:

$$E=\frac{number\ of\ acidic\ hydrogens}{M}=\frac{M}{3}$$

2. Correct option: (A)

3. Correct option: (B)

$$Equivalentweight=\frac{Molecular\,Weight}{Net\ positive\,valence}$$

The equivalent weight of a substance is the molecular weight divided by the number of equivalents.
For (calcium carbonate), we need to consider the number of equivalents of the substance.

The molecular formula of $CaCO_3$ is:

Atomic weight of calcium (Ca): 40.08
Atomic weight of carbon (C): 12.01
Atomic weight of oxygen (O): 16.00 (there are three oxygen atoms in CO_3

Molecular weight of calcium carbonate = $40.8+12.01+3\times16.00=100.09$

$$Equivalent\ weight\ of\ CaCO_3=\frac{Number\ of\ equivalents}{Number\ of\ equivalents}$$

$$Equivalent\ weight\ of\ CaCO_3=\frac{100.9}{2}=50.045$$

4. Correct option: (B)

The balanced chemical equation for the reaction is as follows:

$$K_2Cr_2O_7 + 6H^+ + 6e^- \rightarrow 2K^+ + 2Cr^{3+} + 7H_2O$$

$$EW = \frac{Molecular\ weight}{Change\ in\ oxidation\ state}$$

$$EW = \frac{294}{6}\ g/mol = 49\ g/equivalent$$

5. Correct option: (B)

$$EW = \frac{Molecular\ weight}{Change\ in\ oxidation\ state}$$

Change in oxidation on state of $KMnO_4$ in basic medium =3

Molecular weight of $KMnO_4$ is 158.

$$EW = \frac{158}{3} = 52.66$$

6. Correct option: (C)

Equivalent weight=12

$Molecular\ weight = 12 \times Valency\ factor = 24u$

Its oxide is XO where X is the bivalent metal.

molecular weight of oxide is 24+16 = 40.

7. Correct option: (A)

Equivalent weight of Sulphur in $SCl_2 = \frac{Atomic\ mass\ of S}{valency\ of S} = \frac{32}{2} = 16$

Equivalent weight of Sulphur in $S_2Cl_2 = \frac{Atomic\ mass\ of S}{valency\ of S} = \frac{32}{1} = 32$

Ratio of equivalent weight of S is $SCl_2\ and\ S_2Cl_2\ is\ 16:32 = 1:2$

8. Correct option: (A)

$(+2)CuSO_4 \rightarrow Cu_2I_2(+1)$

Change in oxidation state of copper is +1.

Equivalent weight = M/1.

9. Correct option: (B)

The equivalent weight of a substance is defined as the molecular weight divided by the number of equivalents of protons or ions it can donate or accept. For boric acid (H_3BO_3), the equivalent weight is equal to its molecular weight divided by the number of replaceable hydrogen ions.

The molecular weight of boric acid is

$1(1.01) + 3(10.81) + 1(16.00) = 61.84\ g/mol$

Boric acid can donate one proton (H^+), so the equivalent weight is

$61.84\ g/mol \div 1 = 61.84\ g/equiv \approx 62$

10. Correct option: (D)

$2Ca_3(PO_4)_2 \rightarrow 6Ca^{+2} + 4(PO_4)^{-3}$

Oxidation number of calcium is +2.

Molecular mass of $Ca_3(PO_4)_2$ is 310grams.

$$Equivalent Weight = \frac{Molecular weight of salt}{ch\arg e}$$

$$= \frac{2 \times 310}{6 \times 2} = \frac{310}{6}$$

11. Correct option: (A)

$CuSO_4(+2) \rightarrow Cu_2I_2(+1)$

Change of oxidation state of copper is +1.

$$Gram equivalent weight = \frac{M}{1}$$

12. Correct option: (C)

$H_3PO_4 + 2KOH \rightarrow K_2HPO_4 + H_2O$

In the given reaction only two H-atoms of H_3PO_4 are replaced by metal atoms. H_3PO_4 acts like a di basic acid.

$$EW = \frac{Molecular weight}{2}$$

$$EW = \frac{98}{2} = 49$$

13. Correct option: (A)

Given acid is phosphoric acid.

It's molar mass is $98g/mol$.

Equivalent weight of acids is calculated by dividing the molar mass of the acids with number of hydrogen ions donated by acids.

Here Phosphoric acid has donated $2H^+$ ions.

$$Equivalent\ weight = \frac{98g/mol}{2} = 49g/mol$$

14. Correct option: (D)

$2KMnO_4 + 3H_2SO_4 + 5H_2C_2O_4 \rightarrow K_2SO_4 + 2MnSO_4 + 8H_2O + 10CO_2$

This equation shows that 2 moles of $KMnO_4$ are reacting. The equivalent weight (Eq. Wt.) is determined by the number of moles of electrons transferred or involved in the reaction.

For $KMnO_4$, it can donate 5 moles of electrons in a redox reaction (Mn goes from +7 to +2 oxidation state). Therefore, the equivalent weight is given by the formula:

$$Equivalent Weight = \frac{Molecular Weight}{Number of\ moles of\ electrons transferred}$$

Oxidation state of Mn changes from +7 to +2, so the valency factor is 5. Equivalent weight would be M/5.

15. Correct option: (A)

Oxidation state of Mn in $KMnO_4$ is +7 and in K_2MnO_4 is +6. N-factor value is 1, therefore equivalent weight is equal to M/1 or M.

16. Correct option: (B)

$Na_2S_2O_3(+2) \rightarrow Na_2SO_4(+6)$

$S_2 + 2e^- \rightarrow S^{+6}$

$2(+2) \rightarrow +6 + 2e^-$

$$Equivalent\ weight = \frac{M}{2}$$

17. Correct option: (A)

Mohr's salt $Fe^{+2}, KMnO_4$ oxidizes ferrous to ferric in acidic medium.

$$Fe^{+2} \rightarrow Fe^{+3} + e^{\rightarrow}$$

$$Equivalentweight = \frac{M}{1} = M$$

18. Correct option: (A)

$$Equivalent\ mass\ of\ CH_4 = \frac{atomic\ mass\ of\ C}{Valency} + \frac{atomic\ mass\ ofH}{Valency}$$

$$Equivalent\ mass\ of\ CH_4 = \frac{12}{4} + \frac{1}{2}$$

$$Equivalent\ mass\ of\ CH_4 = 3 + 1 = 4$$

As the molecular mass of CH_4 is 16 u.

equivalent mass of CH_4 = 16/4=4=M/4.

19. Correct option: (B)

Oxidation state of iodine in I_2 is zero and in NaI, it is -1. As there are two moles of NaI, therefore -1 becomes -2. Equivalent weight is M/2.

20. Correct option: (C)

$$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$$

Oxidation number of carbon atom changes from 0 to +4.

$$\frac{Total\ increase\ in\ Oxidation.No}{molecule\ of\ Glucose} = 6 \times 4 = 24$$

$$Equivalentweight = \frac{M}{24}$$

LEVEL - II

1. Correct option: (C)

Equivalent weight of rust is M/4.

2. Correct option: (B)

The empirical formula represents the simplest, most reduced ratio of elements in a compound. It gives the relative number of atoms of each element in the compound in terms of the smallest whole numbers. While it provides the ratio, it does not give the exact number of atoms; that information is provided by the molecular formula or the structural formula.

3. Correct option: (D)

The chemical formula of Glucose is $C_6H_{12}O_6$. The ration of atoms in glucose is CH_2O that shows the empirical formula of glucose.

Empirical formula gives the simplest ratio of atoms in a molecule or compound.

4. Correct option: (A)

The empirical formula CH corresponds to ethyne.

5. Correct option: (A)

$$Empirical\ mass\ formula = 12 + 2 + 16 = 30$$

$$n = \frac{MolarMass}{EmpericalMass} = \frac{90}{30} = 3$$

$$MolarFormula = (CH_2O)_n = C_3H_6O_3$$

6. Correct option: (B)

$$Molecularmass = 2 \times vapordensity$$

$$molecularmass = 2 \times 17 = 34$$

$$n = \frac{34}{17} = 2$$

$$Molecularformula = (HO)_2 = H_2O_2$$

7. Correct option: (B)

Molecular weight = 2 × Vapor density

Molecular weight =2 × 45 = 90

Let the molecular formula be $C_xH_{2x}O_x$

$$90 = 12x + 2x + 16x = 30x$$

$$30x = 90$$

$$x = 3$$

the molecular formula is $C_xH_{2x}O_x = C_3H_6O_3$

8. Correct option: (A)

The empirical formula of acetylene (C_2H_2) is CH.

A) C_6H_6 - The empirical formula is CH, not the same as acetylene.

B) C_3H_4 - The empirical formula is CH_2, not the same as acetylene.

C) C_2H_4 - The empirical formula is CH_2, not the same as acetylene.

D) C_2H_6 - The empirical formula is CH_3, not the same as acetylene.

9. Correct option: (A)

HNO_3 *reduced as* NH_4NO_3

$$Equivalentweight = \frac{Molecularweight}{8}$$

10. Correct option: (A,C)

$$C_2O_4^{-2} + MnO_4^- = CO_2 + Mn^{2+}$$

oxidation state change in one c atom is +1 so for 2carbon atoms it is +2.

$$equivalentweight = \frac{88}{2} = 44$$

11. Correct option: (B)

H_3PO_2 (Phosphorous Acid):

$$Molecularweight = 3 \times \mathrm{H} + 1 \times \mathrm{P} + 2 \times \mathrm{O} = 3 + 31 + 32 = 66\,\mathrm{g/mol}$$

$$Equivalentweight = \frac{66}{1} = 66\,\mathrm{g/mol}$$

12. Correct option: (B)

The correct relationship between the molecular formula and the empirical formula is given by: $Molecular\ formula = Empirical\ formula \times n$ .n is equal to the ratio of the molecular mass to the empirical formula mass.

13. Correct option: (A)

Glucose ($C_6H_{12}O_6$) and fructose ($C_6H_{12}O_6$) have the same empirical formula (CH_2O), which means they have the same ratio of elements.

14. Correct option: (C)

To find the empirical formula from the given ratio of atoms (1:0.5:1), we need to normalize the ratio to whole numbers. In this case, we can multiply each part by 2 to get whole numbers:.

The given ratio $=1:0.5:1$

Multiply each part by 2 the normalized ratio $2:1:2$

So, the empirical formula would be A_2BC_2

15. Correct option: (B)

To determine the empirical formula from the molecular formula $X_3Y_6O_3$, we need to find the simplest whole-number ratio of the atoms.

In this case: There are 3 atoms of X.

There are 6 atoms of Y.

There are 3 atoms of O.

To find the simplest whole-number ratio, we can divide each of these numbers by their greatest common factor, which are 3 in this case. Empirical Formula = $X_1Y_2O_1$

16. Correct option: (B)

The empirical formula of hydrogen peroxide is H_2O_2.

17. Correct Matching is: a-S; b-R; c-P; d- R

a) Acidic	S)31.6
b) Neutral	R) 52.6
c) Strongly basic	P) 158
d) Weakly basic	R) 52.6

18. Correct Matching is: a-Q; b-S; c-R; d- T

a) Glucose -	Q) CH_2O
b) Oxalic acid -	S) CHO_2
c) Inorganic Benzene -	R) BNH_2
d) Oxygenated water -	T) OH

19. Correct Answer is:

The balanced chemical equation for the oxidation of coppers (I) sulfide (Cu_2S) to copper (II) ions (Cu^{+2}) and sulfur dioxide (SO_2) is as follows:

$2Cu_2S + 3O_2 \rightarrow 2Cu^{+2} + 2SO_2$

From the balanced equation, you can see those 2 moles of Cu_2S react with 3 moles of O_2 to produce 2 moles of Cu^{+2} and 2 moles of SO_2.

Molecular weight of $Cu_2S = 63.55(\text{Cu}) \times 2 + 32.07(\text{S}) \times 2 = 127.1 \frac{\text{g}}{mol}$

Molecular weight of $Cu^{+2} = 63.55(\text{Cu}) \times 2 = 127.1\,\text{g/mol}$

Molecular weight of $SO_2 = 32.07(\text{S}) + 2 \times 16.00(\text{O}) = 64.07\,\text{g/mol}$

$$Mol.wt\ ratio = \frac{Molecular\ weight\ of\ Cu_2S}{Molecular\ weight\ of\ Cu^{+2} + Molecular\ weight\ of\ SO_2}$$

$$Mol.wt\ ratio = \frac{127.1g/mol}{127.1g/mol + 127.1g/mol + 64.07\,\text{g/ mol}}$$

$$Mol.wt\ ratio = \frac{127.1}{318.27}$$

$$Mol.wt\ ratio \approx 0.399$$

20. Correct Answer is:

The balanced equation for the reaction of hydrogen peroxide in a basic medium is

$H_2O_2 + 2OH^- \rightarrow O_2 + 2H_2O$

From the balanced equation, you can see that one mole of hydrogen peroxide reacts with two moles of hydroxide ions.

$Eq.wt\ of\ H_2O_2 = \frac{Mol.wt\ of\ H_2O_2}{2}$

$Mol.wt\ of\ H_2O_2 = 2 \times Mol.wt\ of\ H + 2 \times Mol.wt\ of\ O$

$Mol.wt\ of\ H_2O_2 = 2 \times 1 + 2 \times 16$

$Mol.wt\ of\ H_2O_2 = 2 + 32 = 34\frac{g}{mol}$

$$equivalent\ weight\ = \frac{sum\ of\ molecular\ weights\ of\ molecules\ in\ a\ compound}{basicity} = \frac{34}{2} = 17.$$

Previous Year Questions

1. Correct Answer: D) II, III only

Balance the redox equation:

This is a disproportionation reaction of phosphorus in a basic medium.

- **Oxidation half-reaction:** $P_4 \rightarrow H_2PO_2^-$

  Balancing P atoms: $P_4 \rightarrow 4H_2PO_2^-$

  Balancing oxidation state ($P^0 \rightarrow P^{+1}$, 1 electron per P, so 4 electrons total): $P_4 \rightarrow 4H_2PO_2^- + 4e^-$
- **Reduction half-reaction:** $P_4 \rightarrow PH_3$

  Balancing P atoms: $P_4 \rightarrow 4PH_3$

  Balancing oxidation state ($P^0 \rightarrow P^{-3}$, 3 electrons per P, so 12 electrons total): $P_4 + 12e^- \rightarrow 4PH_3$

  To equalize electrons, multiply the oxidation half-reaction by 3:

  $3P_4 \rightarrow 12H_2PO_2^- + 12e^-$

  Combine both half-reactions:

  $4P_4 \rightarrow 4PH_3 + 12H_2PO_2^-$

  Divide by 4 to simplify:

  $P_4 \rightarrow PH_3 + 3H_2PO_2^-$

  Now, balance charges using OH^- (since the right side has a -3 charge):

  $P_4 + 3OH^- \rightarrow PH_3 + 3H_2PO_2^-$

  Finally, balance H and O using H_2O on the left side:

  $P_4 + 3OH^- + 3H_2O \rightarrow PH_3 + 3H_2PO_2^-$

  Comparing coefficients: $a = 1, b = 3, c = 3, d = 1, e = 3$.

Verify statements:

- **I:** $a + b + c = 1 + 3 + 3 = 7$ (Incorrect, statement says 5).
- **II:** $b + c - e = 3 + 3 - 3 = 3$ (**Correct**).

- **III:** Let the oxidation state of P in $H_2PO_2^-$ be x:
  $2(+1) + x + 2(-2) = -1 \Rightarrow 2 + x - 4 = -1 \Rightarrow x = +1$ (**Correct**)

2. **Correct Answer: C) Statement 1 is correct but Statement 2 is incorrect**

Statement I is correct: Hydrogen peroxide (H_2O_2) contains oxygen in the -1 oxidation state, which is intermediate between 0 (in O_2) and -2 (in H_2O). Therefore, it can undergo both oxidation and reduction in both acidic and basic mediums.

Statement II is incorrect:

Volume strength = $11.2 \times$ Molarity (M)

For a 10V sample: $10 = 11.2 \times M \Rightarrow M \approx 0.893$ mol/L

Strength in g/L = Molarity × Molar mass of $H_2O_2 = 0.893 \times 34 \approx 30.36$ g/L

$\%(W/V) = \frac{g/L}{10} = \frac{30.36}{10} \approx 3.03\%$ (not 6%)

3. **Correct Answer: D) 3**

By definition, the equivalent weight of an element is the mass that reacts with 8 g of oxygen.

$$\text{Equivalent weight} = \frac{\text{Mass of element}}{\text{Mass of oxygen}} \times 8$$

$$\text{Equivalent weight} = \frac{12}{32} \times 8 = 3$$

4. **Correct Answer: C) In reaction I water is reduced and in reaction II water is oxidized**

Solution:

- **In Reaction I:** Hydrogen in H_2O goes from an oxidation state of +1 to 0 in H_2. Since the oxidation number decreases, **water is reduced**.
- **In Reaction II:** Oxygen in H_2O goes from an oxidation state of -2 to 0 in O_2. Since the oxidation number increases, **water is oxidized**.

5. **Correct Answer: C) Tl**

Down Group 13 (B, Al, Ga, In, Tl), the stability of the lower oxidation state (+1) increases due to the **inert pair effect** (reluctance of the valence *s*-subshell electrons to participate in bonding). For Thallium (Tl), the +1 oxidation state is significantly more stable than its +3 oxidation state.

6. **Correct Answer: A) +2, 0, +2**

The structural formula of carbon suboxide is:

O = C = C = C = O

- The two terminal carbon atoms are bonded to oxygen (more electronegative), giving each terminal carbon an oxidation state of +2.
- The central carbon atom is bonded only to other carbon atoms, giving it an oxidation state of 0.

Therefore, the states are $+2, 0, +2$.

7. **Correct Answer: A)** $2H_2O_2 \rightarrow 2H_2O + O_2$ **and** $Cl_2 + 2NaOH \rightarrow NaCl + NaOCl + H_2O$

A disproportionation reaction is a redox reaction where the same element is simultaneously oxidized and reduced.

- **Reaction 1:** $2H_2O_2 \rightarrow 2H_2O + O_2$
  Oxygen goes from -1 (in H_2O_2) to -2 (reduced in H_2O) and 0 (oxidized in O_2). → **Disproportionation**
- **Reaction 2:** $Cl_2 + 2NaOH \rightarrow NaCl + NaOCl + H_2O$
  Chlorine goes from 0 (in Cl_2) to -1 (reduced in NaCl) and +1 (oxidized in NaOCl). → **Disproportionation**

Both reactions in option A satisfy the definition perfectly.

8. **Correct Answer: B)** $F_2(g) + 2OH^-(aq) \rightarrow OF_2(g) + 2F^-(aq) + H_2O(l)$

Fluorine (F) in F_2 is in the 0 oxidation state.

In OF_2, Fluorine is more electronegative than Oxygen, so F is in the -1 oxidation state.

In F^-, F is also in the -1 oxidation state.

Since Fluorine only decreases its oxidation state from 0 to -1 in both products, it is only reduced. Oxygen in OH^- (-2) changes to $+2$ in OF_2 (Oxidized).